

Neur 340 Brain Metabolism and Mental Health

Numbered Transcript

NEUR 340 • Dr. Chris Palmer

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Lecture 1: Rethinking Mental Illness

At a glance — Palmer reframes mental illness, introducing the metabolic view of psychiatric disorders.

- 1.1 Hi. My name is Chris Palmer. I am a psychiatrist at McLean Hospital and Harvard Medical School. I am here to talk with all of you about a topic that is of deep interest to me. It revolves around mental health and the mental health epidemic, but it ultimately leads to something called metabolism or brain metabolism. It, in fact, is the only way that we can really understand the connections between mental health and physical health.
- 1.2 Before I get started, I just want to be clear that I have no financial relationships with any companies related to the content of this presentation. The only conflict of interest I have, really, is that I get royalties from a book called Brain Energy.
- 1.3 Let me tell you what is in store. First, we're going to talk about the mental health epidemic, in case you didn't know about it. But really, the next eight hours are going to focus on this overarching question: what causes mental illness? This is a question that has perplexed people for millennia, and still to this day, many people can't answer it. That is going to lead us to this concept of metabolism and metabolic health.
- 1.4 We're going to end up talking about treatment and prevention strategies. I want all of that to truly blow your minds. I want you to come away from this course thinking about mental health and mental illness in profoundly different ways than you probably do right now. But I also want this to be useful and practical for all of you. We're going to make this personal. As I talk to you about the science, I'm going to ask you questions so that you can try to assess your own mental health and metabolic health, and hopefully come away from this course with tools and strategies of things that you might try to improve your life and your overall health.
- 1.5 On that front, I'm going to jump right in and we're going to get started. The first question I want all of you to answer is: how would you rate your current mental health on a scale of 1 to 10? One is the worst mental health you can imagine. Ten is the best mental health you can imagine. The next question I'm going to ask you: if you did not give yourself a 10, what could you do to get yourself one point higher? Now, for those of you watching from home, if you need a little more time to think about this, pause and then restart.
- 1.6 The next question I'm going to ask you is, I want you to assess your metabolic health, whatever those two words mean to you. How do you think your metabolic health is on a scale of 1 to 10? One is the worst metabolic health you can imagine. Ten is the best metabolic health you can imagine. Then the same question: what can you do to get your metabolic health one point higher? If you gave yourself a seven, what would it take to get you to an eight?

- 1.7** Now, that's where you all begin. You all have different ranges of knowledge and understanding. What I'm really hopeful for is that over this course, your understanding of those concepts is going to be refined, and your understanding of what you can do to get yourself one or two or even three points higher is going to be greatly expanded.
- 1.8** Now I'm going to jump right in. In case you didn't know, we have a mental health epidemic. Mental and substance use disorders affected about 1 billion people prior to the pandemic. This represents about 13% of the world's population. In the United States and other Western countries, the rates are even higher, affecting about one in five people. But since the pandemic, the rates have gone even higher. Mental disorders are one of the leading causes of disease burden and disability worldwide. The problem is not always a lack of treatment. For some people, it's that our treatments fail to work.
- 1.9** There's this curious thing. The rates of mental illness across a wide range of diagnostic categories have been increasing for decades now. The rates of autism spectrum disorder have quadrupled in the last 25 years in the United States. The rates of ADHD have tripled in the United States since 2010. The rates of major depressive disorder have reached an all-time, ever-recorded high just last year in 2023, both in terms of current prevalence and lifetime prevalence.
- 1.10** In fact, probably no population is more impacted and talked about than our youth. The government does an annual survey of high school students looking at numerous health metrics, including their mental health. They asked high school students, "Have you had persistent feelings of sadness or hopelessness during the past year?" Since 2011, rates have steadily been rising. They tend to be much higher in girls or young women than they are boys or young men.
- 1.11** But in 2021, granted, that was in the peak of the pandemic, in 2021, 57% of high school girls said, "Yes, I have had persistent feelings of sadness or hopelessness over the past year." When these same high school students were asked, "Have you seriously considered ending your life in the past year?" 30% of high school girls said yes. 14% of high school boys said yes. When they went even further and asked, "Have you actually tried to kill yourself? Have you attempted suicide?" In the past year, in 2021, 13% of high school girls said yes.
- 1.12** That leads me to another question, which is this question of who is most profoundly impacted by mental health challenges. I want to take a poll right now just to see, what do you think? Which age group in the United States is the most likely to be prescribed an antidepressant? Of all of the people in that age group, which age group has the highest percentage of people taking antidepressants? Is it children and adolescents up to age 17? 18- to 39-year-olds? 40- to 59-year-olds? Or 60 years old and older?
- 1.13** I want to take a quick survey right here in the studio audience. How many of you think it's children and adolescents up to age 17? How many of you think it's 18- to 39-year-olds? How many of you think it's 40- to 59-year-olds? How many of you think it's

people 60 years old and over? All right. We have wide variation. We have people thinking it's any of those age groups. Over the course of this lecture, I hope that I'm going to inform those statistics and those facts, and I hope that you're going to think about mental health in a very different way.

- 1.14** But all of this begs the question: why do we have a mental health epidemic? There are experts and pundits who have wide-ranging views on this. Some will say, "It's just recognition and diagnosis." It's not that the rates are actually going up, it's that everybody's talking about mental health now. Celebrities are coming out with their mental health challenges. Clinicians are being trained to screen for mental health challenges. Pediatricians are looking for autism spectrum disorder. That's why the rates have quadrupled. It's not because the rates are actually going up. It's just because we are recognizing it and diagnosing it more.
- 1.15** Now, other people say, "Well, no, psychiatry has gotten way out of hand, and it's overdiagnosis." Psychiatrists and mental health professionals are labeling every little human condition, every anxiety, every insecurity as a diagnosis that needs to be treated with a pill or with psychotherapy. It is rampant. Of course, everybody's got ADHD; the rates have tripled. Well, yeah, who doesn't have inattention? Who can pay attention to a book perfectly? Not many of us. We are just overdiagnosing this.
- 1.16** Others will say, "It's the pharmaceutical companies." They are advertising. They are asking, "Have you had these symptoms? Are you a little inattentive? Maybe you need to talk to your doctor about our drug, which will fix you." So it's pharmaceutical companies. They are clearly manipulating the masses. They are selling their products. In order to sell their products, they need higher rates of diagnoses.
- 1.17** Some people just blame Americans and people in general. "Everybody's lazy and whiny." Like, Gosh. Why don't you know Angela Duckworth? Talks about grit. We need a little more grit in our children. People just need to suck it up. Just behave yourselves, folks. Be responsible. Stop complaining. Stop whining.
- 1.18** Others blame social media and screens. The surgeon general of the United States is targeting loneliness. "There is an epidemic of loneliness in the United States." That's what's causing our mental health epidemic. Everybody's lonely. We're all on social media, we're all on screens. Nobody's interacting with each other. Nobody ever picks up a phone and actually talks. If you're lucky, somebody might text you back. That is our social interaction.
- 1.19** Now, some will say, "mental illness, mental health, that's really about trauma." There are experts who will say all mental health conditions are trauma reactions. Some will go so far as to say that everybody with a mental health condition probably has some trauma in their past, and they just haven't dealt with it. And that these are all normal trauma reactions, and that trauma can cause ADHD, trauma can cause bipolar, trauma can cause depression and PTSD, and even schizophrenia. That begs the question: do we

have more trauma in the world? If trauma is at the root of this, does that mean more and more people are being abused and tormented or neglected?

- 1.20** In case you're in the camp that thinks, "Yeah, some of that sounds right. It's overdiagnosis, or maybe it's just better recognition, or maybe it is the pharmaceutical companies." So if any of that sounds right, I want to point out a few more statistics—statistics that are undeniable. Hospital admissions for severe eating disorders—these are the eating disorders in which women are starving themselves. You don't get hospitalized for just a tiny eating disorder. You get hospitalized because you are on the verge of death. In the United Kingdom, the rates are up fivefold in the last 25 years. The suicide rates since 2000: slowly but steadily rising. Overdose deaths in the United States, largely driven by the opioid epidemic, are up exponentially—up over fivefold in the last 25 years.
- 1.21** Now, I want to be clear: we've long known how to recognize death. There's no ambiguity in diagnosing suicide, or diagnosing an overdose, or diagnosing anorexia that results in starvation. Those rates, unequivocally, are going up exponentially. This graph actually tracks a statistic called "deaths of despair." Deaths of despair put together all of the deaths from suicide, alcohol-related deaths, and deaths related to drugs. That can be an overdose, but alcohol-related deaths also include DUIs and killing somebody, or somebody getting killed because they were intoxicated while driving.
- 1.22** This chart shows us the deaths of despair from the year 1900 to 2017. Around the turn of the century, the 1900s, early 1900s, we had very high rates of deaths of despair. That was because of economic turmoil. There were no social supports in place. There was nothing called Social Security. If you were old and you hadn't saved for retirement, good luck. Good luck. You better have kids and if you don't, if you don't have kids who are going to take care of you, good luck to you. You may starve, you may decide life's over. You may have a death of despair from suicide or from alcoholism or something else. If you're disabled, there's no supports for you.
- 1.23** So the rates of deaths of despair were actually pretty high in the early 1900s. They came down significantly. You see a bump around the Great Depression and World War II. The rates went up again. Understandably— the Great Depression, World War II. Record levels of unemployment, one in four people unemployed. People not able to take care of their children. The shame and humiliation of that.
- 1.24** Since about the 1950s, the rates of deaths of despair have slowly but surely been rising—exponentially. We now have over double the rate of deaths of despair that were present during the Great Depression and World War II. That is undeniable. There is, in fact, a mental health epidemic.
- 1.25** So what are we doing about this? We have treatment. The great news is that we have a plethora of treatments. We have numerous medications— we've got antidepressants, antipsychotics, mood stabilizers, anxiolytic medications. We've got lots of different

forms of psychotherapy. Maybe you talk with your therapist about your past. Maybe you do cognitive behavioral therapy. Maybe you do an addiction treatment program, an eating disorder-type therapy. We've got hospitalization and group therapies. In severe cases, we've got electroconvulsive therapy where they will put electricity through your brain. We've got transcranial magnetic stimulation. We've got ketamine. We've got psychedelics.

- 1.26** All of these treatments can work for some people. I want to be clear, there is no doubt these treatments save millions of lives every year. To support this point, I just want to tell you a quick case of somebody for whom this actually works. Let me tell you about John. John is a middle-aged man. He's got a wife and two kids. He works as an engineer. He actually probably overworks, isn't paying as much attention to his wife as maybe he should be, isn't paying as much attention to his kids as maybe he should be. But otherwise, he thinks he's got a really good life.
- 1.27** Then all of a sudden, his wife is having an affair. He is completely distraught. He recognizes, "This is somewhat on me—I have some ownership of this." And he is begging and pleading with her, "Please, give us another chance. Please leave him." She says, "I'm done. It's over. That was long ago. There's no way. I don't want you back. I'm moving on. I found the love of my life. I'm in love. I'm infatuated."
- 1.28** He develops what we would diagnose as severe major depressive disorder, along with a lot of anxiety. He cannot sleep. He's literally getting two hours or less sleep every night. Friends and family are talking to him. They're trying to comfort him. That's worthless. He is going down the tubes fast. Within a month, work is starting to question, "What's going on with you? You're not productive like you used to be. You were our star employee." Friends and family are still trying to help out. Nothing is working.
- 1.29** Within a couple of months, somebody from his family says, "John, you've got to get help." He goes to a psychiatrist. He is prescribed an antidepressant. He is prescribed a sleeping pill, and he starts talk therapy. Within days, the sleeping pill starts working, and he's able to sleep for the first time in months. That helps him get just a little bit of ground under him—he can at least get through the day thinking coherently. Within about three, four weeks, his mood starts improving. He still recognizes this is a catastrophe. "My wife is leaving me. My family is going to be broken. I'm not going to be able to see my daughters all the time." But he starts dealing with it. He takes the steps needed: get a divorce lawyer, gets a divorce.
- 1.30** Within a couple of months, he no longer needs a sleeping pill to sleep. He decides, "I'm going to get in shape." He was a little overweight. He recognized, "I'm single now." He's going to get in shape. He's going to start losing some weight, working out a little. Within a year, the divorce is finalized. They've set up separate households. They're managing the custody okay. He's able to taper off his antidepressant and remain well. He's now lost about 50 pounds, and he's dating again. Many in the mental health field will point to him and say, "This is the success of mental health treatment in 2024." This

man's life, potentially, was saved. He was completely a mess and potentially on the verge of suicidal thinking.

- 1.31 All of this treatment, an antidepressant, a sleeping pill, and psychotherapy, potentially saved his life. It certainly changed his life and improved a lot of other people's lives: his daughters' lives, even his ex's life. It allowed him to move on and get a divorce and be reasonable about it. This gets touted as "Here's what the mental health field has to offer. We can save lives. We've got treatments, we've got pills, we've got therapy."
- 1.32 In fact, a lot of people in the United States are getting mental health treatment. These statistics, published from the CDC, show us the overarching trend. Women are about twice as likely to get mental health treatment as men. Women are more likely to be diagnosed with mental health conditions—in particular depression, anxiety, eating disorders, and others. In any given year, about one in four women is getting some form of mental health treatment. One in four, that means either taking a pill—an antidepressant, an anti-anxiety medicine, or doing talk therapy. About 13% of men are getting some form of treatment.
- 1.33 The sad news is that access to care is abysmal. A lot of people who desperately need help aren't getting help. 93% of people with substance use disorders do not get any help. Over half of adults with a diagnosable mental illness in the United States do not receive any treatment—and we're the United States. Think about what it's like in other countries. 42% of adults with severe mental illness—these are the illnesses like schizophrenia, bipolar disorder, chronic unrelenting depression— 42% report that they are unable to receive necessary care because they can't afford it. If you've got a mental health condition and you want treatment and you want to use your insurance, good luck. Good luck finding a therapist or a psychiatrist who will take your insurance.
- 1.34 The really tragic statistic is that when researchers look at all of this treatment being delivered, a lot of the people who are getting treatment, even if they're paying out-of-pocket, are not getting evidence-based, adequate treatment. They're not getting the right pills. They're not getting the right doses of the pills. They're not getting evidence-based psychotherapy.
- 1.35 But unfortunately, even if you are getting the best mental health care possible, our treatments often don't work. About 20% to 60% of patients—now that's a wide range, and I'll explain that in a little bit— 20% to 60% of people with mental health conditions are labeled treatment-resistant, meaning they don't respond to the treatments that we have to offer. So why such a wide range? How is that even possible? Well, it's possible because, one, it's difficult to define treatment resistance. How many pills do you have to try before we call it treatment resistance? How much psychotherapy do you have to try before? Which diagnoses are we studying? Are we looking at people who are on Medicare and Medicaid? Are we looking at people who are paying out-of-pocket? Are we looking at commercial insurance? All of those things make a massive difference.

- 1.36** But what is largely agreed on in the mental health field is that our treatments do not cure mental illness. Our therapies and treatments are not disease-modifying therapies. That when we are lucky, they reduce symptoms. If you're really lucky, they might put your illness in remission, but far too few people get remission.
- 1.37** If that doesn't really sound believable to you, let me tell you about the largest depression treatment trial ever conducted. It's called the STAR*D Trial. It included about 4,000 outpatients who had major depression. This was a national study—23 different psychiatric and 18 different primary care sites. The trial involved four different levels of treatment. Patients would come in, they would get the first level of treatment, and if that treatment didn't work, then they would get the second level of treatment. If that didn't work, the third level of treatment. If that didn't work, the fourth level of treatment.
- 1.38** How well do our treatments work? With the first treatment, which was an antidepressant, citalopram, about one-third of patients achieved remission. Remission means you no longer meet criteria for major depression. That means two-thirds of people were still clinically depressed despite getting treatment.
- 1.39** Now, when the researchers looked at how many people got better over all four treatment levels, they said, "Well, it looks like it's 67%." 67% of people in this trial got remission after all four treatment levels. That statistic, however, has been challenged. Some other researchers noticed that, "Wait, they changed the criteria for how they were categorizing remission in the middle of the study. You're not supposed to do that in research. And that if they had actually used their initial criteria for remission, only 35% of patients would have achieved remission.
- 1.40** Now, even if you take the 67% at face value, that means one-third of patients after trying four different levels of treatment—that means four different drugs, basically, for your depression— one in three are still depressed. Fully depressed, full-on depression. They have five out of the nine criteria. If you take the other, the revised statistics that says, "Wait, you guys changed your criteria. Let's do what you said you were going to do." That means two-thirds of people who are getting four different levels of treatment for depression are still clinically depressed.
- 1.41** Even if you get a remission, that doesn't mean you keep your remission. The long-term treatment studies indicate that 50% to 85% of individuals who have depression once will get it again, even if you're taking your pills. Even if you're taking your Prozac, Prozac puts you into remission, you can keep taking it, and then four years later, out of the blue, your depression can come back, and everybody's asking, "Well, what happened?" I'm taking my medicine. I'm doing what I'm being told. Why did it come back? Everybody says, "No one knows."
- 1.42** There are other long-term depression treatment studies. This study was one that followed people for 12 years—over 400 people being treated at academic medical

centers for their depression. The researchers basically assessed levels of depression every single week. They asked them, "Are you still having symptoms of depression," every single week for 12 years.

- 1.43 What they found was that 59% of the weeks, people were symptomatically ill. You're getting treatment at academic medical centers. You're getting the best treatment available. You're in a research study, so everybody knows they're watching you. The clinicians are going to be on guard. They're going to be like, "Oh, I better take really good care of you because you're part of this research trial." 60% of the time you are ill. It's hard to have a job. It's hard to go on when your depression symptoms keep coming back or they just never go away. Patients with double depression or low-grade depression that gets worse have worse outcomes.
- 1.44 The reality, the sad reality, is that at the end of that 12 years, they looked at who got remission and stayed in remission. How many of those patients just were the ideal that we all want? Take Prozac, get better, and then stay better. How many got that? 10%. 10% of the people being treated for depression at academic medical centers, 10% got a lasting, durable remission.
- 1.45 What about other diagnosis? That's sad and depressing about depression. What about other diagnosis? Well, for any of you familiar with the mental health field, if you've got bipolar disorder, if you know somebody with schizophrenia, is actually dramatically worse. Bipolar disorder, schizophrenia, they affect anywhere from 1% of the population for schizophrenia, 2% to 5% for bipolar disorder.
- 1.46 One large research trial tracked the outcomes of people with schizophrenia in particular and looked at how many of them actually get better. They tracked over 6,000 patients over just three years. They asked how many of them got symptomatic remission, meaning their hallucinations and delusions stopped. Only one in three got even a period of time where their hallucinations and delusions stopped.
- 1.47 Did you have a decent quality of life? Because that should be part of treatment, right? You should have a good quality of life if you're getting mental health treatment and it's working. Well, that was only one in four. One in four people had a decent quality of life. Are you able to work or function in society? Because that should be part of the treatment. You should be able to get back to your life, take care of yourself. Only 13% were able to do that.
- 1.48 If you put those three things together, the ability to work, symptomatic remission, and quality of life, if you put all three of them and say, "Well, that's the gold standard. That's what we're looking for. We want all of those." Only 4% of people with schizophrenia got that. That's what we would call recovery. It is clear that we desperately need better treatment solutions.
- 1.49 All of this begs the question: why can't we do better? What's the problem? I am going to argue it's because our field has been unable to answer two fairly simple questions.

The first one is, "What exactly causes mental illness?" If you talk to the leading psychiatrist and neuroscientist today, they will say, "No one knows. It's way too complicated." All we know are risk factors, and they will say, they will tell you about the biopsychosocial model, which says that there are biological, psychological, and social factors that all come together to cause mental illness. Well, exactly how does that work? "Oh, nobody knows. It's just too complicated. No one can figure it out."

- 1.50** The second common sense question that our field has been unable to answer is, "What exactly is mental illness? What is it?" Although that probably sounds surprising to many of you, because we have DSM 5, the Diagnostic and Statistical Manual. This is like the Bible of Psychiatry. It's got all of these labels that are household words. It's got diagnoses like schizophrenia, bipolar disorder, and depression, and anorexia nervosa, and alcohol use disorder, which is the new word for alcoholism.
- 1.51** When I say these things, you probably all know what those things are, or at least have a sense. "Well, yeah, I know someone with that. I know what that means." The reality is that those labels are really good at describing symptoms. But there are three problems with our diagnostic labels. The three problems are heterogeneity, comorbidity, and a lack of validity.
- 1.52** Heterogeneity says that two people with the same diagnostic label can be wildly different from each other. An example I talk about a lot because it's just such a glaring example is the example of autism spectrum disorder. You can have a child with autism spectrum disorder who lives in a group home, even at 12 years old, because his parents aren't able to care for him safely. He has profound cognitive impairment. He bangs his head against the wall to the point of bleeding, so the group home puts a football helmet on him to keep him safe. When you ask the best professionals, "What's wrong with this child?" They'll say, "He has autism."
- 1.53** Another person with that exact same diagnostic label can be a billionaire businessman that we see in the news every day. Does anybody really believe that those two human beings have the same brain condition? It certainly doesn't impact their lives in the same ways. I mean, maybe they're both a little socially off, but the variation is profound.
- 1.54** I could go on with people with depression. People with depression—some overeat and oversleep when they're depressed. Others do the exact opposite. They undereat, they lose their appetite, they lose a tremendous amount of weight, and they can't sleep. Some are suicidal, some aren't suicidal. Some are sensitive to rejection, others aren't not so much. I could go diagnosis by diagnosis. They can be wildly different from each other because all of the diagnostic criteria, you get a menu of symptom options, and all you need are 60, 70% of that menu in order to make the diagnosis. That leaves for wide-ranging variability.
- 1.55** The second problem is something called comorbidity, which says that if you have one mental disorder, you're more likely to have another. The National Statistics of

Population Health says that if you get diagnosed with one mental illness, there's a 50% likelihood that you have more than one. But if you're actually seeking treatment for mental illness, on average, people have about three to four different diagnostic labels.

- 1.56** That means that if you have depression, you are more likely to also have OCD and a substance use disorder, for example. If you have an eating disorder, you're more likely to have OCD and a substance use disorder. If you have autism, you're more likely to have OCD and a substance use disorder. On the surface, although depression, eating disorders, and autism are wildly different from each other, when you actually start to look at real human beings who have these diagnostic labels and their comorbidity and all of the different symptoms that they have, they start to look maybe a little more similar than they do differently from each other.
- 1.57** That leads to the final problem with DSM, which is a lack of validity. This comes from none other than Dr. Tom Insel who is the former director of the NIMH. He has basically said our field is never going to make progress as long as we cling to these diagnostic labels because they are not valid disease constructs. The reason we're not making progress is because we cling to these labels. These labels describe symptoms, but otherwise, they are not valid disease constructs.
- 1.58** When you look at the biology of them, when you look at the family history, when you look at the genetics, when you look at the neurotransmitter systems, when you look at any of it, they seem to have more in common than they do different. All of that says, wow, the mental health field is really a mess. Based on the statistics I shared with you, it's a tragic mess. People are dying. People are suffering.
- 1.59** But there are several lines of research now that suggest something both shocking and, in my mind, filled with hope. And that is that mental disorders may, in fact, share a common pathway or a common pathophysiology. This was first proposed by Dr. Leahy in 2012, who looked at all of the internalizing and externalizing mental disorders. These are things like depression, anxiety, ADHD, substance use disorders. He looked at all of them and they looked at everything known about them. They looked at all of the neuroimaging studies, all of the genetics, all of the risk factors, trauma history, how does it all fit together?
- 1.60** What he found was that they appeared to all share something in common with each other. We called this the general factor that was leading to these mental disorders. Now, Caspi and Moffitt expanded on this in 2018, and they looked at all of the different labels in DSM, and they found the same thing. They said, "Whoa, it's not just the internalizing and externalizing mental disorders; it's all of the diagnostic labels in DSM. They all appear to share some common pathophysiology."
- 1.61** They looked at genetics, family histories, upbringing, trauma, all of that, and they all appeared to converge in some way. They called this the p factor. Then, just last year, Brent et al. said, "Wait, it's not just these mental illnesses; it's actually a lot of physical

illnesses, too. They appear to share this common pathophysiology." Now, the problem with all of that research is that none of it was designed to tell us what the common pathophysiology is.

- 1.62** These were just mathematical models looking at everything known, looking at all of the risk factors, all of the neuroimaging, all of the genetics, everything known, and seeing how does it all sort? Which genes increase risk for which mental disorders? When they do that, it all converges on: it's all of them. It's all of them. How is that possible? Their research couldn't tell us what this common pathophysiology is. It could only tell us that there appears to be a common pathophysiology.
- 1.63** In some ways, if this sounds bold and audacious, just so you know, in 2009, the National Institute of Mental Health proposed the Research Domain Criteria, which said, "We've got to get rid of our diagnostic labels like schizophrenia and bipolar and depression. We have to get rid of them. We have to go to a transdiagnostic model." Transdiagnostic model means we need to think about psychopathology in broad terms, across all of the different diagnostic labels.
- 1.64** There are competing models. There's something called the Hierarchical Taxonomy of Psychopathology, HiTOP. That's a competing model. These models get really complicated fast. My goal isn't to go into them. My goal is simply to share with you that leading authorities in the mental health field have come to this realization decades ago, that mental illnesses share more in common than they do different, even though we give them different labels and even though the symptoms can be wildly different from each other.
- 1.65** But I shared with you the example of the autism spectrum. Even when you get the same label, the symptoms can be wildly different in two different individuals with the same diagnostic label. Again, that leads us with, gosh, the mental health field is a mess. Let me be clear, that is precisely where I was in 2016. I had been a psychiatrist since 1995 at that point. I'd been practicing for over 20 years at one of the leading psychiatric hospitals in the world: Harvard Medical School, the flagship psychiatric hospital of Harvard Medical School.
- 1.66** I was leading our education. I had access to the best minds possible. I was inviting these expert speakers, neuroscientists, psychiatrists, therapists, others. I was inviting these world-renowned experts from around the world to speak at conferences that I was organizing and hosting. I had access to the best minds. I have to tell you, at that point, I was pretty hopeless. I was very frustrated. I wasn't hopeless for my patients. I was still doing the best I could. But I had patients often ask me, "Dr. Palmer, what caused this mental illness? Why do I have this? What happened to my son? Why is my son hallucinating now? He was normal four months ago, and now he's hallucinating. Do something. Why aren't you doing something?"

- 1.67** I would just have to stand there and say, We don't know. We don't know what's causing this. Tragically, I recognized—I'm going to try lots of treatments. I really am, and I'm going to try one after another, and I'm really going to cross my fingers and hope one of them works. But I knew the statistics. I knew there's a good chance we're not going to find a treatment that's going to save your son. Once we diagnose somebody with schizophrenia, it really is a life sentence. It's a life sentence for an innocent person.
- 1.68** All of that changed in late 2016 when one of my patients, who was diagnosed with schizoaffective disorder, which is a cross between schizophrenia and bipolar disorder, asked me to help him lose weight. This man had been my patient for eight years at that point. He had been seen by the best psychiatrists. I'm not even counting myself. I'm okay, but I'm not counting myself in that. He had been seen literally by some of the best psychiatrists that Harvard had to offer. He had tried numerous medications. He had been in a private pay residential treatment program, and nothing worked.
- 1.69** When I first started working with him, he was on three different antipsychotic medications at higher than the FDA-approved dose. He was on antidepressants, stimulants, anti-anxiety medicines. He was sleeping 16 hours a day. He was still psychotic. Hallucinating, delusional all day, every day.
- 1.70** He was a near hermit. He was terrified to go out in public. When he went out in public, he was convinced everybody was part of the conspiracy. They were all trying to hurt him. They were mocking him. Somebody might try to kill him at any moment. He couldn't ride on a bus. He couldn't go to a movie theater because he was convinced they would be there in the dark waiting for him. So he stayed home.
- 1.71** But unfortunately, staying home doesn't free you from your mind. Your mind still torments you. And he lived at home alone, tormented. And for some reason, he gets it in his head, "Dr. Palmer, I'm never going to get a girlfriend if I don't lose some weight." Now, he weighed 340 pounds. He had gained a massive amount of weight from the medications we put him on.
- 1.72** So I'm thinking, "Wow, okay, you want to lose some weight. Great. It's a goal. Maybe it'll help you in some way." He had actually tried several other methods of weight loss. He had tried calorie restriction. He had tried a couple of other different diets. None of them worked. So, at this point, I recommended the ketogenic diet to him, which is a diet that is low in carbohydrates, moderate in protein, and high in fat.
- 1.73** I'm really thinking, I just want to help the poor guy lose some weight. I'm really not all that hopeful that it's even going to work for weight loss, and I have no expectations that it's going to do anything else for him. Within two weeks, not only is he losing weight, but he is becoming a lot less depressed. He's less of a zombie. He's actually making eye contact with me in a significant way for the first time ever in the eight years I've seen him. He's starting to talk more, and I'm thinking, "What's gotten into you? You

have a lot to say. Usually, you just sit there like a bump on a log, and I have to pull words from you just to get you to communicate with me."

- 1.74** He's laughing and making jokes, and his jokes are really funny. I'm thinking, "What the hell? This is amazing." But he's still hallucinating. He's still delusional, "They're still after me. I'm not going to take it. They can't humiliate me. They can't torment me like this." But he can at least make some jokes every now and then. He stays on the diet. About two months go by, and now he's lost a significant amount of weight.
- 1.75** He spontaneously reports to me, "Dr. Palmer, you know those voices that I hear? They're going away." Another two weeks go by and he says, "Dr. Palmer, you know how I talked about those families that were targeting me and how they were tormenting me and how they could control my mind and all of that?" I'm initially thinking, "Yeah, we've talked about that a million times. We're going to talk about that again." He says, "I don't think that's true. Now that I say it out loud, it sounds crazy. Maybe it never was." He says, "Maybe I have schizophrenia, like everybody has been trying to tell me for the last 10 years, and it's going away."
- 1.76** That sent me on a journey to understand what in the hell just happened. That journey is going to be the remainder of this course.

Lecture 2: Brain Energy Theory

At a glance — The brain energy theory, seeded by a patient's dramatic recovery.

- 2.1 Mike's phenomenal recovery led me, initially, to treat other patients with schizoaffective disorder. When I saw the exact same response, an even better response in some of those patients, I began publishing, began collaborating with researchers around the world. That whole journey led me to develop what I'm calling the brain energy theory, which suggests that mental disorders are metabolic disorders of the brain, that when we look at all of these massive, perplexing questions facing the mental health field, what is the common pathway?
- 2.2 What is the common pathophysiology? Why do all of these disorders share the same risk factors? When we look at all of that, the only way to unite all of it is through this thing that we call metabolism. And that metabolic dysfunction in cells can actually explain the symptoms of mental illness. It is the only way to integrate the biopsychosocial risk factors for mental illness. And once and for all, we need to integrate what we call mental illness and physical illness. And this entire course is going to take you on that journey, lots of detail along the way, in a wide range of the biopsychosocial topics.
- 2.3 But before I do that, I just want to read a quote from Albert Einstein. "Creating a new theory is not like destroying an old barn and erecting a skyscraper in its place. It is rather like climbing a mountain, gaining new and wider views, discovering unexpected connections between our starting point and its rich environment. But the point from which we started out still exists and can be seen, although it appears smaller and forms a tiny part of our broad view gained by the mastery of the obstacles on our adventurous way up."
- 2.4 So, in case you wandered off while I was reading some of that, what he means is that, when we create a new theory, it's like we're all at the bottom of a mountain, and we all know what we know, and we believe what we believe. And then when you create a new theory, it doesn't replace that. It can't replace that, because we all know what we know, and we observe what we observe, and other theories are out there. So creating a new theory has to unify; it has to integrate it. You have to climb that mountain and see how this theory connects with this theory, connects with this theory. Only when you're at the top of the mountain can you see that.
- 2.5 When it comes to a theory of mental illness, that means all of the biopsychosocial risk factors are still in play. A new theory doesn't replace them. A new theory helps us understand how do they fit together. That is the journey I'm going to try to take you on now. We have to start this journey with a simpler model of thinking about mental illness. The easiest model I can think of to demonstrate this concept is the concept of pain.

- 2.6** Pain is a normal human sensation. When we injure ourselves, we will have pain. There's no way around it. It's uncomfortable. We can treat it. We sometimes take medications or we might put wound healing or things on it. It's not that we ignore pain. We pay attention to it. But pain is normal. It is not a disorder. It is not a disease. And yet you can have pain disorders in which the pain system is malfunctioning. The pain system is firing when it shouldn't and producing the sensation of pain.
- 2.7** In other cases, the nerves that produce pain aren't working at all. This happens in people with severe diabetes in their feet. They can step on a nail and not even feel it. That, too, is a disorder of the pain system. The pain system isn't working. So pain can be normal. It can be extreme. If you get surgery, you're going to have extreme pain, but that's not a disorder or a disease. But you can have disorders of the pain system in which the pain system isn't functioning properly, and it either causes suffering or causes impairment, that I can't feel a nail in my foot. And those people are going to have higher rates of injury, even though they can't feel anything.
- 2.8** What I'm arguing is that mental disorders fall into the exact same camp. Every symptom listed in DSM can be normal under the right circumstances. Depression. If your wife leaves you, you're going to be depressed. If you're not, you didn't really love her. Anxiety. If people are bullying you, teasing you, mocking you, you're going to have anxiety. That's normal. Even hallucinations. Hallucinations aren't normal. Well, we all have hallucinations. At night, when we dream, we hear things and see things that aren't there. Our brains hallucinate.
- 2.9** What about paranoia? That's not normal. Well, yeah, it is. You watch a scary movie, like a really scary movie, and then try walking through the dark alley and see how paranoid you might be. Every sound is going to terrify you, and you're going to be imagining there's the bogeyman right behind that wall. What I'm arguing is that all of the symptoms of mental illness are, in fact, normal brain functions that have a time and a place when they are supposed to be operating; and that mental disorders are those same brain functions gone awry, that they are not working. Just like the pain system can fail to work, the brain symptoms or the brain is producing mental disorders.
- 2.10** To be more explicit, what is a mental disorder? One, the brain is not working properly. There's something wrong with the way it's working. Two, has to result in symptoms. Three, this malfunction has to occur over a period of time. It can't just be a one-time deal because we all hear things every now and then. You're in the shower and you think somebody was just yelling at you and you're like, "Did you say something?" "No. What are you talking about?" We all hear things. You might even go so far as to say that was an illusion or a hallucination, but that's not a mental illness. It has to occur over a prolonged period of time, and the symptoms have to cause suffering or impairment in function.
- 2.11** So, brain disorders. How can we even conceive of that? As I was developing this theory, I recognized, OK, I just need to take a step back and think, if there really is a common

pathway or a common pathophysiology, there's a common pathophysiology between autism and Alzheimer's, between anorexia and schizophrenia, and Alzheimer's, and autism, how on earth is that even possible? How can one thing connect all of those? Because some of them occur in neurodevelopment and others occur in old age.

- 2.12** In order to be classified as a brain disorder, I would argue that there are five buckets of dysfunction that can account for all of the mental disorders. The brain, some parts of the brain, have to become overactive or hyperexcitable. That's just like the pain nerve. If that nerve that controls the sensation of pain is hyperexcitable or hyperactive, you might experience pain when you wouldn't otherwise. What happens if your anxiety pathways become hyperexcitable? You might experience anxiety for no reason, out of the blue.
- 2.13** Other brain regions can become underactive, like that diabetic foot. They're not working. I'm supposed to be paying attention in class. The parts of my brain that should be working, helping me pay attention, aren't working. I get labeled ADHD. Or the parts of my brain that are supposed to be helping store and retrieve memories aren't functioning properly, and I might get labeled with dementia or cognitive impairment or concentration difficulties from depression. Overactive and underactive brain regions, those account for the bulk of the symptoms that we see in mental illness.
- 2.14** But there are some brain functions that become absent, either due to neurodevelopmental problems or neurodegenerative problems. Absent would be the diabetic foot that, no matter how hard you try, you cannot get it to feel any pain. A stroke victim might have that. The nerve doesn't work at all. So neurodevelopmental disorders can do that, that the regions of the brain, the connections in the brain that are supposed to produce certain traits or certain behaviors or certain functions just never developed normally. So these people do not have the capacity to actually perform those functions.
- 2.15** We see this in learning disorders. We see this somewhat in autism. These traits, these absent functions, they don't wax and wane. Most symptoms of psychiatric disorders wax and wane. If you have hallucinations you don't have hallucinations all day, every day. You have them some of the time. If you have depression, you're not actually depressed 100% of the time in the same way. It waxes and wanes. You can have good moments and bad moments.
- 2.16** But with neurodevelopmental and neurodegenerative traits, those can become permanent. Somebody with Alzheimer's has permanent deficits in their memory. They don't wax and wane. They can still have good days and bad days, but some of the memory deficits become permanent. That's because those brain cells just aren't there anymore to perform those functions. Then there's this broad category called maintenance problems, in which the brain cells are supposed to maintain themselves, meaning get rid of waste products, get rid of things like tau and amyloid.

- 2.17** We know from a broad range of neuropsychiatric disorders that there are mild maintenance problems and sometimes significant maintenance problems in brain cells. Sometimes we can identify these under a microscope, and usually those are neurological disorders, and sometimes we can't. When developing this theory, I recognized, OK, I have to somehow try to explain all of this. This is where our field is at. I have to explain, why would parts of the brain become overactive? Why would other parts of the brain become underactive?
- 2.18** Why would sometimes people have absent brain functions due to neurodevelopmental or neurodegenerative disorders? What would account for all of these maintenance problems in the cells? This is not a small task to try to unite, but that's where we're at. And what I'm arguing is that metabolism can unite all of those. What exactly is metabolism? When you say metabolism, Dr. Palmer, what is that? A lot of people have this simplistic notion of metabolism. They think it's burning calories. If you burn a lot of calories, then you're going to be thin. If you don't burn many calories as we get older, you're going to gain some weight no matter what you do. That's what metabolism is, burning calories.
- 2.19** There is no doubt it involves burning calories. Some people think about it as metabolic syndrome, high blood pressure, obesity, diabetes. That metabolism has something to do with those things. Well, yes, it does have something to do with those things. But in fact, metabolism is so much more than all of that. Metabolism is the process that all living organisms use to convert food into energy or building blocks used to maintain or grow cells. It also involves the management of waste products. Metabolism is fundamental to the definition of a living organism. It is extraordinarily complicated.
- 2.20** So, now let me ask the question, but we talk about metabolic health. I've heard that term. I asked you that question. Metabolic health. What is metabolic health? Again, a perplexing question when we really get down to it. I'll tell you what we know. There is no clear definition. Some equate it with being overweight or obese. That's the metabolic thing. If you're overweight or obese, then you've got a metabolic problem, and if you're normal weight, then you clearly don't.
- 2.21** Others equate it with metabolic syndrome. Metabolic syndrome includes three or more of the following. Abdominal obesity. If you're a man, maybe more than 40 inches around your waist. If you're a woman, maybe more than 35 inches around your waist. High blood pressure, high triglycerides, low HDL cholesterol, and high fasting glucose. So glucose, blood pressure, lipids, and abdominal obesity. If you have three out of five of those things, you have metabolic syndrome. Metabolic syndrome increases your risk for developing type 2 diabetes, obesity, cardiovascular disease. It shortens your life.
- 2.22** In case you didn't know, we have a metabolic health crisis in the world. Let's start with obesity and overweight. In 2020, 2.6 billion people were overweight or obese. That is 38% of the world's population. This includes countries, third-world countries, where everybody's starving to death. Throw them in, 38% of the world's population,

overweight or obese. By 2035, the projections are that over 4 billion people will be affected. One in two people on the planet will be overweight or obese.

- 2.23** The prevalence of obesity itself is anticipated to rise from 14% to 24% of the population over this same period of time. That means one in four people will be labeled obese, not just overweight. The rising prevalence is expected to be steepest among children and adolescents. Our children and adolescents, the rates are going up much more sharply than they are in adults. The global cost of overweight and obesity will hit \$4.3 trillion per year by 2035. This chart just shows us the obesity rates in children.
- 2.24** What about diabetes and prediabetes? Diabetes and prediabetes is typically diagnosed based on your fasting blood glucose level. The chart that I'm showing on the screen gives people some specific numbers and different tests that people can do to diagnose whether they have prediabetes or diabetes. But here's the shocking news. About 50% of United States adults currently has prediabetes or diabetes. One in two US adults.
- 2.25** In fact, this study looked at all of the biomarkers of metabolic syndrome. How many US adults don't have any of the five biomarkers of metabolic syndrome? 7%. 7% of US adults are metabolically healthy. 93% have at least one of the symptoms. Now, that doesn't give them the full-on metabolic syndrome because you have to have three out of five to get metabolic syndrome, but they're on their way. 93% are on their way if they're not there already.
- 2.26** But in fact, metabolism, metabolic health, is so much more than just obesity and diabetes. Poor metabolic health shows up in every organ, tissue, and cell. Poor metabolic health is associated with all causes of death, premature mortality. It is associated with obesity. Paradoxically, it's associated with starvation and malnutrition. If you are severely underweight, you can have a metabolic problem due to malnutrition.
- 2.27** Type 2 diabetes, high blood pressure, poor lipids, cardiovascular disease, gall bladder disease, liver disease, kidney disease, our joints, osteoarthritis, sleep apnea, many types of cancer, autoimmune disorders, eye conditions, skin conditions, polycystic ovary syndrome and infertility, bacterial and fungal infections, the immune system is affected. Foot health, oral health, hearing loss, pain disorders. All of these are signs and symptoms, diagnoses associated with poor metabolic health.
- 2.28** What about the brain? We almost never talk about the human brain. Now I'm going to take you on a little whirlwind tour to try to persuade you that the brain is an organ, too, and that the brain is affected just like the liver and the kidneys and the joints and the immune system and every other cell, organ, and tissue. But when the brain malfunctions, what are the signs and symptoms of that? What I am arguing is the signs and symptoms are all laid out nice and plain as day in this book called DSM-5-TR. The signs and symptoms of neuropsychiatric disorders are the signs of brain dysfunction.

- 2.29** I'm first going to start with some correlational data. Now, correlation doesn't equal causation. But when we see strong bi-directional correlations, it at least raises the possibility that there might be a common pathophysiology between different diagnoses. In fact, almost all of the mental illnesses have strong bi-directional relationships with obesity, diabetes, cardiovascular disease, and Alzheimer's disease.
- 2.30** What do I mean when I say bi-directional? I mean that if I go to a psychiatric hospital and look at a group of patients there for their psychiatric conditions, they are more likely to currently have or in the future be diagnosed with obesity, diabetes, cardiovascular disease, and Alzheimer's disease. But it works the other way. If I go to a Weight Watchers group, they're more likely to be diagnosed with a mental illness. If I go to a cardiovascular disease unit, they're more likely to have some signs and symptoms of mental illnesses. If I go to a nursing home and look at Alzheimer's patients, they're more likely.
- 2.31** Let me walk you through some of those statistics. Major depression, bipolar disorder, schizophrenia are all associated with higher rates of obesity. People with depression, even though some people lose their appetite and lose weight with depression, on average, they have a 20% increased risk for obesity. But people with schizophrenia and bipolar disorder have dramatically higher rates, about three times the rates.
- 2.32** One study followed people first diagnosed with bipolar and schizophrenia, followed them for 20 years. The overwhelming majority were not overweight or obese when they got diagnosed. Two-thirds of patients with schizophrenia and over half of those with bipolar disorder at the end of 20 years had become obese. Individuals with a higher BMI exhibit alterations in brain function.
- 2.33** This large population study, over nine million patients, followed longitudinally, based on hospital admission stays—if you are diagnosed with obesity, you have increased risk for a large spectrum of psychiatric disorders across the life span. This includes nicotine addiction, schizophrenia, schizoaffective disorder, depression, anxiety disorders, personality disorders, eating disorders, somatization disorders, essentially all of the mental illnesses. If you first get diagnosed with obesity, you're at higher risk for essentially all of the mental illnesses.
- 2.34** In obesity itself, even if we look at one diagnostic category, we look at people with bipolar disorder, if one person is thin and the other is obese, the one with obesity is more likely to have mood episodes. The obesity appears to be affecting their mood symptoms. What about diabetes? The connection between mental illness and diabetes has been known since the 1800s. In some ways, this is nothing new. It's about rediscovering what people knew over a century ago.
- 2.35** People diagnosed with schizophrenia are three times more likely to develop diabetes as the general population. People diagnosed with depression, 60% more likely. Now, a lot of people say, "Well, it's the pills. It's the pills that we're prescribing." They cause

weight gain, they cause diabetes. And in fact, they do. We're going to talk about that later in the course. But some researchers actually looked at people when they first showed up at the emergency room the very first time, no pills prescribed yet. Do they have signs of insulin resistance or diabetes? Sure enough, they do.

- 2.36** What about cardiovascular disease? Depression has long been known to be a risk factor for heart attacks and strokes; and healthy people, if you have depression, it increases your risk by about 50 to 100%. If you've already had a heart attack and you have depression, it about doubles your chances of another heart attack. What about the other way around? People who have coronary artery disease, about 20% of them have major depression. People with congestive heart failure, about 33% of them have major depression.
- 2.37** Now, a lot of people say, "Well, Dr. Palmer, that's just because having a heart attack really sucks. Wouldn't you be depressed if you had a heart attack? Congestive heart failure, that really sucks. And so, of course, that's why they have depression." No, a lot of these people were depressed before they had their heart attacks. Depression is part of the metabolic pathology. This study looked at increased risk of cardiovascular events in youth with mental disorders. And across the board, across all diagnostic categories, the rates were increased anywhere from 50% to over 200%. All of the categories, depression, anxiety, eating disorders, personality disorders, PTSD, all of them are associated with increased risk for heart attacks.
- 2.38** What about Alzheimer's disease? What does that have to do with anything? Well, when you think about Alzheimer's disease, Alzheimer's disease is almost certainly a metabolic brain disorder, and there's an overwhelming amount of evidence for that. You can look at all the risk factors for Alzheimer's. Who's more likely to develop Alzheimer's? People with type 2 diabetes or even type 1 diabetes. If you have obesity in midlife, if you have cardiovascular disease or any of the risk factors for cardiovascular disease, a lack of exercise, poor sleep.
- 2.39** Then there's this gene, APOE4. If you have one allele, you're much more likely to develop Alzheimer's. If you have two alleles, you're a lot more likely to develop. What does that gene have to do with anything? Well, that gene is associated with the transportation and metabolism of fat and cholesterol. It is a metabolic gene. Brain glucose dysregulation has been described in people with Alzheimer's disease so that some people are calling Alzheimer's type 3 diabetes.
- 2.40** What does Alzheimer's have to do with mental illness? Everything, believe it or not. So depression is a significant risk factor for Alzheimer's disease. If you have a history of depression, that increases your risk of dementia by about two and a half fold. If you are older and you develop new-onset depression, if you're 70 years old and you develop new-onset depression, for the most part, you should assume that is the first sign of your dementia. It is not your loneliness.

- 2.41** Everybody thinks, "Oh, Dad's lonely. Oh, Dad's struggling with his retirement." No. You can think of those things, and those things might be true, and we can talk about that. But the first thing you should think of is, "Oh, my gosh, is Dad starting to have dementia?" Because in fact, there is a greater than 15-fold increased risk of dementia diagnosed six months later. Depression and other psychiatric symptoms are often the first sign of dementia in older adults.
- 2.42** What about schizophrenia? A large population study of over 8 million people found that, if you have schizophrenia and live to the relatively young age of 66, you have over a 20-fold increased risk of also being diagnosed with dementia on top of your schizophrenia. Another study looked at, well, OK, that's depression and schizophrenia. What about everything else? Large population study, over 6 million people. Oh, it turns out, all of the mental disorders listed in DSM-5 are associated with increased risk for dementia later in life. Anywhere from a 50% increased risk up to a 20-fold increased risk, as I just shared with you.
- 2.43** What about the other way around? What if you develop Alzheimer's disease? What are the chances that you'll have mental health symptoms? Essentially, close to 100%. People with mental—one study found 97% of patients with Alzheimer's disease have some symptoms of mental illness. Now, that can be depression, anxiety, agitation, ruminations, insomnia. Forty to 50% of people with Alzheimer's disease will have hallucinations and delusions. Those are the hallmark symptoms of what we call schizophrenia.
- 2.44** I want you to think about two of those statistics. If you're diagnosed with schizophrenia and you live to 66, 20-fold increased risk of new-onset dementia. If you get diagnosed with dementia, 50-50 chance that you're going to have the hallmark symptoms of schizophrenia. But in fact, again, metabolism affects the entire body, and mental disorders increase your risk for a wide range of physical health conditions. This study looked at over 6 million people, and there are numerous physical health conditions, all of the metabolic ones that I described, but also autoimmune disorders and other types of disorders.
- 2.45** Some researchers looked into this. They took a population of 85,000 adults who were diagnosed with neuropsychiatric disorders, and they had a wide range of lab tests. They all had brain imaging scans. They had all this stuff done to them. They compared it to 87,000 healthy controls. What they found was that, actually, there aren't that many differences in the brains of people with mental illness compared to the healthy controls. But there are stark differences in body health, between people with mental illness and healthy controls, particularly of the metabolic, hepatic, and immune systems.
- 2.46** So when the researchers looked at massive numbers of people, the most striking differences were abnormalities in metabolic and immune system function. And hepatic just means your liver. What does your liver have to do with your brain? By the end of

this course, I hope you're going to realize, a lot more than anyone probably thinks. Now, in some ways, this is nothing new. People with mental illness are dying early deaths. We have long known this for people with chronic severe mental illness—schizophrenia, bipolar disorder, chronic depression.

- 2.47** Those people die anywhere from 7 to 30 years early deaths, depending on what study you look at and what population they're looking at. That can be like one-third of your life that you're losing, just because you've got this mental illness. But in fact, large population study published in The Lancet a couple of years ago—oh, wait, it's not just those disorders. It's all of them. Every label in DSM-5 is associated with premature mortality. On average, men are losing 10 years of life, women are losing seven years of life. What are they dying of?
- 2.48** Everybody always thinks suicide, and yes, the suicide rates are higher in people with mental illness, and yes, that plays a role in the statistics, but the primary cause of death, and the mentally ill, are heart attacks. They are dying early deaths of metabolic disease. Why? It's not that this has gone unnoticed, and it's not that people don't care. The World Health Organization put together a task force, and in 2016, they issued this report. It appears to be factors related to the individual's behavior, meaning the mentally ill, overeat and underexercise.
- 2.49** Or their disorders. Maybe it's the medications or the symptoms of mental illness itself. Those mentally ill are smoking and sitting on the couches watching TV. They're not doing anything. Problems at the health system level. The mentally ill don't go to their primary care doctors and take their pills like they should. They're not following up. Wider societal issues. The mentally ill are more likely to be poor and unemployed. We know from large population studies, and I'll get into this a little bit in another talk later in the course, people who are poor and unemployed are more likely to die early deaths. So maybe that's it.
- 2.50** I actually believe this is about blaming the victim. The World Health Organization is essentially saying the mentally ill are lazy slobs. They're overeating, they're under-exercising, they don't go their doctors, and they're not working, and if they just pulled themselves up by the bootstraps, if they just ate a little less, if they just tried, if they just went out for some walks, if they just did this, everything would be fine. I am arguing, we are, in fact, missing the elephant in the room, because metabolic disorders, metabolic problems, have been identified in the brains and bodies of people with mental illness for over a century. For over a century, this data has been accumulating.
- 2.51** The basic science research dates to the 1930s, when researchers figured out that there were abnormalities in glucose metabolism in people with schizophrenia and bipolar disorder. There is a plethora of basic science, neuroimaging, and other studies. You know all those neuroimaging studies that we're doing? Guess what they're measuring? They're measuring brain metabolism. That's what functional MRIs, spec scans, PET

scans, that's what they're measuring. They're measuring brain metabolism. We are seeing differences in the mentally ill.

2.52 The diagnoses are all over the map—depression, anxiety, PTSD, schizophrenia, bipolar, anorexia, alcohol use disorders. What do I mean when I say metabolic problems? This is where things get so complicated like that metabolism graph. Differences in lactate, differences in glucose metabolism, differences in the levels of ATP in cells, differences in reactive oxygen species, levels of NAD to NADH, inflammation, cortisol, brain insulin, all of these different things.

2.53 I don't need you to worry about the details of that. I mentioned that, if you're a scientist and a nerd, and you really want to go just Google it, and you'll see literally thousands and thousands of studies documenting those types of metabolic abnormalities in the brains and bodies of people with mental illness. All I need you to take away is that that is metabolism. There are problems in this complex network of metabolism in the brains and bodies of people with mental illness. That is the take-home message.

2.54 Now, some people really have trouble with this. Some people are like, "Dr. Palmer, my brother has schizophrenia. The pills made him gain 70 pounds. He was thin his whole life. He went on the pills and he gained 70 pounds in a year. It's the pill's fault." Others will say, "Well, he also sits on the couch and eats junk food. He binge-eats junk food all day. He doesn't exercise. I try to get him to go out, and he won't do it."

2.55 The pills play a role, and we're going to talk about that. But not having the energy to exercise, having an insatiable appetite, those are symptoms of metabolic dysfunction. It is not his fault. He is not a lazy slob. He is metabolically impaired, and that is why he can't get off the couch. That is why he's eating so much.

2.56 This study in children followed children from age one to 24, and it tells us it is unequivocally not just the pills. The pills are adding insult to injury. This study followed over 5,000 children. For 24 years, those with the highest levels of insulin resistance beginning at age 9 were 500% more likely to have a psychosis at-risk mental state when they turned 24, and they were 300% more likely to already be diagnosed with bipolar disorder or schizophrenia. This says that, in some people, the metabolic problems can start first, and then the mental health brain symptoms can come next.

2.57 In fact, it is not a linear relationship. A lot of people struggle with this. They say, "So, Dr. Palmer, are you trying to say that all mentally ill people are fat and overweight and obese?" No, not saying that. "Are you saying that all fat people are mentally ill?" No, I'm not saying that. What I'm saying is that mental health and metabolic health are in a cycle, and they feed each other. Normally, we think about mental health in two simplistic ways that align with the treatments that we have to deliver.

2.58 Mental illness is due to one of two big buckets. It's either a chemical imbalance. You've got a chemical imbalance, and what do you do for that? Well, you take a pill. You need

a pill. Go to a psychiatrist and take your pill. Or it's trauma, stress, adversity. What do we need for that? Psychotherapy. A little bit of psychotherapy, we'll fix that right up. That's mental illness. Chemical imbalances, neurotransmitters, or stress, childhood adversity, trauma, loneliness. Go to your psychotherapy, that'll fix you up. That's mental illness. That describes the whole thing.

- 2.59** Now, what causes metabolic problems? What causes obesity and diabetes? Well, that's a diet and exercise problem, Dr. Palmer. If everybody just did the right diet, ate a little less, exercised a little more, we wouldn't have this obesity epidemic and diabetes epidemic that we have. That's a diet and exercise problem. Neither is so simple. In fact, we have long known that there's a both category, factors that influence both mental and metabolic health, genetics, epigenetics, hormones, inflammation, sleep, the gut microbiome—all of those things play a role.
- 2.60** But what I'm here to tell you is that childhood adversity, trauma, increase your risk for obesity. They increase your risk for type 2 diabetes. They increase your risk for cardiovascular disease. This is well established. We call these social determinants of health. But what about the other way around? Does diet and exercise have anything to do with mental illness? Well, that is just nonsense, right? Well, that can't possibly play a role. There is an emerging field called nutritional psychiatry, metabolic psychiatry, the intersection of metabolism and mental health. I am here to argue that, in fact, diet and exercise and lots of other factors can play a profound role in mental illness.
- 2.61** Now, I want to give all of you a chance to assess your own metabolic health. I want you to take out your pads of paper. If you're at home, take out a pad of paper or get something to take notes. Do you have metabolic syndrome? Now, again, only 7% of Americans can currently say no to all of these questions. Here are the questions. Do you have abdominal obesity? If you're a man, is your waist circumference, right around your belly button—it's not your hips where you put your belt; it's up above, the gut that sticks out—is that greater than 40 inches? If you're a woman, is it greater than 35 inches?
- 2.62** Do you have high blood pressure? Is your blood pressure greater than 130, the top number, greater than 85, the bottom number? Or do you take medications for blood pressure, in which case you do have high blood pressure? Do you have high blood sugar? Are your fasting blood glucose levels greater than 100? Or do you take medications for diabetes? If you take any medications like Metformin or a diabetes medicine, you have to say yes. Do you have high triglycerides? Are your triglycerides greater than 150? Or do you take medications to try to manage triglycerides?
- 2.63** And do you have low HDL cholesterol? Your HDL cholesterol levels, if they are lower than 40 in men, if they are lower than 50 in women, that is considered abnormally low. And/or if you take any medications that might help you manage HDL cholesterol, you have to count yourself. If you have not had these labs done, most people will have these labs done at their primary care doctor. But I want you to be an advocate for your

own metabolic health. If your doctor doesn't see the need to order these labs, I at least want you to start a conversation with the doctor and say, "Hey, could you please draw these labs on me?"

- 2.64** Some of them are going to overlap with what I just said. I want you to get a fasting blood glucose, a hemoglobin A1C, which will measure an average of your blood glucose levels over the last few months. I want you to get a lipid profile, so we should know your total cholesterol, LDL cholesterol, HDL cholesterol, and triglycerides. I want you to get a C-reactive protein, which is an inflammatory biomarker. If you have high levels of inflammation, those are associated with increased risk for both metabolic health disorders and mental health disorders.
- 2.65** I would love for people to get a fasting insulin. This is the one your primary care doctor is probably going to put up the most stink about. But if you can get the person to do a fasting insulin at the same time that they get the fasting glucose, then you can calculate something called the HOMA-IR. You just go to a website, Google HOMA-IR, you'll get a calculator. You plug in those two numbers—fasting blood glucose, fasting insulin—and that will tell you whether you have insulin resistance. It tells you how much insulin your body needs to make in order to manage a glucose level.
- 2.66** Liver function tests, kidney function tests, vitamin B12, folate, magnesium, homocysteine. The B12, folate, magnesium, homocysteine are essential for metabolic health and mitochondrial function. And thyroid hormones, so TSH, thyroid stimulating hormone, T4 and T3. Thyroid hormone is the master regulator of metabolism. If you have low levels due to an autoimmune disorder, you may be gaining weight for no clear reason. You might be depressed for no clear reason, and it's actually associated with lots more.
- 2.67** Other assessments that you might consider, these are not as essential, but a lot of people have wearables, and you might be able to get your resting heart rate or heart rate variability. So your resting heart rate, as long as you're not having dizziness or lightheadedness, the lower the better. So runners' heart rates are often in the 50s or 60s, and that's good. That's a good thing. Again, as long as you're not fainting or about to faint. Then heart rate variability, surprisingly, you want wider variability. You want more variability because it means that your heart rate is ready to skyrocket when you go for a run, and it's ready to calm down and go back to 50 when you're not running.
- 2.68** You might consider a DEXA scan. This will tell you your body fat percentage. More importantly, it will tell you how much visceral fat you have. If you have abdominal fat, if it's mostly on the outside, if it's fat just right under your skin, that's actually probably not very dangerous fat. It's what we all do. We store some fat, especially if you live in a cold climate and you're going to hibernate and your body is trying to put on a little few extra pounds to get you through what might be food scarcity. Some fat on your abdomen right under your skin is perfectly fine and normal. Usually not a health risk.

- 2.69** What's really a serious health risk is when fat is accumulating around your liver and your other internal organs, and a DEXA scan can help you determine that. Then the last thing that you might consider is something called the VO₂ max. VO₂ max, basically, they will have you exercise as hard and fast as you can and measure how much oxygen can you use. And VO₂ max, that's been associated with longevity. It's been associated with all sorts of good health outcomes. It's an assessment, actually, of mitochondrial capacity, and we'll be talking about mitochondria a lot more soon enough.
- 2.70** There's a lot more to this story. Mental health is not just what we've talked about. We just talked about metabolic health, but make no mistake. Everything that we just talked about affects the brain. Even more surprising to some people, the brain may, in fact, be regulating those biomarkers. If you have high blood glucose, it might actually be a brain problem that is leading to high blood glucose. And so high blood glucose may actually be a symptom of dysregulation somewhere in brain cells.
- 2.71** Nonetheless, we all know there's a lot more to mental health than just metabolic health and blood glucose and lipids. And so we will continue this journey in a little bit.
- 2.72** Hi. When you were talking about the metabolic measures and lab work that we can get done, I'm wondering, like, from a heart health perspective, what about apoB or Lp(a), or some of these other biomarkers of cardiovascular health? Do you think those are also important? And I ask in the context of some of the dietary interventions that I think we're going to talk about. Do you think that's important to look at?
- 2.73** It is important to look at. As a rule of thumb, I actually think if you can get those labs, if you have access to those labs, if your primary care doctor will draw those labs, or if you have access to, usually a self-pay service that will do those labs for you, they're invaluable. If you're going to start on a dietary intervention that is known to impact cholesterol, potentially in an adverse way, and you're going to do it anyway for other health benefits, absolutely. I'm going to be a big fan of more detailed biomarker assessments.
- 2.74** But the list that I just gave right now are just generic health recommendations for every human being in the United States. I think every human being, including children, should get a fasting insulin. Why? Because, again, that large population study of 5,000 kids says that if your child has high levels of insulin resistance at age nine, and we can only know that by getting fasting insulin, your child is at high risk for developing a psychotic disorder. You might want to know that and intervene. But absolutely. If people can get access to apoB and others, go for it.
- 2.75** The controversy gets into, well, what does this one mean? What does that one mean? And then the reality is, I don't have all of the answers for people because it ends up becoming a personalized health care decision, in terms of, if this one's a little elevated, does this mean I have to take a statin? Does this mean I have to stop this diet? Does this mean I have to—and so I don't have generic advice to give you on what to do with

that information. So that's one of the reasons I didn't include it in the generic overall recommendation, too.

- 2.76** I don't know if you're going to answer my question later through the class, but what is the difference between Mike? Why did, after the eight years of seeing you, did he finally, like, "Oh, I'm going to start the gym and start seeing change." What changed for him?
- 2.77** Initially, it wasn't even the gym. He later got into the gym, but it was—I think what changed for him is that he was chronically tormented. He was getting older. He was recognizing that he was about to turn 30 in a couple of years. For him, that was a major milestone. He was living with his father in his father's basement. He was diagnosed with schizoaffective disorder. He was disabled. He had never worked and earned a salary. He had so much going against him, but he was also focused on, "I am so lonely. I want a girlfriend." For whatever reason, what stuck in his mind is, "I'm about to turn 30. If I don't do something, it's going to become too late." And for whatever reason he got it in his head, "I want to lose some weight."
- 2.78** Could it be a self-control thing? We lose control. Like you said, all the symptoms are normal for all of us as humans. Do you think that it's more like, we lose self-control, and then we just start diving deeper into these states of mind? Because even the other guy, John, I think was his name, with the wife, it seems to me like he used the medication as a tool to take responsibility and deal with those emotions of maybe the grief of losing his wife or whatever; or the process, which, like you said, it's human to feel pain when we're going through those processes.
- 2.79** And it's human to go through the process of grief, maybe denial and all of that stuff through these life situations. Do you think from your personal experience that it's like, he used it as a tool to take responsibility and maybe Mike, it took him a little more time to take response, to use it as a tool to take responsibility, the medication? Does that make sense?
- 2.80** If I understand you correctly, your question is, could the mental illness really just be not taking responsibility, not somehow being able to organize yourself, not being just—
- 2.81** Like your emotions or what you're going through in your life?
- 2.82** And that if we can just get people to organize themselves, become a little more disciplined, become a little more responsible, that things will start to fall into place, or no?
- 2.83** Hmm, because I think also discipline comes from self-love. It might sound like woo-woo.
- 2.84** No. I'm going to talk about self-love in a little bit. It does come.
- 2.85** For me, it's more about teaching them to love themselves or accept the human process because I think a lot of it is, like, we're supposed to be perfect or there's supposed to

be a perfect way to do things, and we're not perfect. We're human, and we have processes, and we have—and we make mistakes, and we have to take responsibility for those mistakes so that we can get better. So I guess my question is, do you think that they need to be taught, maybe no more discipline, because discipline, I do believe, comes from self-love and self-acceptance. But yeah, maybe to love themselves and accept themselves more?

- 2.86** So there's no doubt that what you're talking about plays a role in recovery. Zero doubt in my mind that having purpose, organizing yourself, having self-love, plays a role in mental health. And again, I'm going to be talking about that later in the course. However, I still believe it's up for debate, because there are some experts who will 100% agree with what you just said and say that that's what it all is. It's all self-love. If we can just get people to love themselves, everything will be fine, and all of these mental illnesses, including schizophrenia, will start to go away.
- 2.87** I don't believe it's that simple. One of the reasons I don't believe it's that simple is, as you'll learn from this course, I'm going to talk about all sorts of other risk factors that increase the risk, or that cause or exacerbate all sorts of mental illnesses. They have nothing to do with self-love. They have to do with biology and biological factors, or social factors that are beyond our control, which could influence self-love. But the biological factors for me definitively say that I can give you a pill and within two weeks, you could be clinically depressed, no matter how much you self-love yourself right now.
- 2.88** I really could do that. I could manipulate your biology and make you clinically depressed. That makes me convinced that, for at least some people, it is a brain dysfunction problem. And that for Mike, what the diet ended up doing, and I'll get into the science of this later, but what the diet ended up doing is, it restored his brain function. It restored his brain metabolism, so that his hallucinations and delusions started going away, and so that he could start to organize himself and function better. And there's no doubt all of that increased his self-love.
- 2.89** So I'm not saying self-love is not part of it, but I think if we just focus on self-love, like the cliché, that some people love and that some people hate, are the self-affirmations. There are gurus who will say, "If you just do your self-affirmations enough, that will take care of whatever mental illness you've got." There are some people who are on that bandwagon and will say, "Yeah, that worked for me." I can point you to thousands of people who say that didn't work for them. I think it can be part of it, but it's not the whole story, if that makes sense.
- 2.90** I'd like to ask a follow-up question about your patient, Mike. I've heard you tell that story many times, and today was with great detail, but you've never spoken about his diet. What version of the ketogenic diet was he doing, or what did his diet look like at the time when he changed it?

- 2.91** So early on, we were really focused on the biomarker of ketosis. And so, practically speaking, he would have bulletproof coffee often in the morning. That's coffee with some butter and MCT oil added to it, so a high-fat beverage first thing in the morning. He would sometimes have eggs or other protein, but usually not. Usually, it was just bulletproof coffee. Then for lunch and dinner, he would have some type of a protein—steak, chicken, something—and low-carb vegetables. So, broccoli, asparagus, other things. He would have some dairy. He would use heavy cream at times, some cheese. And that was pretty much it.
- 2.92** And that was enough to get him into high levels of ketosis. One of the interesting things that happened to him that I'll just put out there, since you've asked this question, is that one of the lessons he and I learned along the way is that, after many months of doing the diet, he was getting bored with it, wanted some new foods, wanted some treats, and found keto chocolate bars. Lily's keto chocolate bars, keto right on it. And so he went ahead and bought them and assumed he could have as much as he wanted. And so he ate a whole Lily's keto chocolate bar. And the next day, paged me in crisis.
- 2.93** He came into the office. His hallucinations were back with a vengeance. His delusions were back with a vengeance. This man hated the hospital, and he was saying, "I think I need to go to the hospital." He hated the hospital. It was traumatic for him to be in the hospital. And he was saying, "I think I need to be there. You're going to have to put me in the hospital. I can't take this. These voices, I can't take it. I'm going to do something. I'm going to do something bad. You've got to make this stop."
- 2.94** And I had him get on the scale, and he weighed 10 pounds more than he had the prior week. Then I checked his blood ketones, and they were negative. I checked his blood glucose, and it was abnormally high. I said, "You've gone off the diet. What did you do?" He's red in the face, yelling at me, "I did not go off the diet. You cannot say that. I'm doing this stupid diet. Don't you dare accuse me of that." I'm like, "Whoa, no shame, no guilt. But you're off the diet. You've gained 10 pounds. You don't have ketones. Your glucose is higher. Something's gone wrong." He's still insisting, "I haven't changed anything. I haven't done anything wrong."
- 2.95** And I'm like, "OK, calm down. I'm not here to judge. I'm not here to blame. But we're talking about a crisis. You're talking about going in the hospital. I'd like to avoid that if we can." We systematically go through. Turns out, he discloses, "Oh, I bought some chocolate bars, but they're keto. It says keto right on it." For him, that was everything. The sugar alcohols, or some other artificial ingredients in the keto chocolate, completely set him off and threw him out of ketosis, exacerbated his psychotic symptoms.
- 2.96** Fortunately, we were able to just get him back on the diet. I actually had him fast for 24 hours, which he was willing to do. I saw him every single day in person because I was like, you said something bad is going to happen. I don't want something bad to happen

on my watch. So, monitoring him very closely. But within 48 hours, his psychosis was remitted again. He was feeling fine. His ketones came back up, and he was fine.

Lecture 3: Metabolism & Mind

At a glance — Metabolism and mind — mental disorders as metabolic disorders of the brain.

- 3.1 The brain energy theory proposes that mental disorders are metabolic disorders of the brain. I've already told you that metabolism is everything when it comes to biology. Metabolism involves hundreds, if not thousands, of different metabolic pathways. There are so many factors that go into it. On the surface, saying that mental disorders are metabolic disorders to some people, even some scientists, seems pointless. It's like saying, well, mental illness exists in the universe. So what? Yeah, it exists in the universe. So what?
- 3.2 Brain function somehow depends on metabolism. Well, yeah, the function of every organ depends on metabolism. Is that really helpful? Is it proof that, in fact, mental disorders are fundamentally metabolic disorders? It's not. But if we ask a commonsense question, well, what exactly controls metabolism? Then we actually get some answers that start to connect more dots. We are led to these tiny things in our cells called mitochondria.
- 3.3 Now, most people, if they remember from biology class, if they remember anything about mitochondria, they remember one phrase, the powerhouse of the cell. Mitochondria are the powerhouse, or powerhouses, of the cell. What does that mean? It means that mitochondria take food and they convert it into ATP, which is the unit of energy for living cells. In fact, that is what mitochondria do. That is one of their functions. But cutting-edge research over the last 20 years in particular has completely shattered this simplistic definition.
- 3.4 Mitochondria are so much more than just powerhouses. They are primary regulators of what we call metabolism. There are hundreds or thousands of them in each cell. They are highly dynamic. They aren't just sitting there in a stationary way, creating ATP. They actually form a little bit of a biological universe within each cell. They are moving around the cell. They have a traffic system that they use. They merge with each other. They fuse with each other, and then they break off from each other. They send signals. They are involved in numerous cellular processes.
- 3.5 In fact, when we get into the details of mitochondrial biology, it gets overwhelmingly complicated fast. It's like, if you ask some blind scientists who are all surrounding an elephant to describe the elephant, but they have blindfolds on, depending on what part of the elephant they were touching, they might describe very different things. When you talk to mitochondrial researchers, they can study very different aspects of mitochondrial biology, and that makes them use different words, different language, different concepts to describe mitochondria.
- 3.6 Some of them will say it's a powerhouse, the traditional definition. Others will say it's a bean-shaped organelle, and that's no different than the other organelles. Others will

say it's a dynamic network, that mitochondria in one cell are actually communicating with mitochondria in other cells. Mitochondria in one cell are communicating certainly with mitochondria within that cell. They are a signaling organelle. Some will say they induce cell death. So when it's time for a cell to die, because it has become too old or because it's starting to turn into a cancerous cell, mitochondria perform the primary function of killing that cell through a process called apoptosis.

- 3.7 Some will say it's an endosymbiont bacterium, which really speaks to the origins of mitochondria. Mitochondria were once independent living organisms, a bacterium that was incorporated into another cell. The predominant theory of evolution right now is that it was this organism. Two cells, one on the inside, one on the outside, two single-celled organisms merged, and they became all of multicellular life. Now, for people who believe in God, you might say that was the process that God used to create life. For people who don't believe in God and want to turn to science and evolution, you can use that process as well.
- 3.8 Whatever you think, that process was nothing short of miraculous because, as far as we can tell, it's never happened again. That single process of two living organisms merging and becoming one symbiotic organism led to all of the life that we can see with our eyes today. Again, when you do a deep dive into the science of mitochondria, it becomes overwhelmingly complicated fast. They actually do different things in different cells. Even within the same cells, some of them are doing signaling to the nucleus, while others are sequestering calcium, while others are releasing neurotransmitters, while others are just cranking out ATP.
- 3.9 Many research articles, literally tens of thousands of research articles, have been published in the leading science journals on the roles of mitochondria and a wide range of functions of cells. To make it simple, some will say that mitochondria are the workers within cells. You can think of a cell almost like a factory, and that factory makes things like hormones or neurotransmitters, or it does things, like it's a muscle cell that contracts and helps us move. And that mitochondria are these tiny little workers within that cell, making everything happen. That's how some describe it.
- 3.10 There are some mitochondrial biological researchers who will say, if you think of the cell like a computer, most people think mitochondria are the power cord to that computer because they produce the power. And in fact, they are the power cord. But much more important than power, they are the motherboard of that computer. They are directing and allocating resources. They are actually making things happen or not happen. Mitochondria primarily sense our environment and help us respond and adapt. They sense our blood glucose levels and help the pancreas release insulin or not. They form a network within cells and between most of our cells to coordinate metabolism and life.
- 3.11 When you look at all of the known biopsychosocial factors that play a role in mental illness, you can connect all of them through mitochondrial functions. They play a role in

the production, regulation, and release of some key neurotransmitters, and I'll talk a little more about those. They play a role in epigenetics. They are actually responsible for most of the signals that get sent to the cell nucleus that turn genes on or off. They play a role in inflammation, turning it both on and off. They actually impact the gut microbiome, but the gut microbiome impacts them. Diet and exercise impact their function, and we'll talk a lot more about that.

- 3.12** But actually, mitochondrial function in specific brain cells help regulate whether we feel hungry or not. They determine whether we have energy to go out and exercise or not. Adverse childhood experiences, stress and trauma, impact their function directly. Drugs and alcohol do, sleep, and hormones. They play a key role in the synthesis of some hormones— cortisol, estrogen, testosterone, progesterone. They actually control the first step in the synthesis of those hormones. As you probably already know, those hormones play a big role in mental health.
- 3.13** But those hormones then go on to affect mitochondria throughout the brain and body. We're going to dive into more of that science as we go through the course. But I want to be the first to say, I am not alone. I went out on a limb early on to propose this theory, but I am not alone. Other researchers are now citing my work, publishing their own work, talking about how mitochondria are the unifying link of all of the biopsychosocial risk factors.
- 3.14** The basic proposal is that if you have enough mitochondria, and those mitochondria are healthy in all of your cells, you will be a perfectly healthy, happy human being. And that if some cells somewhere in your brain or body don't have enough mitochondria, or if there's something wrong with those mitochondria, those cells may begin to malfunction, and that can lead to what we call diseases or illnesses. If it's really extreme, it might lead to death.
- 3.15** It begs the question, well, what exactly is mitochondrial dysfunction? This is a term that gets used a lot. I use it. Many other researchers use it, and it's been used for decades. I will tell you, the cutting-edge mitochondrial researchers hate this term because it's too generic. It's too broad. They're like, "Mitochondria are doing so many things. How could we possibly just use dysfunction to describe one way that they can dysfunction? They can malfunction in numerous ways." And that's the difficulty, is that they have so many different roles that they can malfunction in many different ways.
- 3.16** But in fact, research on mitochondrial dysfunction has been around for decades. Initially, in the 1950s, there was this free radical theory of aging. Researchers were trying to figure out what makes humans age, and more importantly, what causes all the diseases of aging. What causes obesity, diabetes, cardiovascular disease, and Alzheimer's disease, and cancer? Those are historically the diseases of aging. And so researchers want to know, what causes them and what can we do about it? And they were led to, initially, this free radical theory of aging, which says that high levels of

reactive oxygen species, or oxidative stress, are associated with all of those diseases of aging.

- 3.17** Now, in 1972, that theory was amended by Dr. Denham Harman, who proposed the free radical theory of aging, and he revised it in 1972, saying it has to come from mitochondria. Mitochondria are the things that are producing these free radicals. That led to a wave, an epidemic, if you will, in the healthcare community, of using antioxidants and anti-inflammatory agents. Everybody was told, "Take vitamin E, take vitamin C. It will do you good. Everybody take vitamins. Eat lots of fruits and vegetables to get your antioxidants in." Antioxidants were supposed to save the day.
- 3.18** All of that, that you're probably all familiar with, that all descended from this oxidative stress theory of aging and disease. But in fact, that hasn't panned out. You can't just take a vitamin or a supplement. You can't reduce your oxidative stress to get all the benefits that we want. The studies that have been done show marginal benefit at best, and many of them no benefit. Some of them have even shown harm. High doses of vitamin E, for example, increase mortality. So how does that make sense?
- 3.19** Well, it turns out that this oxidative stress, these free radicals, are actually not just toxic waste products. They're there for a reason. They are there to send signals to the cell, to make things happen. When you completely suppress them with high doses of antioxidants, you are actually suppressing normal biology, and that can actually be harmful.
- 3.20** But in fact, the metabolic and mitochondrial theory of mental illness, this has been around for over a century. In 1879, Sir Henry Maudsley observed, "Diabetes is a disease that often shows itself in families in which insanity prevails." Families that have high rates of insanity also have high rates of diabetes for some reason. This was before we had any medications. We weren't prescribing lots of medicines, psychiatric meds, to patients. They weren't even developed yet.
- 3.21** Within 10 years of the discovery of the molecule insulin, we were injecting it in large doses in the mentally ill, in a treatment called insulin coma therapy. The reports were that it was actually highly effective for some people. It was also sometimes lethal because you can really overdose on insulin quickly and easily. They were giving massive doses of insulin to patients with severe mental illness.
- 3.22** It was actually the most widely used treatment in the Western world for severe mental illness. If you had schizophrenia or bipolar disorder or even chronic crippling depression, there was a good chance you were getting insulin coma therapy if you were being treated at one of the best hospitals. Insulin coma therapy was the standard of care.
- 3.23** Interestingly, I've talked to clinicians and researchers who were around during that time, even the researcher who was in charge of the NIH insulin coma therapy ward. Before he died, he was actually working on trying to resurrect it because he said, "It

really worked. I saw it work. People were getting better." Abnormalities in lactate and glucose metabolism were identified in the bodies of people with bipolar disorder and schizophrenia in the 1930s. The mitochondrial theory of autism was first proposed in 1985. Then in the 1990s, we had all these functional neuroimaging studies, PET scans, spec scans, all these others. Again, as I've said, those are all measuring brain metabolism.

- 3.24** The mitochondrial theory of bipolar disorder was first proposed in 2000, and since then, the mitochondrial theory of numerous neuropsychiatric disorders has been pursued. But it begs the question, well, if it's all mitochondrial dysfunction, what exactly causes mitochondrial dysfunction? Numerous things, including all of the biopsychosocial risk factors that we know play a role in mental illness, all of them. The problem is that we often don't include some highly relevant factors in the biopsychosocial risk factors for mental illness. We almost never talk about diet and its role in brain function.
- 3.25** Let's try to go through just a little bit of a whirlwind tour through some of these different factors. Neurotransmitters. There is no doubt, neurotransmitters play a role in mental illness. The big ones are serotonin, dopamine, GABA, and glutamate. Here's most people's simplistic understanding of those. Serotonin, low levels of serotonin make people depressed and anxious. The treatment are SSRIs, things like Prozac, Zoloft, Paxil. What do those do? Those inhibit the reuptake of serotonin. They increase the amount of serotonin at the synapse, and somehow that helps.
- 3.26** Is it directly serotonin? No, we know it's not directly the effects of serotonin, at least not immediately, because within 24 hours of taking your first dose of Prozac, you already have elevated levels of serotonin in your brain. But you don't have an antidepressant effect yet. It takes weeks to get the antidepressant effect. Why? We don't know. Mental illness is just way too complicated, Dr. Palmer. Nobody can figure that out. But that's the serotonin theory of depression and anxiety. It is pretty much the most predominant theory that people still believe in, clinicians will still talk about. If you come in with depression, and the doctor tries to prescribe you Zoloft, he or she is probably going to say, "It will boost your serotonin levels, and that will help solve your depression."
- 3.27** Dopamine. Dopamine is known to play a role in mental illness. High levels are speculated to play a role in schizophrenia. Dysregulation of dopamine. Dopamine is often known as the pleasure neurotransmitter, the reward neurotransmitter. So that plays a role in addiction. So we know dopamine is somehow involved. GABA and glutamate. GABA and glutamate are actually the two most predominant neurotransmitters in the human nervous system. Glutamate is excitatory, meaning it makes things happen. GABA is inhibitory, meaning it slows down the function of cells. With almost all of the neuropsychiatric disorders, it is speculated that maybe there is somehow an imbalance of GABA and glutamate.

- 3.28** What does that mean? How do we arrive at this imbalance? Oh, no one knows. What causes the imbalance? What causes low or high levels of serotonin? No, nobody knows. What can we do about it? Well, yeah, we prescribe pills. All of the pills we prescribe are supposedly regulating these neurotransmitters, and that is why we're prescribing the pills. There is no doubt neurotransmitters play a role in brain function. We can't get around that. So again, we're climbing Einstein's mountain and trying to see, well, how do neurotransmitters fit into this bigger theory? What do they have to do with trauma and stress, and how does that result in mental illness?
- 3.29** And in fact, what I am here to propose is that all of these neurotransmitters directly impact brain metabolism, and more importantly, they directly impact mitochondrial function. I am not going to spend our time going through each and every hormone or neurotransmitter, but I am going to at least share with you just a tiny snapshot of a couple of lines of evidence on serotonin. Serotonin is a neurotransmitter that's actually found across a wide range of organisms, all the way to worms. Worms have lots of serotonin. And largely, it is not a happy neurotransmitter in worms. It is a metabolism neurotransmitter in worms.
- 3.30** Serotonin plays a profound role in metabolism and mitochondrial function. We have direct evidence from studies published in important leading journals that serotonin regulates the production of new mitochondria, and it regulates their function in brain cells, cortical neurons. Serotonin is playing a role in brain metabolism through mitochondria. Serotonin is actually flowing through our bloodstream. It is taken up by almost all mitochondria. As far as we can tell, all mitochondria—it's hard to say, though, because mitochondria are different in different cells, and not every cell has been studied. But seemingly, every cell that's been studied, mitochondria seem to take up serotonin.
- 3.31** What do they do with it? They turn it into another molecule that you've probably heard of, called melatonin. Melatonin ends up being an antioxidant within mitochondria. Serotonin is flowing through the human body, being taken up by mitochondria, being turned into an antioxidant. Shockingly, serotonin is not really so much a brain neurotransmitter. It is much more a gut neurotransmitter. Ninety to 95% of the serotonin in the human body is produced within our digestive tract, either through cells there and/or through the bacteria in the gut.
- 3.32** We will talk about this a little later, but the gut and the brain have connections that have profound impact on brain function. When you take something like Prozac, it's not just working in the brain, it's also working in your gut and everywhere else. I want you to think differently about these neurotransmitters, and more importantly, about the medications that we take that modulate these neurotransmitters. There is no doubt, I'm not here to say that SSRIs don't work for some people. They do work for some people. They really do. I have seen people, I know people, who swear that the SSRIs

made them better, and I've seen it as a clinician. I've seen antipsychotics work. I've seen lithium work.

- 3.33** What I'm arguing is not that they don't work. What I'm arguing is that they are altering brain metabolism and mitochondrial function. That when they work, that is how they work. But let me remind you of the dire statistics I shared with you earlier about, how often exactly do they work? And how well do they work? And how long-lasting are those effects? We have a lot of room for improvement. But I want to make the point that there is abundant evidence that the current psychiatric medications that we are prescribing are actually altering brain metabolism.
- 3.34** So this study asks the question, how does Prozac really work? The researchers looked at a lot of different animal models of Prozac, and when they saw beneficial effects from the Prozac, they looked at all of the data that was accumulated and tried to figure out what exactly is it doing in the brain. These researchers already know it's not a serotonin imbalance. That's not what it's doing. They're trying to figure out, well, what exactly then is it doing? What they figured out, they looked at 27 different brain regions, and initially, they looked at gene expression and whether gene expression was upregulated or downregulated in all these different brain areas.
- 3.35** And initially, they saw this really confusing, complicated picture of different brain regions have different genes upregulated, and some of them have genes downregulated, and it's just a confusing mess. It is just a confusing mess, and that is metabolism. It is a confusing mess, hundreds if not thousands of pathways. When the researchers took a step back and analyzed it all and said, "Is there any theme here? Is there any unifying pattern?" What they wrote, specifically, "Across diverse brain regions, integrative omics analysis consistently suggested increased energy metabolism via oxidative phosphorylation and mitochondrial changes."
- 3.36** What that means is that when Prozac works, it is changing brain metabolism through mitochondria. That is how Prozac works. How does lithium work? If you talk to some of the leading psychiatrists and neuroscientists, "How does lithium work?" They'll say, "Oh, no one knows. It's really complicated. We know it works, but how? Who knows?" Well, these researchers, a couple of years, looked at everything that was known about lithium because we do have a lot of science on, well, it seems to affect this molecule. It seems to affect that hormone or that enzyme. It seems to do these different things. We've got decades of research looking at how does lithium work.
- 3.37** These researchers put it all together and said, "Wow, it all converges on this thing called the insulin signaling pathway, which converges ultimately right, squarely on mitochondria." So how does lithium work? Lithium works by modulating insulin signaling, which modulates mitochondrial function. We have two broad classes of medications that we often use in psychiatry, stimulants and sedatives. Stimulants are things that perk you up. Sedatives are things that slow you down. What exactly are we

perking up? What exactly are we slowing down? Some will say brain function, but what exactly do you mean, Dr. Palmer? What exactly are we stimulating?

- 3.38** If you really look into the details, we are stimulating brain metabolism, or we are suppressing brain metabolism. In fact, some researchers have looked at the effects of psychotropic medications directly on mitochondria, and there is a wide range of effects directly on mitochondria, even without the neurotransmitter receptors on the outside of the cell and other signaling pathways. Put the medicine in a petri dish with mitochondria, and it directly impacts their function. Here's the dilemma. It's all over the place. Sometimes, it enhances mitochondrial function; sometimes, it suppresses mitochondrial function, and sometimes, it does a little bit of both.
- 3.39** When I was first developing this theory, this was actually one of the biggest dilemmas that I had. It was like, well, if something's suppressing mitochondrial function, how could that be beneficial? Why would that reduce symptoms? Because I know they work. I do. I've seen antipsychotics work. And across the board, antipsychotics are known to impair mitochondrial function. They impair metabolism. People gain weight. They develop type 2 diabetes. They develop all of the risk factors for cardiovascular disease within 30 days of starting these antipsychotics.
- 3.40** We know that antipsychotics, in particular, impair mitochondrial function. That is the proposed mechanism of action for all the neurological side effects, like Parkinson's disease, tremors, tardive dyskinesia, and other things. I really struggled with, well, how does that make sense then? If I'm saying that mitochondria and mitochondrial dysfunction is a cause of mental illness, why would impairing them further be helpful? And that comes back to this concept of what is a brain disorder? How can the brain malfunction? Brain regions can become overactive or underactive.
- 3.41** That is the most critical piece, is that mitochondrial dysfunction can result in hyperexcitability, and that results in some symptoms like anxiety or psychotic symptoms or others. Think of it like pain. It's producing sensations that should not be there. They're not occurring in the right way at the right time and place. So panic attacks, anxiety, other types of symptoms. If you suppress brain cell function even further, you can stop hyperexcitability. By suppressing mitochondrial function and metabolism, you can stop hyperexcitability.
- 3.42** The problem is that you're going to make that cell even more vulnerable over time. That cell is going to become even more metabolically compromised. The mitochondria in that cell are going to be more compromised. And what would that suggest? If that was true, what would that mean, clinically? It would mean that patients who are taking these medications will develop chronic lifelong disorders. They will develop comorbid disorders, like obesity, diabetes, and cardiovascular disease, and ultimately, they will die early deaths. And is that true? It is. That is exactly what mental health care in 2024 gets us.

- 3.43** I want to tell you a story about how I think about medications. Before I do this, I just have to say, I know that what I just said is really tragic and sad and alarming. If I was on an antipsychotic and listening to this lecture, I would want to just go out and stop it, cold turkey. Please do not stop your psychotropic medications on your own, even based on what I just said. I can tell you now, if somebody is taking antipsychotics or mood stabilizers or antidepressants and they stop them cold turkey, it can become a life-threatening crisis within days.
- 3.44** They can become floridly manic, psychotic. They can become suicidal, homicidal. They can get arrested. They can end up involuntarily hospitalized, or they can end up dead. So despite what I just told you, I want you to work with your health care professional and do it right. But I will tell you the story that exemplifies what I'm arguing. When I first started my career, I actually was a consultant in nursing homes, and I was called to see this woman that I will call Jane. The nurses said, "Dr. Palmer, Jane is really agitated, and we need you to prescribe something to stop her agitation."
- 3.45** I went to see Jane. She's sitting in the dining hall at the time that I see her. She's in this wheelchair with a little tray because it's dinner time and she's getting food. She's got food on the tray. She's slumped over. She can't actually have a coherent conversation. She can't even make eye contact. She clearly is unaware that I'm even there. She's smearing food all over herself and trying to smear it on me. She's just writhing and groaning and moaning. She says some words, but they don't make any sense. We call that a "word salad" in psychiatry. She's just saying random nonsensical words.
- 3.46** That was about all I needed. I was done with my assessment. What did I diagnose her with? She's delirious. I go and look at her medication list, and she's on several antipsychotics, mood stabilizers, and benzodiazepines, all prescribed over the last six months to try to calm Jane down. Jane has been a problem for several months. And so they prescribe medicines and they work for a little bit. It calms her down. But then she gets agitated again. So they give her more or they increase the dose. And they've just progressively been doing this.
- 3.47** And now they're like, "Dr. Palmer, you're the new psychiatrist on the block. Tell us what you want to prescribe next because she's agitated and she's up all night and she's yelling and screaming and she's disturbing the other residents and you've got to do something." Essentially, I say, "Jane is delirious. It is from all of these medications. Please stop these medications as quickly as possible." I'm just a consultant, so that's all I get to do. I just get to make my recommendation and I go away and I don't know if they listen to me or not.
- 3.48** I come back three weeks later, because that was when I was scheduled to come back. Come back three weeks later, I'm walking down the hall, and this old woman that I've never met before rushes up to me, and she's like, "Are you Dr. Palmer?" I'm like, "Yeah." She reaches out, grabs me, hugs me. "Thank you so much for saving my sister." I said, "I don't know you. Who are you? What are you talking about? I'm the

psychiatrist. I'm just here consulting." She said, "You saved my sister Jane." I'm like, "What?" I barely even remembered Jane, honestly, because I see tons of patients. I'm going to other nursing homes in between that three-week visit. I'm like, "Huh?" I have to refresh my memory.

- 3.49** The doctor who was treating her took her off everything. Within a week or two, she was completely non-agitated. She still had dementia. That's why she was in the nursing home. Getting her off all those meds didn't make her dementia go away. But she was calm, cooperative, could have nice visits, could feed herself, could help dress herself, everything. The treatment was to get her off psychiatric meds. Now, geriatric psychiatrists will not at all challenge this story that I just said. They know that this is true.
- 3.50** The question I'm here to ask is, how often does this happen in young people? How often do we see 15-year-old kids with autism spectrum disorder, and we just keep pumping them full of meds, and their agitation gets worse, and so what do we do? We prescribe more. We increase the dose. How often do we do this to people with schizophrenia? How often do we do this to people with bipolar disorder? This is perhaps the most radical, disruptive concept of brain energy because it completely challenges the status quo and challenges what we're doing.
- 3.51** Again, if you are on medications, do not stop them on your own. Do not stop them cold turkey. You will get in trouble. I promise you, you will, and you'll probably end up involuntarily hospitalized, arrested, or dead. But working with a health care professional to adjust your medications to optimize your metabolic and mitochondrial health might be a path to recovery.
- 3.52** In the biopsychosocial model, we talk about genetics. Genetics play a role in mental illness. Some people think, "Oh, no, it's all genetic. 78% of ADHD is genetic. That's what we hear. The parents have it, so of course, the kids have it. That's how it works, Dr. Palmer, and it's just that simple." How does that fit in with the mitochondrial or metabolic theory of mental illness? Many of the genes that are known to play a role in mental illness directly regulate brain metabolism. We have good evidence of that. They are directly regulating mitochondrial function or other enzymes or pathways or ion channels that influence whether cells can become hyperexcitable or not. And that makes them more vulnerable to hyperexcitability.
- 3.53** And then if you pair it with additional mitochondrial dysfunction, you can end up that way. The good news is that, for most of the neuropsychiatric disorders, there are not genes that confer high levels of risk. The genetics suggests that one gene confers risk for a wide range of mental disorders. There is no schizophrenia gene, there is no bipolar gene. People think that and they talk about that, but it's not true. We mapped the human genome over 25 years ago. We know a lot about it. What we know is that there is not a schizophrenia gene. It doesn't exist. It's not there. It's not for a lack of looking. We looked. It's just not there. We need to move on.

- 3.54** What we know is that the levels of risk from genetics are very tiny. More than likely, a lot of the inheritance patterns of mental illness are probably epigenetic. They are not occurring through the genes themselves, but they are occurring more through the regulation of gene expression, which can be transmitted from parents to children. Now, let's talk about hormones. I've told you already that cortisol, estrogen, and testosterone, the first step in the synthesis of those hormones occurs within mitochondria. They are key regulators of whether those hormones are being produced, when those hormones are not being produced, or when the regulation of those hormones becomes dysregulated. You have to consider mitochondrial dysfunction as the primary cause of that dysregulation.
- 3.55** Cortisol, we'll talk about a little bit more in the next lecture when we talk about trauma and stress. Estrogen plays a profound role in women's reproductive health, everybody knows that. Guess what else it plays a profound role in? Brain metabolism. If you look at the role of estrogen, there have been books, scientific papers published, on the role of estrogen in brain metabolism. It's not just women either. Men have estrogen; men have estrogen receptors. It plays an equally powerful role for men. The difference between men and women is that men are constantly producing it. We have the good luck and privilege to have nice, steady levels of our steroid hormones.
- 3.56** Women, unfortunately, do not. Women have cycles in which they have higher levels and lower levels. Stress can impact it, sleep can impact it, diet can impact it, and stages of life impact it. If you get pregnant, that's going to change the cycles of those hormone production. When postpartum, that is a very high-risk period for women, for all sorts of metabolic and mental health conditions. Unfortunately, women get the short side of the stick when it comes to estrogen, even though estrogen is also a male hormone.
- 3.57** Thyroid hormone. Thyroid hormone is not produced directly by mitochondria, but thyroid hormone's primary target is mitochondria. Thyroid hormone is regulating metabolism. It is regulating both the production of ATP and the production of heat. I didn't say this already, but that's another thing that mitochondria do. We are warm-blooded animals. Where does our warmth come from? It comes from one and only one place, mitochondria. Mitochondria take food, they fuel it, and it goes through this uncoupling protein in order to produce heat. So, thyroid hormone regulates our metabolism; it regulates our body temperature. If thyroid hormone is off, then your mental health and your metabolic health can be off.
- 3.58** Insulin is another important hormone. Most people know insulin as it relates to diabetes. Some people know insulin in terms of the implications for insulin signaling or insulin resistance with obesity. Here's the shocker. Insulin is produced in the human brain. It is not serving the same function in the human brain. Insulin appears to be more like a neurotransmitter in the human brain. More than likely, the insulin signaling in the brain is actually controlling insulin release and regulation and glucose regulation in the whole body, including from the pancreas.

- 3.59** There is abundant evidence that abnormalities of brain insulin signaling play a key role in numerous neuropsychiatric disorders, including bipolar disorder, schizophrenia, chronic depression, and Alzheimer's disease. This is why Alzheimer's disease is considered type 3 diabetes, abnormal insulin signaling and glucose metabolism in the brain. I could go on and on. What does all of that have to do with mitochondria? Everything.
- 3.60** This is one paper, but if you just Google, if you really are skeptical, and you're like, "There's no way all these hormones could have something to do with mitochondria," actually, mitochondria are directly controlling the regulation and release of most key hormones. But then those hormones travel through the brain and body and impact other cells, and when they impact other cells, they are impacting the function of those cells, which ultimately means the mitochondrial function of those cells.
- 3.61** Now, let's turn our attention to inflammation. Getting an infection increases your risk for neuropsychiatric disorders. I'm going to take the example of COVID. When people get COVID, most of us get over it, but many people don't. Some estimates are 20 to 25% of people don't fully recover within a month of getting COVID, and they have what's often referred to as long COVID. What are the symptoms of long COVID? Well, some people have breathing problems, sure. Some people have weakness. Maybe the muscles are affected, sure. The brain is where most of the symptoms are coming from. The brain are the symptoms that are most debilitating.
- 3.62** Exactly which symptoms can happen from long COVID? Brain fog gets talked about a lot. But guess what? It's all of the symptoms. It's depression, anxiety, psychosis. Psychosis has been described from COVID and long COVID. Mood instability, impulsivity, all sorts of things. In fact, high levels of inflammation—so an infection increases inflammation, and that can produce brain symptoms. We have a whole body of literature suggesting that people with chronic mental illnesses have high levels of chronic inflammation. Not necessarily from an infection, they just have high levels of inflammation.
- 3.63** The three biomarkers, in particular, are C-reactive protein, interleukin 6, and tumor necrosis factor-alpha. Those three biomarkers have been found in many studies, numerous studies, to be elevated in people with diagnoses like depression, anxiety, bipolar, schizophrenia, PTSD, autism, OCD, and substance use disorders. What else, you ask? Oh, those same identical biomarkers are elevated in people with obesity, diabetes, cardiovascular disease, and Alzheimer's disease. Shocking. Shocking.
- 3.64** How does that work? What does that have to do with mitochondria? Well, if you dive into the science, it actually has everything to do with mitochondria. This article, review article, published in one of the Nature journals, the caption says it all. "Mitochondrial control of inflammation." Mitochondria are controlling whether inflammation is getting turned on or off. If mitochondrial function becomes impaired, you can have impairment in immune system function and inflammation signaling.

- 3.65** How does it work? Here's a simplistic graph to just help you understand how this works. You can get inflammation from two clear sources, infections and allergens. Allergens can be from anything. Hay fever, mold, all sorts of things can cause higher levels of inflammation. Infections and allergens can increase inflammation. We have direct evidence that tumor necrosis factor, interleukin 6, and some other inflammatory biomarkers are directly impairing mitochondrial function. So inflammation itself can impair mitochondrial function. And if the mitochondrial dysfunction becomes significant enough, that can lead to a whole host of physical and mental health conditions.
- 3.66** Now, the problem is that you can get this feedback loop because mitochondrial dysfunction in cells can then make those cells become—basically, puts them in a state of disrepair, which then causes even more inflammation. Now you've got inflammation causing mitochondrial dysfunction. Mitochondrial dysfunction causes more inflammation, and both of those things are leading to illnesses.
- 3.67** What types of illnesses would those be, given that kind of a feedback cycle? Chronic illnesses. We're talking about chronic mental health conditions, chronic obesity, chronic type 2 diabetes, chronic Alzheimer's disease. We're talking about chronic illnesses that are not going to get better because now they are stuck in this feedback cycle. But inflammation is not the only thing that can cause mitochondrial dysfunction. I want to remind you, all of the biopsychosocial risk factors that we're talking about can also cause mitochondrial dysfunction.
- 3.68** So somebody may never have an infection or an allergen. They can have mitochondrial dysfunction from something else, say, trauma or a hormone imbalance or a poor diet. That can then cause inflammation, which then causes more mitochondrial dysfunction, which then causes more inflammation, and that leads to chronic, progressive neuropsychiatric and metabolic health conditions.
- 3.69** We've now talked about neurotransmitters, genetics, hormones, inflammation, and infections. We're going to be coming to drugs and alcohol, sleep, circadian rhythms. In the next lecture, we're going to talk about psychological and social factors. I'm going to continue on our journey about neuropsychiatric disorders. How do all of the biopsychosocial factors fit together through this lens of metabolism in mitochondria? But since I'm talking about mitochondria, I want to take a step back and let you know, mitochondria play a role in human health, not just mental health, all the way to aging.
- 3.70** If you ask some of the leading experts in the world, what causes aging? What do we know about aging? Many of them will say, well, there are nine hallmarks of aging. There are nine biological cellular things happening that appear to be associated with aging. Those are gene instability, so genes can become dysregulated in their expression and/or mutations. Telomere attrition. The end caps of chromosomes are called telomeres, and they get shorter and shorter. Epigenetic alteration. So alterations in which genes are getting expressed.

- 3.71** Loss of proteostasis. What that means is that proteins can become abnormal; they can become misfolded, and they're not getting recycled. They're not getting handled in the way they should. That helps us understand like, why do people with Alzheimer's disease have higher levels of beta amyloid accumulation in their brains? Dysregulated nutrient sensing. There's something about nutrient sensing that is off. It gets dysregulated in aging. Mitochondrial dysfunction is actually one of the key hallmarks of aging.
- 3.72** Cellular senescence, so some cells become zombie cells, and they usually cause inflammation and other things. There's something called stem cell exhaustion. We have stem cells, and as we get older, we lose some of them. If we lose them, maybe we can't stay youthful and vibrant. An altered intercellular communication. The cells usually communicate with each other through a wide range of mechanisms, and that becomes impaired.
- 3.73** This study done by none other than David Sinclair's group, who's a leading aging guru, published in one of the Nature Reviews, actually says, "Wait. When you look at all that we know about these nine hallmarks of aging, cellular metabolism interconnects all of them." Then when we look at everything that's known about aging, or proven strategies to slow aging, they all revolve around metabolism. Calorie restriction and other methods, exercise, other things, are all about metabolism.
- 3.74** In fact, he and his group are not the only ones. Again, thousands of research articles published. This one looks at the effects of mitochondrial dysfunction in heart disease, Alzheimer's disease, lung disease, atherosclerosis, diabetes, fatty liver disease, osteoarthritis, and others. Remember, I shared with you that mental disorders share strong bidirectional relationships with obesity, diabetes, cardiovascular disease, and Alzheimer's disease. Those are diseases of aging. What I am proposing is that when people have mental disorders, for some, that is the first sign that they have accelerated aging.
- 3.75** Now, I want to come back to the question that I asked you all in that first lecture. Which age group is more likely to use psychiatric medications? I asked you this poll, which age group is more likely to take an antidepressant? Children, adolescents, 18 to 39, 40 to 59, 60 and over? Based on everything that I've just told you, does anybody want to modify their answer? How many of you think the age group is still children and adolescents up to age 17? How many of you think it's 18 to 39? 40 to 59? 60 and over? We've got a lot of 60 and overs now.
- 3.76** So this theory would suggest, if what I'm proposing is true, and it's a big if, but if what I'm proposing is true, older people should have more mental illness than younger people. Do we have any evidence for that? This data from the CDC looks at the number of Americans between 2015 and 2018 who used an antidepressant in the past 30 days. There is a clear linear relationship, with adults over 60 years old being the highest category of users of antidepressants.

- 3.77** What about other medications? What about other psychotropic meds? Benzodiazepine use, by age, it goes up. Over the life span, it comes down slightly between the category of 50 to 64, which is the highest category of use, and those over 65. I am here to argue that the reason it comes down in those over 65 is because there are guidelines in place, that clinicians should not prescribe benzodiazepines to older adults for fear of falls. Clinicians are not prescribing them, not because they're not warranted, but because they want to use something else.
- 3.78** What else might they use? If a patient is anxious, if a patient is agitated, if a patient can't sleep, what else might clinicians use? That leads us to antipsychotic use. This chart—it is somewhat dated. I couldn't find a more recent chart, from 2010—looked at the rates of antipsychotic use by age. You see that it's actually the lowest in the 20 to 22 age group, even though that's the peak onset for schizophrenia and bipolar. That's not when people are, that's not when most of the population by any age group is using antipsychotics.
- 3.79** It steadily increases. You'll notice in this chart that there is a mini peak, in particular in women, between the ages of 50 and 54. What do you think is happening to women between the ages of 50 to 54 that we're prescribing so many antipsychotics to them? Menopause, estrogen. Their brain metabolism is being dysregulated because they are acutely losing a primary driver of brain metabolism.
- 3.80** Then the rates come down, and then they go up exponentially, so that the highest age group most likely to take an antipsychotic in the United States is the oldest age recorded at 84. Could it go higher? It almost certainly goes higher. That is, you even get older than that. What I am here to propose is that mental disorders are, in fact, diseases of aging.

Lecture 4: Stress & Cellular Health

At a glance — Stress and cellular health — psychological and social causes tied to metabolism.

- 4.1** Now we're going to, in some ways, maybe shift gears and talk about psychological and social causes of mental and metabolic disorders. We're going to come back to some more science and how drugs and alcohol can affect mitochondria and how they play a role in mental illness. But I decided to put these psychological and social factors more toward the front. The reason is because I know a lot of people are really skeptical about this theory, and a lot of people are really fixated on, "Come on, Chris, psychological and social causes are the primary drivers of mental illness. You can't get around that."
- 4.2** I want to dive into at least a little bit of the science on this. Then I also want to give you some self-assessment questions that you can use to maybe improve your mental and metabolic health. Again, I want to first start and reiterate this basic concept. Psychological, social, mental health, mental disorders follow the same pattern as pain. Pain can be normal. Pain can be extreme, and pain can be a disorder. If I stub my toe, I might have a little bit of pain that lasts 30 minutes and then goes completely away and I never think about it again. That's normal pain. That's not a disorder. Normal pain.
- 4.3** If I have surgery on my toe, I might have pain for the next month, as that surgical wound is healing, as the tissue is healing. That's not a disorder. That is the pain system doing exactly what it is designed to do and tell me, "Hey, you've got an injury of your foot. Don't walk too much on it. Limp. Don't put any pressure on it," because the body is trying to heal your toe. It doesn't want you touching it. It doesn't want you stepping on it because you can injure it again. It's vulnerable, it's weak, it needs to be healed.
- 4.4** All of that is adaptive. That pain that we are experiencing, it's painful, yes. It's a nuisance, yes. Maybe we have to take medications to try to calm it down, yes. It is adaptive. It is there for a reason. It is there to try to tell me, "Don't walk on it. It's injured. It's damaged. You're going to injure it more. It's never going to be able to heal if you keep walking on it." That's extreme. But then I can have a pain disorder in which my foot is hurting with numbness, tingling, chronic pain, for no reason, because I have diabetes, and the pain system is malfunctioning.
- 4.5** So I want to say, at the outset, psychological, social, same deal. We all have anxiety. That is not a disorder. It's not a mental illness. Everybody has anxiety. What causes it? You already know, I hope. I don't have to go through that. We can have extreme anxiety. What causes that? Well, the obvious things. A kid getting unrelentlessly bullied and teased on the playground. Everybody in the class is mocking him, making fun of him, making him feel unsafe. His parents aren't supporting him. He has no friends. Does he have a mental illness? Is his brain malfunctioning? I would argue no.
- 4.6** Even if he meets criteria for a mental illness, according to DSM, I would argue no. That kid does not have a disorder of his brain. He is responding appropriately to his

environment. His brain is hardwired to have depression, anxiety, and other symptoms in response to those circumstances. But I can tell you, there are plenty of people that I have treated in my career who have anxiety disorders. They have chronic, unrelenting anxiety for no reason. They can be in tears saying, "I don't know what's wrong with me. There's no reason. I have talked in therapy for years about this. There is no reasonable explanation for me to be this anxious. I just can't help it."

- 4.7 Or maybe they get panic attacks out of the blue that end up crippling them and leaving them unable to leave home. They, I would argue, have a disorder. Nonetheless, all of that—normal, extreme disorder—all of that plays a role in mental health and metabolic health. I want to go through some of that. Just to recap, we've already talked about neurotransmitters, genetics, hormones, inflammation, infections. We're going to talk about some other ones later, and now we're going to dive in to the psychological and social factors.
- 4.8 I want to start with a really easy-to-understand example. If any of you in the room have been affected, if any of you watching have been affected, this may be triggering, but we're going to talk about sexual assault in women. The reason we're going to talk about that is because it is a very clear trauma. It is a discrete event that results in a traumatic experience, and in fact, researchers have done a tremendous amount of research on women who have experienced such events, and so we know a lot about it. So what happens to women following sexual assault?
- 4.9 This first study that I'm citing, it's actually all the way from 1992, they looked at women as soon as they could get access, usually within days of a sexual assault, and they enrolled them in a study. They asked them about symptoms of post-traumatic stress disorder. They assessed them four times, separated by about a week for each of those times. At the first assessment, when they ask them the criteria for post-traumatic stress disorder, do you currently have these symptoms?
- 4.10 The symptoms of PTSD, for those of you who don't know, hyper-startle response, hypervigilance. But the opposite of that, too—numbness, just almost dissociation, inability to sleep, constant fear, flashbacks of the event. There's a lot of depression, anxiety symptoms that are in there, but those are the hallmarks of PTSD. When these researchers first approached women, usually within days of a sexual assault, 94% of the women met symptomatic criteria for post-traumatic stress disorder. The way I interpret that is post-traumatic stress symptoms are not always a disorder. I find it difficult to believe that 94% of these women's brains began malfunctioning shortly after an assault. I don't see it that way.
- 4.11 I see it as, this is a normal human adaptive response, that when you are assaulted, when your life is threatened, in order to protect yourself, you better be on high alert. Don't trust anybody. You hear a sound in the middle of the night, wake the hell up, get ready to run, get ready to fight. Nothing is safe. Your life is in danger. You can't trust anyone anymore. That is adaptive immediately after, because 94% of women were

experiencing these symptoms at the first assessment. Again, this was often within the first week of the assault.

- 4.12** What happened at the fourth assessment, which, on average, in this study was 35 days later? 65% of the women still met criteria for PTSD. Now, according to DSM-5, they now have a mental disorder. All you need are symptoms for one month. 65% of the women now have a mental disorder called post-traumatic stress disorder. Does anybody here really think that they all of a sudden all developed brain disorders and that that's the way we should think about it? I want to challenge that.
- 4.13** And again, a lot of experts are already out there. They've been doing this work for decades. They're like, "It doesn't make sense. We already know this, Dr. Palmer. We've been saying this. That is not a disorder. That is a normal adaptation." It is a normal response to trauma. 35% of the women no longer met criteria. What this begins to hint at is that, in some people, the symptoms will start to go away. That also is normal. We are not hardwired and built to be on hypervigilance and reclusive for the rest of our lives after an assault. That is not adaptive.
- 4.14** It is adaptive immediately after a threat because you just never know. Who did the crime? Are they still around? It's fresh. Maybe you were injured. You need to hide from the world. You need to let your body heal. But that shouldn't be the rest of your life. At some point, it's important for those symptoms to be able to go away and that person to get on with her life. So what happens a year later? This study looked prospectively at PTSD symptoms, and it was a meta-analysis of lots of different studies that got done. What they found, they actually measured it over time, at the one-month mark and the 12-month mark.
- 4.15** At the one-month mark, they found similar rates to what I just shared with you. 75% in their study met criteria for PTSD. All you need are symptoms for a month to get the DSM diagnosis of post-traumatic stress disorder. So 75% of women at one month meet criteria after an assault. How many still meet criteria 12 months later, a year later? 41%. About 60% symptoms seem to resolve, or at least recede enough so that they no longer meet full criteria. 41% still meet criteria. How about two years later? 32% still met criteria for PTSD two years later. So one-third of women after an assault are going to meet criteria for PTSD.
- 4.16** Now, what I'm here to argue is, are some of those brain disorders? Are some of them not brain disorders? Are they just normal reactions? I think the overwhelming concept is this. Having PTSD symptoms immediately after a trauma is normal. It is normal and expected. And in fact, you're in a rare minority if you don't have PTSD symptoms. I would argue the human brain is hardwired to have PTSD symptoms after a traumatic event. But what seems to be more adaptive is that, over time, those symptoms should go down.

- 4.17 But this assumes the woman is safe, right? What if that sexual assault was from a partner? What if she's still with that partner? Are her symptoms supposed to go away? Does she have a brain disorder if she's in a chronically abusive relationship and she's having PTSD symptoms over years? Is her brain disordered? Is that the way we think about it? Does she have a neurochemical imbalance? Does she have a neurotransmitter imbalance? Maybe too little serotonin? I would argue no. She is having normal adaptive responses.
- 4.18 And yet, a Vietnam veteran 50 years later, who cannot sleep through the night still, is plagued by flashbacks, can't function in the world. I would say that person does have a brain disorder. That person should have been able to find a way to move on, and that their symptoms are actually tormenting them. Their symptoms are reducing their ability to live in the world and function in society, and that those symptoms, I can't make a good argument that those symptoms are adaptive.
- 4.19 Now, here's the complicated thing. What I'm here to say is, all of it plays a role in mitochondrial function, and all of it plays a role in both metabolic and mental health, even the normal. When we look broadly at trauma, stress, and adversity, they increase the risk for mental and metabolic health disorders. Which ones, you ask? Essentially, all of them. The best evidence for this comes from numerous studies now that have been done on what are called adverse childhood experiences, commonly referred to as ACEs.
- 4.20 Adverse childhood experiences include abuse and neglect. Just in case you're not familiar with what's the difference, abuse is somebody actively hurting you. Neglect is somebody not taking care of your needs. If you're a young child, if you have parents who aren't doing laundry, you don't have clean clothes, you get hurt, you get bullied and teased at school, and they could care less, they're not around, they don't have time for you, they're working, they're busy, they're doing whatever they're doing—that is neglect. Emotional, physical sexual abuse, emotional, physical neglect are all considered adverse childhood experiences.
- 4.21 Other things that get counted as adverse childhood experiences— having somebody in the household, usually a parent, but maybe a brother or sister or an uncle or someone else who lives with you, who misuses substances or who has a mental illness, including attempting suicide. If your parents get divorced or are separated, if you have a parent who is incarcerated for any reason, it doesn't matter how long, incarceration is an adverse childhood experience for that child. It's not a great experience for the parent either going through it, but—
- 4.22 Intimate partner violence, domestic violence. Whether you are the victim or not, if you grow up seeing your mother get beaten up by her string of boyfriends or your father, that is an adverse childhood experience. Other things that count, bullying, community violence. What if you live in a neighborhood where people are getting shot at least once a week? You live in a neighborhood where you hear gunshots several times a week. You've seen dead bodies on the street, and the ambulances and police pull up to

take them away. That is an adverse childhood experience. It's an adverse adult experience as well, but we're talking about children.

- 4.23 Natural disasters, refugee, wartime experiences. People in Ukraine right now, people in Israel, across the board, almost regardless of who they are, where they live, they are all having adverse experiences simply by being close to war zones, even if they're not directly seeing it, directly affected. Witnessing or experiencing acts of terrorism. How many people in the United States have adverse childhood experiences? 64% of US adults will report at least one type of ACE. 17% report experiencing four or more ACEs.
- 4.24 Here are more statistics on ACEs from the government. One in six adults experiences four or more. I shared that statistic with you. Five out of the top 10 leading causes of death are associated with ACEs. 44%, preventing ACEs could reduce the number of adults with depression by as much as 44%. These statistics estimate that if we could eliminate ACEs—that's not going to happen—but if we could make the world a happy, peaceful, cooperative place, no more trauma and abuse, up to 21 million cases of depression would probably not occur. Up to 1.9 million cases of heart disease would not occur. Up to 2.5 million cases of overweight and obesity would probably not occur.
- 4.25 This study looked at what impact do ACEs have on the development of specific psychiatric disorders. It's not just post-traumatic stress disorder. It's also depression, anxiety disorders, substance use disorders. This study found on average for each additional ACE that you experience, you have another 50% increased risk for developing one of those disorders—depression, PTSD, anxiety, or substance use disorders.
- 4.26 But in fact, this study only looked at these four diagnostic labels. What other diagnostic labels are associated with adverse childhood experiences? You'll never guess. All of them. Adverse childhood experiences, if they occur at a young enough age, are associated with autism, learning disorders, and all of the other neurodevelopmental disorders. Look at all the studies done in the Romanian orphans. These were infants who were severely neglected, and they developed a whole range of neuropsychiatric disorders, including all of the neurodevelopmental disorders.
- 4.27 But high levels of ACEs are also associated with increased risk for bipolar disorder, increased risk for schizophrenia, increased risk for learning disorders, ADHD, increased risk for dementia, later in life. It's all of them. This study published in Science in just last year, 2023, looked at the social determinants of health and mortality. What they found was that, not unexpectedly, people who are really poor have a lower life span, a shorter life span, than people who are wealthier. Your social network plays a role. If you are less integrated with friends and family, you have a higher risk of mortality, early death, than if you have more friends and family.
- 4.28 The number of ACEs. People who had six or more ACEs on average, lived 20 years shorter than people who had zero ACEs. It is clear that ACEs are somehow affecting

human health, almost all of the neuropsychiatric disorders, but also all of the metabolic health disorders. How exactly does that work? At first, we're led to what's called the HPA axis, the hypothalamic-pituitary-adrenal axis. We know that trauma, stress, triggers the fight-or-flight response. And the fight-or-flight response is associated with cortisol being released, catecholamines being released. Those are things like adrenaline, epinephrine, norepinephrine.

- 4.29** But the stress response also increases your level of inflammatory biomarkers, and it ends up changing gene expression. Now, what does any of that have to do with metabolism or mitochondria? In fact, it has everything to do with metabolism and mitochondria. I've already told you a little bit of the story on inflammation, how that is directly impacted and directly impacts mitochondria and metabolism. Glucocorticoids are actually directly activating mitochondria. Initially, this is adaptive. They are making your mitochondria in your muscles and your brain overactive. They are stimulating ATP production. They are getting you prepared to run, to fight, to flee, to whatever you're going to do to survive. They are trying to rev up your metabolism.
- 4.30** At the same time, you're getting something called gluconeogenesis, where your liver is producing more glucose and just shuttling it out. Again, in an effort, we need all hands on deck, all the fuel sources we can get. So your fat cells are releasing fatty acids. Protein is actually getting degraded. Any fuel source will do. Like, your body is trying to survive. It recognizes when you have the perception, "I am in danger," it really is, all hands on deck. Don't worry about sleep, don't worry about rest, don't worry about health and wellness. Get ready to fight or flee.
- 4.31** Catecholamines, what are they doing? They're increasing blood pressure and heart rate. They're increasing blood glucose, and this is diverting blood from the GI tract to the muscles and brain. So all adaptive. In fact, cortisol affects the entire body. If we look at the longer-term effects of cortisol, what if this system is upregulated over an extended period of time? Initially, cortisol is actually beneficial. It is anti-inflammatory. That's why we give cortisone injections into joints that are inflamed or damaged.
- 4.32** But over time, cortisol takes a toll on the entire body. It suppresses your immune system function so you can get infections. It has problems for your bones. You can get osteoporosis. It dramatically impacts the brain. It can cause weight gain and type 2 diabetes. It can cause ulcers. It can cause cataracts, and it can cause heart disease. But does it have anything directly to do with mitochondria? I've already said so. Several lines of research have looked directly at this.
- 4.33** This particular research was actually looking at mitochondria in a petri dish, and they added cortisol, and the mitochondria starts speeding up. They start producing more ATP immediately. Cortisol is firing up your mitochondria. Again, adaptive. That is not pathological. The problem is that when you keep stimulating them with this cortisol, eventually they develop high levels of what's called oxidative stress, and it begins to damage them. And that when mitochondria are exposed to high levels of cortisol over

time, they become dysfunctional, maladaptive. You may lose the number of mitochondria in your cells. Your body is trying to adapt, but it essentially burns out, if you will.

- 4.34** Now, there are many research studies published. This is another one, getting very granular in terms of the pathways that are affected. When you have high levels of stress, they directly impact mitochondria. That directly leads to cell dysfunction, which can result in all sorts of metabolic and mental health diseases and accelerated biological aging.
- 4.35** What about something less severe than trauma? What about something less severe than all of that? What about just relationships? What about somebody who's isolated socially, maybe by choice, maybe not by choice, versus somebody who's got lots of friends and family? If you don't know the statistics, we have good evidence that the more relationships you have, the better those relationships are, the more connected you are to those people, the better your health and happiness and longevity.
- 4.36** Happiness, in particular, is probably most strongly correlated to the number of relationships you have in your life. If you want to be happy, reach out, call people, invite people over, go out on outings, do something, join a club, do hobbies with people, something. Be connected. Is there any evidence that that— so we have good, strong epidemiological evidence, even from the Harvard, it's often called the Harvard Happiness Study, where they studied these medical students from Harvard for decades since the 1940s. John F. Kennedy was actually part of this cohort. They studied them for decades, and they're still studying them to this day.
- 4.37** What they found is, the more relationships these people had, the better health outcomes. It was probably the strongest predictor of good health and good lives, were good solid relationships. Does that have anything to do with mitochondria? That seems far-fetched, right? That can't have anything to do with it. This study literally just published in the last month in a leading journal called the Proceedings of the National Academy of Sciences, looked at 400 postmortem brains. They had lots of data on these people before they passed away.
- 4.38** They looked at, were these people, did they have positive psychosocial experiences, or did they tend to have negative psychosocial experiences? They included trauma stress, but a lot of what they were looking at were, how connected were these people? How many relationships did they have? They specifically looked at, is there any correlation with brain mitochondria? What they found is that the more positive psychosocial experiences people had, they had greater abundance of the mitochondrial energy transformation machinery. Meaning, their brain mitochondria were producing more energy. If you had high levels of negative experiences, that was directly correlated with a lower abundance of brain energy machinery within mitochondria.

- 4.39 Now, this study was correlational. They just found the correlation, and the correlation could go either way, or the cause could go either way, or it could go both ways. It could be that people who have a lot of good psychosocial experiences have these effects that result in better metabolic mitochondrial health, including brain mitochondrial health. But it could also be, it's also conceivable that poor brain mitochondrial health makes people more socially awkward or anxious or depressed, and so they don't go out.
- 4.40 I would actually argue, my speculation, based on the brain energy theory, is it's both. Both of those things are highly plausible. So that if somebody has really poor brain mitochondrial function, they are more likely to be depressed and anxious, and that is absolutely going to play a role in whether they have positive psychosocial experiences. But if somebody is having no psychosocial experiences, and they want them, and they say, "I'm really lonely," that is going to produce a stress response which is going to adversely affect their metabolic and mitochondrial health.
- 4.41 Now, it's not just mitochondria, it's all of these other factors. We have evidence that childhood trauma influences how much inflammation you have in your body. This study looked at people who are all depressed, and they sorted them by, have you had any trauma, sexual abuse in particular? They put them into categories, none or low, moderate, and severe. And it was a linear relationship. The more exposure to trauma you had and the more severe the trauma was, the higher the levels of interleukin 6 and tumor necrosis factor alpha.
- 4.42 I've talked a little bit about telomeres. Telomeres are these end caps on chromosomes, and they are one of the common ways that we measure aging in people. Do we have any evidence that they are impacted by trauma, childhood adverse experiences? Yes, we do. The more childhood maltreatment you have, the greater the likelihood that your telomeres are shorter, meaning you have accelerated aging.
- 4.43 What about something not so dire? This study looked at mothers. They asked the mothers, "How stressed are you from being a mother?" Now, they specifically chose some mothers who were mothers to disabled children. They chose some other mothers who were mothers to normal, healthy, thriving children, hoping that they would see a signal. But at the end of the day, they really focused on, how stressed are you at being a mother? Sure enough, they saw a correlation between those, but there were some resilient women who had disabled children, and they were happy as clams.
- 4.44 There were other women who had normal, thriving, healthy children, and they were really highly stressed, maybe because they were being abused or they had a childhood trauma history or other things. But what they found was that there was a direct correlation between how much stress the women reported and their telomere length, meaning high levels of stress are correlated with accelerated aging.
- 4.45 Now I want to do a quick self-assessment, and then we'll get into what are some things that you might be able to do about this. Here's the self-assessment. Take notes if you

want, or even mental notes, or jot things down. Do you feel safe most of the time? Now, that could be yes, no; it could be a more nuanced answer. Do you feel that your life has meaning and purpose? Do you have people in your life that you care about or love? Do you feel cared for or loved by other people? Do you feel respected? Do you respect yourself?

- 4.46 Are you able to take care of yourself? Do you contribute anything worthwhile to your family, community, or the world? Do you feel bored more than a few times per week? Do you feel lonely more than a few times per week? I want to talk a little bit about this, so the safety issue I hope you get from what we've already discussed. If you feel unsafe, that is taking a metabolic toll. There is no way around it. Your perception of a lack of safety is enough to trigger the response.
- 4.47 Even if everybody around you says, "Stop worrying. Why are you so worried? What's wrong with you? Why are you on edge? That trauma was years ago. Get over it. Move on." It doesn't matter whether other people think you should feel safe or not. It is simply dependent on whether you feel safe or not. If you don't feel safe, that is a problem. If you do feel safe, it can be a benefit. It's not always a benefit because some people feel safe when they shouldn't. Some people are in dangerous war zones or dangerous neighborhoods, and they recklessly have too much confidence and go out and get beaten up or killed because they should have been a little more cautious.
- 4.48 Meaning and purpose in life, having people that care about you, love you, you caring about others, this is all part of purpose. This is all part of feeling connected. I would argue, on so many levels, you can talk about this from a philosophical standpoint. Some people would want to talk about this from a religious standpoint. Their faith says that these are things that we should all be doing. I want to at least consider that this is actually biologically hardwired into humans, that we are actually biologically hardwired to need to stay in a group. Because living with the group keeps us safe.
- 4.49 The group will provide food and resources and protection against other tribes, other people, animals. The group will help create housing. The group will do everything together. We will certainly reproduce together. If you're out alone being a hermit, there's no reproduction. So that's biologically disadvantageous from an evolutionary stance. But what I would argue is that we are hardwired to live in groups and that we need each other and that when we don't feel like we belong or are accepted in the group, we actually develop a stress response. That stress response is actually adaptive. It is trying to get us to be part of the group.
- 4.50 That might mean, groups have hierarchies. Some people get to be the leaders and other people have to be the followers. Some people are the popular ones, others are the not-popular ones, but they're all still part of the group. That stress response is going to get you to hopefully navigate your way. Maybe you have to accept being a follower, even though you don't want to, but it's better to be in the group as a follower than to not have any group.

- 4.51 When we don't have a group, when we don't have other people, when we aren't accepted, when we aren't incorporated into society, it takes a toll. We know this from large epidemiological studies. I showed you the outcome studies of the social determinants of health. The fewer people you have in your life, the more likely you are to die early death. What is the mechanism? I am arguing, the most plausible mechanism for that, metabolism in mitochondria. It increases risk for all of the mental health conditions and all of the metabolic health conditions.
- 4.52 One of the challenges with the questions that I asked you is that I asked you them, and you probably heard them, and if you are otherwise a healthy individual, you're doing a realistic assessment of, do I feel safe? Sure, I do. Do I have meaning and purpose? Yeah, I do. I have a good job. I have community. Yeah, people love me. Yeah, I love other people. Sure, I respect myself. Maybe I'd want to work on that a little bit or whatever, but yeah, I'm good.
- 4.53 The assumption, so that's a good, healthy person, the assumption is that there will be people who can legitimately and honestly say, no, I don't feel safe. Again, we have to go back to this concept of normal, extreme, and disorder. You can feel unsafe for obvious normal reasons. You live in a war zone, you live in an abusive household, you live in a dangerous neighborhood. If that person says, "I feel unsafe," we take that at face value. No further research needed. It's obvious.
- 4.54 There can be extreme examples of that. The challenge as a mental health professional is that I have people who on the surface and who six months ago were perfectly healthy. Six months ago, they would say, "Yeah, I've got everything." And now they say, "I'm so depressed. I'm so worthless. My life has no meaning whatsoever. I'm ready to be dead. I think I should be dead." When you ask them what happened, they will sometimes say, "I don't know. I don't know what's wrong with me. On the surface, I see that everything's fine. I have a loving spouse. I have a good home. I have a good job. We have lots of money saved, but it just feels like none of it matters anymore, and I'm so empty and lost."
- 4.55 Now, some people will go after a trauma history or something. This is somehow like, resurrecting your childhood trauma. I, as a psychiatrist, am immediately thinking, "You have clinical depression." You have major depressive disorder out of the blue that is causing your brain to malfunction, that is causing you to perceive that you don't have worth, you don't have connections when, in fact, you actually do. The people in your life are all saying, "We love you. What are you talking about? Why are you doing this? We still love you. You should be fine. Snap out of it. Stop it." The person is saying, "But I don't feel fine."
- 4.56 That person, I would argue, may have a brain disorder that is causing their brain to malfunction, that is causing them to answer these questions in a very different way than reality-based. That is a reflection of their brain disorder. And fascinatingly, I can share with you countless examples of when that happens. People start focusing on the

psychosocial. They're focusing on, this is psychological, this is social, because that's what they're complaining about. They don't feel connected to their wife. Maybe he fell out of love. Maybe he's thinking about some past love. Who knows? Maybe it's a midlife crisis. We throw that term around a lot. It's a midlife crisis. That's what this is, as though that solves the problem.

- 4.57 I can't tell you how many times I've seen people like that. Then four months later, we find out they have no thyroid hormone. They developed an autoimmune disorder of severe hypothyroidism, and you give them thyroid hormone, and everything's fine. It's like, wow, how did that happen? Thyroid hormone affected the way your brain functioned, which affected the way you perceived your psychosocial environment.
- 4.58 So I give you these questions because I think these questions are actually important reflections. I think these questions, if you are disconnected—you know, the pandemic really took a toll. Being forced to isolate from your friends and family, there are so many people who still say, "I haven't gotten back. I haven't gotten back to what I had before the pandemic. The friends that I had, everybody's just in their homes doing their own thing. We're all just lonely and isolated." If that's you, that's real, that's reality-based. There are solutions.
- 4.59 Call those friends. Tell them, "Come over. We're having a potluck dinner. We're going back to the old days. We're going to have game night every week or whatever. We're going to go out to movies. We're going to go hiking together. We're going to go to the gym together," whatever you do. Some of these are reality-based, but you have to take these questions with a grain of salt because you have to contemplate whether the person is answering them accurately.
- 4.60 But what can you do about it, if you did answer any of these? So, the first step, I've probably talked about this enough, is safety. If a woman is in an abusive relationship, there's no starting any mental health treatment until she gets out of that abusive relationship. Now, that was maybe a little glib because we can do some things to help her. But the priority, the number one priority on everyone's mind, has to be, you've got to get out of that environment. You may end up dead, but we're never going to help you recover. We're never going to help you get over your depression or anxiety or anything else until you are safe. So safety, number one.
- 4.61 Two, self-care. What can you do if you have been traumatized and are trying to get back? Well, you can some basic types of self-care, and we'll get into more details on some of these in a bit. But that includes physical self-care, nutrition, exercise, sleep. It includes mental self-care, mindfulness, maybe therapy, emotional self-care. Some people will do gratitude journals. They'll just journal in general. Some people will do spiritual self-care. Maybe you're going to get back into a faith community or lean more heavily on your faith community. Maybe you're going to do yoga, intellectual self-care. You're going to read, learn, do hobbies.

- 4.62 Environment. You're going to make sure you're in a clean, safe environment. Social. You're going to manage boundaries. You're going to appropriately manage connections with other people. And financial. You may want to work on budget, saving, make sure that you're safe, at least financially. I want to talk a little bit about stress reduction practices or relaxation responses, because this has gotten some interesting science from the medical community. When we say relaxation response, this is a term that researchers use to describe meditation, yoga, and repetitive prayer. So any of those three things, people who are doing meditation, yoga, or repetitive prayer.
- 4.63 The researchers knew that they were using these strategies as part of a mind-body medicine program, and they noticed or they felt confident that these interventions could help people improve blood pressure, could help improve anxiety, could help improve insomnia, and could actually slow aging itself; that people who practice meditation yoga and repetitive prayer seemed to live longer. And so they actually set out to try to identify how.
- 4.64 And the reason they set out to identify it is because they were largely ignored by the medical community. The medical community was like, "Whatever, we don't care about you and your prayer stuff or your yoga stuff. That's not a real intervention. Real interventions are pills. That's not a real intervention, and it's just nonsense. Unless you can demonstrate a biological mechanism of action for all of those things, we're really skeptical about your data. We just don't think it's true."
- 4.65 These researchers actually were very convinced, though. These researchers were like, "No, we know this works. We've got more than enough data. We've got more than enough, again, correlational data," but they didn't have direct causation data. You can't follow people for decades and randomize them. I'm going to randomize you to a meditation group, and you can never meditate for the rest of your life. Then I'm going to see which of these large groups of people live longer on average. You can't do those studies.
- 4.66 They set out on a journey and they actually looked at people who were healthy practitioners with years of relaxation response practices. Then they also looked at a group of novices who had never done any of this stuff, and they trained them how to do it over eight weeks. They did all of these biomarkers, looking for what is changing in the brain or body. What is changing? They were actually just drawing blood from the arm. What they found was that the blood samples were actually demonstrating gene changes related to mitochondria and insulin signaling. Those were the primary changes.
- 4.67 There were some inflammatory biomarker changes as well. But here is the quote. Their quote is, "Our results for the first time indicate that relaxation response elicitation, particularly after long-term practice, may evoke its downstream health benefits by improving mitochondrial energy production and utilization, and thus promoting mitochondrial resiliency through the upregulation of ATPase and insulin function." At

the end of the day, they are pushing relaxation, yoga, and prayer for health through mitochondrial mechanisms.

- 4.68** If you have a trauma history and you are suffering, I want you to start with those things that I just mentioned. But what if those aren't working? Standard treatments can still work. There is no doubt trauma-informed therapy can play a powerful role for some people. So trauma-informed therapy might work. Medications might work. But again, I want to remind you of the complexity of medications. If you are having PTSD symptoms, flashbacks, intrusive memories, that is hyperexcitability. Just like you're getting the sensation of pain when it shouldn't be there, you're getting these intrusive, recurrent memories when you don't want them, when they are no longer adaptive. That represents hyperexcitability.
- 4.69** If you take medications that completely suppress brain metabolism, otherwise known as sedatives, they can reduce symptoms in the short run. That's the trick. That's the trap. "Well, I took a Valium, and it made me feel better for a few hours. So now I'm just going to take Valium every day." It sounds like a great plan, but unfortunately, then when you take Valium every day, usually the Valium starts wearing off and it doesn't work as well. Then over time, we have longitudinal data. Oh, actually, these people go on to develop chronic disorders, and that is because Valium is suppressing brain energy metabolism, which can be great for episodes of hyperexcitable symptoms, but is not good for the long-term health of the brain.
- 4.70** So if you're going to try medications, do it wisely. But I do want to say, there is a road to recovery, even if you have had horrible adversity in life. Whether it's a lot of adverse childhood experiences, whether it is a sexual assault, whether it is anything else, there is a road to recovery. All of the strategies that we're going to be talking about can help people heal and recover. With that, we're going to wrap up this lecture. Next, we'll be talking about lifestyle factors and whether they might have a role in mental health. Thanks.
- 4.71** We speak about mitochondria having a wide variety of function, and there's hundreds in each cell, and there are different mitochondria in different cells in the body. Do they vary? Do mitochondria vary per different body parts, or if there's mitochondrial dysfunction in the brain, would you also see mitochondrial dysfunction in muscle tissue in the arm? Is there a consistency across human body, or is there variation if you have maybe a stronger mindset versus stronger physical connectivity?
- 4.72** That is a really important million-dollar question that a lot of people ask, and I'll answer it. Connected to it is, well, what test can I do for my mitochondrial health? That's what everybody wants. What's the blood test I can do for mitochondrial health? Tell me, Dr. Palmer. And so what you've picked up on is spot on, is that actually it can vary widely. Some cells can have robust mitochondrial health, while other cells are about to die due to poor mitochondrial health.

- 4.73 So I want to give you an extreme example, and then I want to talk a little more about it. The extreme normal human physiology example is that every day we are losing billions of cells. Cells are becoming old, damaged. Their shelf life is over, and they undergo a process called apoptosis, and they die. That process of apoptosis is actually through mitochondrial disintegration. The mitochondria are becoming highly dysfunctional. They have high levels of reactive oxygen species. That makes them begin to degrade, and then their degradation leads to this enzyme called the cytochrome c, go into the cytoplasm, and that results in the cell dying.
- 4.74 So if I took that once, and this is happening every day in all of us in billions of cells. If I took that one cell and analyzed it and analyzed its mitochondrial function, I would say, "Whoa, this person's about to die." The mitochondria are disintegrating on the microscope, and the cell is dying. It's dysfunctional because of the mitochondrial biomarkers. Levels of oxidative stress are through the roof. Levels of ATP are plummeting. This cell is dying. It is dying. That is the process of dying.
- 4.75 At the very same time in that exact same human being, another cell is born. It's a little infant cell, and it is thriving. It is beautiful. It has exquisitely healthy cell parts and mitochondria and proteins and enzymes. Everything is just beautiful and pristine because it's brand new. If I biopsy that cell and look at it, I would say, "Well, this person has perfectly robust, healthy mitochondria, and isn't that beautiful." There's always that interplay, is that there are going to be some cells because of just the natural cycle of life. We have some cells, like our skin is always sloughing off, dying, and we're getting new skin. Hair is growing. Things are happening.
- 4.76 Overall, if you get very granular about specific cells, there will be some that have poor mitochondrial health and others that have robust mitochondrial health. However, all of these disorders go together, even the mental disorders, as I've shared with you. As people get older, their likelihood of psychosis, of depression, of anxiety, is increasing. What that strongly suggests, and what we know from the aging literature, is that, on average, cells in numerous different organs and tissues are all progressing along the same way.
- 4.77 So that if you have dysfunction in brain circuits that lead to OCD, and that could be for lots of reasons, if you can increase the health of your gut cells, if you can increase the health of your muscle cells, liver cells, that actually will help your brain cells. And one of the ways, there are numerous ways that it can help. There are all sorts of signals and molecules, hormones, neuropeptides, inflammatory biomarkers floating around that are all playing a role in that. But one thing that I haven't mentioned yet is that cells actually transfer mitochondria from one cell to another. Some red blood cells, I mean, white blood cells, macrophages, will transfer mitochondria to other cells.
- 4.78 There are astrocytes in the brain that will transfer mitochondria to neurons when neurons are struggling. If somebody or if an animal has a near stroke event, essentially a stroke means your mitochondria are cut off from oxygen and they start dying. If

enough of them are still alive that the cell is barely holding on, but enough have been damaged that the cell may die, it's very vulnerable, astrocytes will actually form these little tunnels with that neuron and send in mitochondria to try to save that neuron. It's mitochondria to the rescue. These inflammatory cells are also going to the region, so they're doing it.

- 4.79** So, the bottom line is, it's complicated. There's no one test for your mitochondrial health, unfortunately. The best test for your mitochondrial health is the basic assessment that I'm walking you through in this course. We don't need a mitochondrial biomarker test. I can tell how healthy or unhealthy someone is based on whether they have metabolic syndrome, based on whether they have signs or symptoms of mental illness, and based on some other things that we're yet to talk about, that we'll be talking about.
- 4.80** So I can tell whether somebody's good health or poor health based on that. Then the solutions actually become pretty easy and obvious. But as a rule of thumb, if you improve mitochondrial health in one organ or tissue, you're more likely to, that will likely spread to other organs and tissues.
- 4.81** You said that mitochondria are responsible for producing body heat through a process of converting food to ATP. My question is, is that the only time or the only way they can do that? Because I practice extended fasting, and sometimes I get very cold without food for a couple of days.
- 4.82** Yeah. So mitochondria are actually— there was a study published a few years ago in a really leading journal that actually measured the temperature of cells and all of the organelles. And mitochondria are slightly warmer than the cell itself, which makes sense, because some people talk about them as little furnaces or batteries. They're transforming chemical energy. They're taking electrons from some molecules, transferring them to other molecules. That process will release a little bit of heat in it. It's just like if you run a battery at home, it will get a little bit hot.
- 4.83** So the transferring energy from one molecule to another often is not perfect. It's not a perfect transfer, and so you get a little bit of heat. Mitochondria actually primarily produce heat for the human body, though, not through ATP production. So it's like those are the two pathways, that mitochondria can use that energy if it's producing— well, I shouldn't even I don't say that. It's more than those two buckets. But mitochondria can produce ATP through the normal powerhouse function.
- 4.84** But there are these things called uncoupling proteins, which uncouple the transfer of electrons to result in ATP. When you do that, you actually don't get any ATP. Instead, you get these electrons flowing through this different thing called the uncoupling protein, and that itself generates heat. So mitochondria, in some ways, almost have to choose between, are we going to make energy, or are we going to use this food for heat?

- 4.85 They are also producing lots of other things. They are producing the building blocks for neurotransmitters. They're producing the building blocks for hormones, for proteins, lipids, for the cell membrane and everything else. But my guess is that, when you were on a ketogenic diet or when people are doing fasting, that your body will actually adapt. Your body senses that, wait, there's food scarcity. The body doesn't like food scarcity. It can be a highly adaptive and beneficial state to be in, and we'll talk more about that in one of the next lectures.
- 4.86 But part of that process is that the body is trying to conserve energy. One of the ways it conserves energy is by reducing body temperature, actually. So when people go on rapid weight-loss diets, for example, not saying that you are doing that, but if people go on a rapid weight-loss diet, their body temperature will actually go down because the body is trying to save energy. It's like, well, we're just going to have to make do. We can't use all this energy for heat production because who knows when you're going to get another meal and we've got to survive. It almost goes into a little bit of a hibernation mode. Usually, people's energy starts going down at that point. Heart rate is going down.
- 4.87 And your brain, the brain is adapting, trying to get you to hide in a cave or something and not use energy. And you certainly don't want to be facing any predators. And you don't want to be getting into fights with other humans, because that will all take energy that you don't have. So when you're in a fasting state or a starvation state, your brain is actually making you feel crappy so that you will go to bed and just lay there and wait until some food comes. Then once you get some food, then you can go out and start exploring the world again.
- 4.88 Hi. The question I have relates to trauma therapy or trauma healing. I was wondering if there's much research looking at what the metabolic biomarker effects are when people have those types of therapies.
- 4.89 It's a great question. Unfortunately, we don't have really good data. I'm hoping, based on this theory, that we will start to pursue data like that, that we know that some of the psychotherapies work. We don't always know how they work. Just like the data that I showed you on antidepressants, we thought they were working by increasing serotonin levels in the brain. Turns out, oh, no, that's not true. Then we ended up finding, oh, maybe they are working through metabolic pathways, in the same way that we found the relaxation response—meditation, mindfulness, prayer— might improve it through metabolic pathways. We don't yet have clear data that psychotherapy is directly producing a metabolic effect.
- 4.90 My strong hunch is that it is, and we do have some correlational data, meaning that— so there was one study, a meta-analysis, looking at people who participated in psychotherapy. If they got better with the psychotherapy, because a lot of people don't, unfortunately, if they got better from the psychotherapy, what happened to

their heart attack risk years later, and it was reduced. If psychotherapy appropriately treats your mental health condition, that reduces your risk for heart attacks later.

- 4.91** Now, some people will argue with that data because you could say, well, maybe those people spontaneously got better, and that's why they didn't have heart attacks, and all sorts of things. But it at least lines up with the theory that, if you have mental illness and it's not remitted, that you remain at higher risk for heart attacks, that is unequivocal. That we have abundant data for. Does effective psychotherapy somehow work through metabolic or mitochondrial mechanisms? We don't know, but I hope we'll find out soon.
- 4.92** I read recently a study. Scientists were studying people living in these blue zones where people live over 100 years. A lot of them, they live on elevations, somewhere in the mountains. They explain it because the lack of oxygen, maybe the reason because the oxidation is low. And I don't really, personally, don't really understand because level of ROC, the free radicals, during the oxidation, during the, like, mitochondria process, it's good when it's high or it's good when it's low? Because I understand if it's low level of oxygen, it means ROCs is also going to go probably low, no? But at the same time, mitochondria also eats ROC. It needs them because, I don't really—can you just explain a little bit more between connection between mitochondria and ROC?
- 4.93** So mitochondria are actually producing a lot of the ROS, so reactive oxygen species. The question about oxygen level is a really important one because we actually have therapies called hyperbaric oxygen, where we're putting people in hyperbaric oxygen chambers. The assumption is that, maybe if we give people more oxygen, it will be good for their mitochondria. Mitochondria are essentially—they are using well over 95% of the oxygen in the human body. Almost all of the oxygen that we breathe in is going to mitochondria. And when we breathe out CO₂, that's a waste product of what mitochondria are doing.
- 4.94** So the level of oxygen in the air is actually a really important thing. There is research recently showing that sometimes hyperbaric oxygen can actually be toxic, and that low levels of oxygen, hypoxia, can actually be health-promoting over time. The first few days or week when you're exposed to hypoxia, it's detrimental. They've done these studies in animals. They expose them to hypoxia chambers. ATP production goes down, energy level goes down. The rodents are just like, lying around like, "Oh, my god, I'm sick. Help me." Because they literally don't have energy in the form of ATP because they don't have adequate oxygen.
- 4.95** What the researchers found is, after about a week or so, the animals normalized, ATP levels normalized, and behavior normalized, still in the hypoxia chamber. When they then let them out of the hypoxia chamber, those animals lived longer and are healthier. My speculation about the mechanism is that low levels of oxygen are actually increasing levels of reactive oxygen species initially, because oxygen is the final receiver

of electrons and reactive oxygen species. And oxygen is trying to detoxify ROS. That's largely what it's being used for.

4.96 And so when you abruptly reduce the amount of oxygen available, the mitochondria are struggling and suffering. But the way the body and the cells respond is to produce more mitochondria. It is a way to stimulate mitochondrial biogenesis or the production of more mitochondria. Then when the animal gets out of that hypoxia chamber, the animal's cells will actually have more mitochondria. And that seems to confer health benefits and longevity benefits.

4.97 I'm actually really interested in seeing more research in humans. Again, there's risk because, early on, people get worse. So you'd have to do it very cautiously and carefully. But it does help us understand some of the blue zone data. It helps us understand some people who—you know, athletes do this routinely for good reason, because one of the other benefits is that you get more hemoglobin, because you get more hemoglobin and that allows you to transfer more oxygen. It gives you more oxygen-carrying capacity, and that may be helpful.

4.98 Again, the end user of hemoglobin and the oxygen on it are mitochondria. That's it. They're the primary users of oxygen. Even if it's through a mechanism like red blood cells or hemoglobin, it's still ultimately conferring mitochondrial benefits. But it is a paradox to think that, wait, we're going to give less oxygen and we're going to get a benefit? But it seems like that may be happening.

Lecture 5: The Six Pillars of Wellness

At a glance — The six pillars of wellness — lifestyle and mental health.

- 5.1** Now we're going to dive into the topic of lifestyle and mental health. I just want to point out there is an entire field of medicine called lifestyle medicine. These are some of their highlights from their website, the American College of Lifestyle Medicine. They basically say that lifestyle medicine is a medical specialty that uses therapeutic lifestyle interventions as a primary modality to treat chronic conditions, including, but not limited to, cardiovascular diseases, type 2 diabetes, and obesity.
- 5.2** They talk about nutrition, physical activity, stress management, restorative sleep, social connection, and avoidance of risky substances. Those are the six pillars of lifestyle medicine. Five out of six of these, I completely agree with the American College of Lifestyle Medicine. The one area that I have some concern about is nutrition. At the end of the day, they prescribe primarily a plant-sourced diet, and that is the generic recommendation. Eat more plants, and that will somehow confer health. My beef with that, so to speak, is that I believe in personalized nutrition plans and recommendations, and that I don't think that there are one-size-fits-all recommendations for all humans with all health conditions.
- 5.3** A clear, easy, indisputable example of this are people with celiac disease. So plant-based nutrition plans say, "Eat lots of whole grains." Well, I hope it comes with a caveat that unless you happen to have celiac disease or sensitivity to gluten, and yet that's somehow not in the recommendation. The biggest glaring deficit of the American College of Lifestyle Medicine is that we're not talking about mental health. We're talking about physical health. Does lifestyle medicine, do lifestyle factors, have anything to do with mental health?
- 5.4** Believe it or not, this is actually highly controversial still, and most mental health professionals will say, "Probably not." Sure, lifestyle factors are important for generic, general health and wellness, but do they play a role in serious mental illnesses, like chronic unrelenting depression, ADHD, autism spectrum disorder, bipolar disorder, schizophrenia? Most mental health professionals will say, "No way, no how." Those are brain disorders, and those are disorders of neurochemicals, or those are disorders of trauma and stress and childhood adversity. They have nothing to do with what people eat.
- 5.5** However, I'm going to walk you through some lines of evidence that I hope will make you reconsider and that I hope will make mental health professionals reconsider this stance. I'll start first with observational data. Observational data just says, if we look at a large group of people, 287,000 participants from the UK Biobank, and we look at lifestyle factors, do they have any correlation with whether those people have depression? In fact, this study suggests, yes, across the board. Alcohol consumption,

healthy diet, regular physical activity, not smoking, healthy sleep, low to moderate sedentary behavior, and frequent social connection are all correlated with improved mental health, less risk for depression.

- 5.6** What I'm arguing is that lifestyle treatments, although a lot of people will consider them light interventions, I actually think they can be life-changing, extraordinarily powerful interventions, because, in fact, they are one of the critical ways to target metabolic health and mitochondrial health. I'm going to go through some of the different lifestyle interventions. This is not going to be an exhaustive list and an exhaustive resource on all of the different things, but I'm going to at least give you some highlights from some of the different pillars of lifestyle medicine.
- 5.7** We've already talked about social connection and purpose. We'll be talking about substance use a little later, but let's dive into sleep. Sleep is critical for human health across the board, both mental health and metabolic health. On average, adults need about seven to nine hours of sleep. There are rare adults who seem to be healthy with six-ish hours of sleep, but most adults need somewhere between seven and eight hours of sleep. Some maybe a little more. The real answer is that one in three adult Americans are not getting adequate sleep. So although this seems like common sense, it's not commonly practiced.
- 5.8** Many lines of research connect sleep with mitochondrial health and function. I'll review just a little bit of that. But there are sleep disorders. There are things that can prevent people from sleeping. Insomnia is actually a disorder. It doesn't give us the exact reasons, but it tells us somebody can't sleep. Then there are concrete reasons why people can't sleep. Sleep apnea is one of the more common ones. People stop breathing in the middle of the night, more often than not because their neck is too thick and their tongue basically collapses within their airway and prevents them from breathing. This is actually pretty common.
- 5.9** There are other types of sleep apnea. There's something called central sleep apnea, where the brain, apparently, isn't giving the signals to continue breathing. There are different treatments for those. There's something called restless leg syndrome, where people, it happens at night, just as they climb into bed. They're perfectly fine all day. Then when they're exhausted and climb into bed to go to sleep, their legs just become restless, fidgety, uncomfortable, and they have to move them, and they're constantly moving, and it can turn into thrashing.
- 5.10** No matter what they do, no matter how many deep breaths they take, no matter how much they try to distract themselves from it, their legs are moving. That is a treatable condition. It's a recognizable condition. Iron deficiency, surprisingly, can play a big role. So getting ferritin checked can actually be invaluable for some people with restless leg syndrome. Medications and drugs can interfere with natural and restorative sleep. So although I'm saying a lot of Americans aren't getting enough sleep, my solution is not

take pills to get you to sleep, and it's not take drugs or alcohol or marijuana to get you to sleep. It is trying to get natural restorative sleep.

- 5.11** When you eat food can interfere with sleep. If you are eating late at night, if you are eating in the middle of the night, snacking, that can actually impact both metabolic and mental health. One of the challenges with talking about sleep is that sleep disruption can be a symptom of mental health disorders. It is also a very common symptom of all of the metabolic health disorders. If you look at the rates of insomnia in people who have cardiac disease, they're much higher than the general population. If you look at the rates of insomnia in people with type 2 diabetes, they're much higher than the general population.
- 5.12** So both metabolic health disorders and mental health disorders appear to be able to cause sleep disruption. But this study suggests sleep disruption in and of itself can result in both metabolic and mental health problems. And so, again, we're talking about a vicious cycle. And you can enter that cycle either way. You can enter that cycle by not getting enough sleep. That can cause a metabolic or mental health condition, which then might make your sleep even worse, which then might make your metabolic and mental health condition even worse. So, you can enter it many ways, but the obvious solution is, try to get good restorative sleep without the use of pills or substances.
- 5.13** Light exposure plays a big role in sleep, but it also plays a big role in circadian rhythms. It can play a powerful role in both mental and metabolic health. It's ideal to get at least 20 to 30 minutes of bright light exposure in the morning. If you live in a warm-ish climate, even if you live in a cold climate, if it's not too terrible and there's light outside, when you wake up in the morning, you could just go for a 20- or 30-minute walk. If you want to do it in nature, that's great. If you want to walk barefoot on the grass, some people love that. They call it grounding and all sorts of things. There are lots of things you can do, but try to get outside and get some natural light.
- 5.14** If you're like me and you live in the northeast where it is freezing cold and maybe there's a blizzard outside and the sun's not up anyway, there's no point in going out in a blizzard. You might use what's called a bright light therapy lamp. These are 10,000 lux. They are safe for the eyes, but you basically sit in front of it, pretty close to your face, about two feet or so away from your face. You're going to be on the computer, reading a book, answering emails. You're going to be doing something useful with this time. You're not going to just sit there and stare at a light. But you're going to have this light in front of you. Maybe you're eating breakfast or having your coffee, whatever. You're going to get this light exposure first thing in the morning.
- 5.15** Probably one of the most important things about light exposure in the morning is that it's training your circadian rhythm. When you train your circadian rhythm, that is going to help your metabolic health, your mitochondrial health, and it will also help you sleep better, which will in turn help your metabolic and mental health. The converse of light

is that you don't want any light at night. Minimize light exposure at night. That means close the curtains, get blackout blinds or drapes. Avoid nightlights. There was a recent study. People who were exposed to light, even from nightlights, a certain amount of light, had higher rates of type 2 diabetes.

5.16 We know that disruption of sleep with light at night can exacerbate essentially all of the neuropsychiatric disorders. Does any of this have anything to do with mitochondria? The reality is that there are hundreds of studies published in leading journals connecting all of these dots, connecting sleep with circadian rhythms, with mitochondria. This is just one example, circadian rhythms and mitochondria connecting the dots. I only show this to you not to go into the scientific details of it in the molecular pathways, but just to let you know, yes, this is about mitochondria and metabolic health.

5.17 Now let's do a sleep self-assessment. Pull out your pads of paper. If you're at home doing this on the computer, get ready to take some notes. How much sleep do you get? On average, how many hours per night? What time do you usually go to bed? What time do you usually wake up? Let me stop right there. One of the most important things is a consistent wake-up time. If you have trouble sleeping and you can't get in a normal rhythm, the first most important rule is start waking up at the same time every day, even if you didn't sleep well. That's the only way you're going to be able to train your circadian rhythm.

5.18 If you want a really important privileged biohack, if you have the privilege and luxury to do this, go to bed really early so that you don't have to set an alarm. Train your circadian rhythm so that you always wake up at least two hours before you have to be anywhere. Then you will always wake up when your body is ready to wake up and you will have the rest that you actually needed. Instead of spending two hours at night in front of the television set, having a glass of wine or a beer and binge-watching Netflix and eating junk food, instead of that being your two hours of free time, you're going to actually forego that. And instead, you're going to go to bed.

5.19 Then you're going to wake up 2 hours early, and you're going to have your free time in the morning, 2 hours early. The huge difference is that when you wake up in the morning after a really good night's sleep, you're not going to want to sit and binge-watch Netflix. You're not going to want to overeat or overdrink. You're actually going to feel great. That's going to be the time that maybe you're going to be creative or productive.

5.20 If you're a mom to a family, and one of your responsibilities is to clean up the house or get things organized, maybe that's a chance for you to work unbothered by everyone else. It's a chance for you to catch up on the day. If you're a business executive and you need to create business strategy plans, that's a great time to be working on that. If you have trouble fitting in a workout, it's a great time for you to work out. Do some meditation, do yoga, whatever.

- 5.21** Do you have trouble sleeping? This is back to the self-assessment. Do you have trouble sleeping? If so, specify how, why. Do you wake up feeling refreshed and rested? If the answer is no, there's a problem. Do you take naps or doze off during the day? It is shocking how many Americans doze off during the day. If you're dozing off during the day, that's a problem. Do you snore? Snoring is a sign of possible sleep apnea. If you are a regular snorer and you're waking up feeling unrefreshed, if you're dozing off during the day, you really should be getting assessed for sleep apnea.
- 5.22** Has anyone ever observed that you stop breathing or gasp for air during your sleep? That is an overt sign of sleep apnea. And do your legs bother you at night? This is a screening questionnaire for restless leg syndrome. If your answer is no, you don't have restless leg syndrome. Don't think twice about it. If your answer is unequivocally, "Oh my gosh, yes," you may have restless leg syndrome and might want to look into it.
- 5.23** What if you do have problems with sleep? What if the responses you already kind of know, yeah, sleep's not so great. Here are some basics from sleep hygiene. First and foremost, prioritize sleep. Understand that sleep is essential for human health. Number two, or really number one in terms of sleep hygiene, fixed wake-up time. That is what I've said, and that's what I'm going to stick by, is get up at the same time. If you have chronic insomnia, the best way to overcome that chronic insomnia is to force yourself to get up at the same time every day.
- 5.24** Then you've got to stay out of bed for the rest of the day, and you're going to say, "Well, I'm too tired to do that. I'm exhausted. That's going to be a lousy day." Yes, it is going to be a lousy day. I'm not going to fight you on that. It's going to be a really lousy few days while you are trying to train your circadian rhythm and train your body to sleep. But what you're going to do is, if the wake-up time that you set is 7 AM, I really don't care what time you want to wake up, but if you want to wake up at 7 AM, get up at 7 AM, force yourself to get out of bed, go through the motions. That will hopefully get you tired enough so that, later that night, the next night, you can sleep.
- 5.25** You're going to wake up at 7 AM the next morning, and do it again and do it again, until your body is basically saying, "OK, seems that the only time I get to sleep is at night, so I'm going to take advantage of that and I'm going to sleep at night." Watch napping. If you really have trouble sleeping at night and you want to fix that, maybe back off on napping. Try to have a consistent routine. Try to wind down 30 minutes before bedtime. Unplug from electronics just before bed.
- 5.26** If you are having trouble sleeping, don't stay in bed tossing and turning all night. Give yourself about 20 minutes. And once you recognize, "I don't think I'm going to be able to fall back to sleep," force yourself out of bed. Get out of bed and go do something really boring and quiet. Read a really boring book. And you'll say, "But I can't pay attention." That's perfect. That's perfect. You're so exhausted. Well, go read a really boring book until you start dozing off and then try to go back to sleep.

- 5.27** Make sure your bedroom is cool, dark, and quiet. Watch your use of alcohol, especially right before bed. It's great to help you fall asleep, and it's great to make you wake up halfway through the night, tossing, turning, sweating. So alcohol is great to initiate sleep. It's not good for sustaining sleep. If you are having trouble sleeping, watch your use of caffeine, especially after noon. So any caffeine. That means diet sodas, tea, anything. If you're having trouble sleeping, you should avoid caffeine use because you want to build up a sleep appetite, is what they call it.
- 5.28** If you're really having trouble sleeping, restrict activity in bed to sleep, and one other thing if you have a partner, but nothing else. Don't be reading in bed. Don't be doing crosswords in bed. Don't be talking on the phone to friends while you're in bed. You want your bed to be associated with healthy, good sleep.
- 5.29** Circadian rhythms. Here's a self-assessment for that. How much exposure do you get to bright light in the morning? Is your bedroom dark? Are there any lights? Is there light coming in through the windows? Do you have a night light? Do you have a really bright alarm clock that just shines in your face? Do you get outside much during the day? Do you get any exposure to sunlight? Do you get natural light in your home or office? Do you work in an environment where you've got windows? Do you work in a place with no windows? Do you eat late at night or in the middle of the night, which can disrupt circadian rhythms?
- 5.30** Now I want to move on to exercise. It is well known that exercise is good for metabolic health and mitochondrial health. Is it good for mental health? We have large epidemiological studies. This one's over a million US residents, looking prospectively, people who exercised more, within reason, but people who exercised tended to have better mental health. So exercise is associated with improved mental health. You can overdo it, and I'll get into some of the science of that. But in general, this is a correlation, and a lot of people will debate why this correlation exists.
- 5.31** Some will say exercise is improving mental health. But others will say, no, it's backwards. People who have good mental health have the energy and motivation to go exercise. People who have poor mental health lack motivation. They lack energy because of their mental health condition. And so, of course, we see this connection between people who exercise and people who have good mental health. Just like we talked about with relationships, it can go either way. My strong guess is that it does go both ways, that both of those relationships are true.
- 5.32** How can we measure physical health or exercise capacity? One simple measure that healthcare clinicians might start doing at some point or another in their practices is hand grip strength. There are some who are arguing hand grip strength should be a vital sign, just like blood pressure and heart rate. Several diseases have shown a strong correlation with hand grip strength—the higher, the better your health. This includes overall mortality, risk for hospitalization, risk for morbidity in terms of health

conditions, physical health conditions, functional disabilities, and at the top, psychological health is associated with how strong your hand grip is.

- 5.33** Can exercise actually be a treatment for mental illness? It is known to improve metabolic and mitochondrial health. One meta-analysis of exercise for depression, just published this year, looked at 218 unique studies, over 14,000 participants, and the conclusion was, yes, exercise appears to be a viable treatment for depression. The exercises that they particularly noted were as simple as walking, jogging, yoga, strength training, tai chi, and qigong.
- 5.34** World Health Organization also agreed in their report. "A review of the evidence for the benefits of physical activity for people with depression, schizophrenia, and dementia indicates improved mood, slowed cognitive decline, and delayed disease onset." If you talk to a lot of mental health professionals about exercise for mental illness, most of them will say, "Yeah, those are nice add-ons. They're good health and wellness information, but it's not a real treatment. It's not a real treatment."
- 5.35** So am I saying that if everybody just exercised a little more, we would solve the mental health epidemic? Let me be clear. No, I am not saying that. And so how does that work? Why not? Why wouldn't exercise work for everyone? There are a few reasons. This is one of them. Exercise and insulin resistance. This was a study published in a very prestigious journal called *Cell*, where they looked at a small number of volunteers, only 36, but they did such extensive batteries of tests on them before and after exercise.
- 5.36** What they found was that there were thousands of molecular changes that occur when people exercise. And in people who have low levels of insulin resistance, these changes are highly beneficial. You see anti-inflammatory effects. You see effects for building muscle and tissue in mitochondrial health. But in the people who had high levels of insulin resistance, sometimes these molecular changes were the exact opposite. People with type 2 diabetes, when they exercise, might actually have more inflammation after exercise than less inflammation.
- 5.37** So we need to take that kind of information into account when we prescribe exercise to people and when we talk about making modifications. It should not be a generic one-size-fits-all. "Everybody go exercise. The more, the better." No, you're going to get some people in trouble. People with type 2 diabetes, for instance, may not be able to tolerate. They may need to build up their tolerance for exercise, and/or they may need to make changes in diet first before they try to add exercise to their routine, because they may actually feel worse after doing exercise than before.
- 5.38** People with chronic fatigue syndrome, in particular, long COVID, there's some evidence that exercise can actually make them feel worse. Their mitochondrial function is already impaired, and there's evidence that it can actually further impair their mitochondrial function. That their mitochondria are so vulnerable that they can't handle exercise. Now, does that mean exercise can't play a role in their treatment? No.

I think exercise can play a role in their treatment, but it needs to be done in a gradual, progressive way, customized to the individual to make sure that it's actually helping the person and not harming them.

- 5.39** What about medications? Can medications impact the effect of exercise? In fact, yes. This study looked at 53 adults, ages 61 to 63, and assigned all of them to 12 weeks of aerobic exercise treatment. Then they did all of these metrics on adaptations and insulin sensitivity and metabolic health and inflammatory biomarkers. Half of the people got Metformin, which is a diabetes medicine, and half of the people got a placebo. The people who got Metformin had impairment in the health benefits. They had reduced health benefits from the same exercise program.
- 5.40** What this suggests is that Metformin may, in fact, be impairing mitochondrial adaptations to exercise. Am I saying nobody should take Metformin? No. If you have type 2 diabetes and you don't have it under control yet, Metformin can be a great medication. But if you can somehow get your diabetes under control in other ways, diet, lifestyle, medicine, then doing without Metformin may actually improve your health even more. And with exercise, the dose matters. So even in healthy people, this study looked at having healthy people do exercise. They found very high levels of benefit from high-intensity training.
- 5.41** But if they push them to keep going further and further, at some point, it actually began to impair mitochondrial function and impair metabolic health. What's a clear example of that? The clearest example of that is someone with anorexia nervosa who is over-exercising and undereating. They are starving themselves to death. They are already malnourished, and the intense level of exercise that they are doing is actually causing harm. And so I think we need to think about metabolic health in a more nuanced and sophisticated way. It's not just a one-size-fits-all recommendation. It's the right dose for the right person.
- 5.42** Let's do a quick self-assessment on exercise. How do you think your physical abilities, compared to most people your age right now? Well, one, not great, to 10, you are very fit. You've got the best physical health, physical abilities you can imagine. As I've asked before, if you did not rate yourself a 10, what can you do to get yourself one point higher? Some other quick self-assessment questions. How active are you in your daily life? Not including what you call exercise, just how active are you? Do you have a desk job? Are you sitting all day? Do you work cleaning houses? You're active all the time. You're walking, scrubbing, carrying vacuum cleaners. Those are very different professions.
- 5.43** Do you exercise? Do you do what you call exercise regularly? If so, what do you do? How much and how often? Here are some concrete ones. Can you hang from a bar for at least 30 seconds? How many pushups can you do? If you're watching this from home and you don't know, push the pause button and try it now and see. If you are a woman, you should be able to do at least five. If you are a man, you should be able to do at

least 10. From large epidemiological studies, if you are a middle-aged man, 40 confers optimal health. 40 pushups will get you in the optimal health category that is associated with reduced risk for cardiovascular disease and premature mortality.

- 5.44 Can you sit in a chair and get back up without using your hands? In other words, can you do an unassisted squat? This may seem silly, and some people actually get a little worked up about it, but I'm telling you, being able to sit down in a chair and get back up without using your hands is a predictor of whether you are going to need assistance later in life getting on and off a toilet. It's a big deal. It's a really big deal. Can you stand for 30 seconds on one foot at a time without falling or needing to hold on to something?
- 5.45 Now I want to move on to one of the biggest pillars of lifestyle medicine, which is diet and nutrition. There is an entire field called nutritional psychiatry that has been around for a few decades. Unfortunately, most traditional mental health practitioners don't really give it much attention. In fairness, one of the reasons is because some nutritional gurus have way overplayed their hand. There were reports decades ago that, if all people with schizophrenia just took niacin supplementation, it would cure everyone with schizophrenia. That turned out to not be true.
- 5.46 We were told that vitamin C and vitamin E, these are antioxidants, oxidative stress. This plays a role in schizophrenia. If we just give people enough antioxidants or have them eat lots of fruits and vegetables, which are high in antioxidants, we will combat schizophrenia. It didn't work out. In fairness to the millions of mental health professionals who are skeptical, that drives their skepticism. They've heard this before, and they have been disappointed and frustrated. They are looking out for the health and welfare of the vulnerable people they serve, and they don't want them getting duped by charlatans.
- 5.47 And so when they hear claims that diet can play a role in severe mental illness, they just remember all of those failed attempts and promises from the past and think, "Oh, here we go again. More quackery." With that said, I hope by the end of this course, you're going to understand that I am a passionate advocate for dietary interventions in mental health, and that, in fact, it is not at all quackery. What are the models to consider? How should we even think about nutrition in psychiatry?
- 5.48 One model is that either there is—the most common perception is that there's something wrong with the diet. You're saying something's wrong with the diet, and that could be one of two things. Either you're getting some bad stuff in your diet that's not good for you, or you're not getting enough good stuff in your diet. You're not getting enough good stuff. That is, you need more antioxidants, eat more fruits and vegetables. That'll fix you up. More fruits and vegetables will give more of the essential nutrients you need. Eat some spices, have some spices on your food. Those are good for you. So increase the good stuff, and the good stuff will take care of the problem.

- 5.49** There's another model that says, no, no, no, let's focus on the bad stuff. There's bad stuff in diet, ultra-processed foods. We'll talk about that in a little bit. Bad stuff can harm people. So just reduce the bad stuff and you'll be all set. Now, there's a whole new field that's focused on the gut microbiome. In fact, it's actually not really new. The researchers have been doing it for decades. They're a little pissed off that people think it's new, but it's getting a lot of new attention, the gut microbiome.
- 5.50** We know that the gut microbiome plays a critical role in both metabolic health disorders and also a wide range of mental health neuropsychiatric disorders, including autism spectrum disorders, schizophrenia, chronic depression, OCD, binge-eating disorder, lots of mental health disorders. Maybe it's the gut microbiome. It's less about the foods that we're putting in, and it's more about the microbes in our gut and what they're doing with the food and how that all connects.
- 5.51** Then I would argue there's another category, and I call them metabolic interventions. They primarily consist of fasting, intermittent fasting, and fasting-mimicking diets like the ketogenic diet. I actually don't see those falling neatly in any of the other categories, although some people do. Some people say, "Well, no, this is the right diet. People should only eat meat, and if they only eat meat, that's the only food that humans were meant to eat. Everything else is bad for you, so we're putting in the good and taking out the bad." And that's one interpretation of a carnivore diet.
- 5.52** I actually don't see it that way. I actually see a carnivore diet as an intervention, a metabolic intervention that shifts metabolism. The primary reason I say this is because these are fasting-mimicking diets. Fasting is the absence of food. Fasting has nothing to do with getting rid of bad stuff or putting in good stuff. I guess you could say it has something to do with getting rid of bad stuff because you're not eating anything, but it certainly doesn't include putting in good stuff because you're not putting in anything. Fasting is no food. And yet fasting has been used in just about every culture, most religions, for healing purposes, for millennia.
- 5.53** And shockingly, maybe there's actually something to it. We, modern medicine, are so skeptical of that, and yet science is coming around to saying, wait, there might be something to this. So now let's do a little bit of a tour through some of the variables. I could talk for a thousand hours and not cover all of the different topics that need to be covered when we are talking about this. There are so many variables, so many foods, so many ingredients, so many different health conditions. I'm only going to give you a tiny snapshot.
- 5.54** Sugar. Does sugar have anything to do with poor metabolic health and poor mental health? We know from correlational studies, unequivocally, yes. People who eat a lot of sugar, consume a lot of sugar, sweetened beverages, have higher rates of poor metabolic health, obesity, type 2 diabetes, cardiovascular disease, premature mortality. Guess what? They also have higher rates of poor mental health. I share with you two observational large cohort studies that confirm this.

- 5.55** That means we should all use artificial sweeteners, right? Splenda, aspartame, all of those. If sugar is bad, then we should use diet soda and products. Well, hold on, unfortunately, World Health Organization and others are coming out. "No, no. Hold on, folks. Looks like these artificial sweeteners are actually not so benign after all." People who consume a lot of these have higher rates of metabolic and mental health disorders.
- 5.56** Ultra-processed foods. This was an umbrella review published just this year in 2024. People who consume more ultra-processed foods, what are the problems we see in them? If you consume a lot of ultra-processed foods, you have higher risk of all-cause mortality. You're more likely to die an early death. Higher rates of certain types of cancer, higher rates of respiratory illness, cardiovascular disease, gastrointestinal disorders, metabolic health disorders, obesity, abnormal lipids, type 2 diabetes, and oh, yeah, mental health makes the list, too. Higher rates of mental illness.
- 5.57** Ultra-processed food and one specific mental illness, depression— we've got lots of lines of evidence. Many studies have been done now. People who consume nine or more servings of ultra-processed foods had about a 50% higher risk of depression compared to people who consumed four or fewer servings of ultra-processed foods a day. And in this particular study, they noticed a link between artificial sweeteners, in particular. Artificial sweeteners stood out.
- 5.58** This was a meta-analysis done of lots of studies that have been done on the connection between ultra-processed foods and mental health, and sure enough, confirm. The more ultra-processed foods you eat, the more likely you are to have a mental health condition. I brought up artificial sweeteners. What's wrong with artificial sweeteners? What's not to love about them? They're zero calories. What is not to love about artificial sweeteners?
- 5.59** This study was published in 2022 in the Proceedings of the National Academy of Sciences, a pretty prestigious journal. Now, it is a study in mice. We can't do these type of studies in humans. Some people will say, "Well, it's a mouse study. It doesn't count." But we can't ethically do these kind of manipulations to humans. We can't dissect brain regions after we give them certain foods. This is the only way we can ever get mechanistics evidence, is through animal studies.
- 5.60** What did the researchers do? They took mice and they gave some of them aspartame, NutraSweet, which is found in nearly 5,000 diet foods and drinks. This is something that if you drink Diet Coca-Cola, you're drinking this, aspartame. They gave the mice doses of aspartame, equivalent to below 15% of the FDA recommended maximum daily intake for humans. So they're not overdosing the mice on aspartame. They're giving them reasonable doses of aspartame. And what did they find? Aspartame induced anxiety symptoms in mice.

- 5.61** These anxiety symptoms were alleviated by Valium, an anti-anxiety pill. The reason that's important is because it suggests this is the anxiety that we're all talking about. This is common run-of-the-mill mental health anxiety. It gets better with Valium. Aspartame, they then looked at the brains to see, what is it doing in the brain that might be resulting in these symptoms? Aspartame changes gene expression, and it changes the activity of the amygdala. The amygdala is often referred to as the fear center of the brain. It's actually a lot more than that. It's probably better termed the salience center, meaning it's the center of the brain that helps us determine whether something's relevant or not.
- 5.62** Life-threatening situations are always relevant. So yes, it is definitely involved in fear and threat perception. But when your amygdala is firing, it might mean that you're having panic attacks. The activity of the amygdala was affected in the mice, which helps us explain why are these mice looking, behaving, in an anxious way? Here is the heartbreaking and tragic news. The anxiety and the brain changes appear for up to two generations of mice after the exposure.
- 5.63** The researchers gave aspartame to these mice. The mice had babies. The babies never got exposed to aspartame, yet they had higher rates of anxiety. They had these same brain changes. Those babies had babies. Those babies were also affected, even though they never got a drop of aspartame. What this suggests is that, if this is true in humans, there's no reason to believe it can't be true in humans.
- 5.64** A mouse study doesn't prove it's true in humans, but it at least raises the question of, could this be happening in humans? If this is happening in humans, what does it mean? It means that if your grandmother drank a lot of Diet Coke, that may be one of the reasons why you have an anxiety disorder today. And think of the millions and millions and millions of people consuming products that contain aspartame.
- 5.65** Now I want to talk about this connection, the gut-brain connection. The gut-brain connection is actually ridiculously complicated. It involves in some ways, maybe three different components. There's the inside of the gut, which most people are focused on the microbiota, all of the different bacteria, fungi, viruses, and other living organisms in our guts, that they are actually serving a useful purpose, it turns out. But then there's the gut itself, and the gut itself is lined with cells, and those cells are absorbing nutrients. When they absorb those nutrients, they create chemicals, neuropeptides. They create signals that turn neurons on or off.
- 5.66** The gut is connected to the brain through the vagus nerve. And so the vagus nerve is actually playing a role in some of this. And then it all leads to the brain. And so this study looked at just some of the different ways that this axis can play out. So there are pathways, the immune system, the neuroendocrine system, the enteric nervous system. There's a whole nervous system around your gut. There's the circulatory system and the vagus nerve. And then there are secretions—peptides, hormones, other molecules.

- 5.67** The enteric nervous system is actually the largest component of the autonomic nervous system. So when you think about the autonomic nervous system outside of the brain, the biggest component is actually the GI tract. And this affects behavior, mood, and cognition. Who and what is in the gut? We all have about one to two kilograms of microbiota, and it's composed of bacteria, viruses, fungi, archaea, inhabiting the gut, about 100 trillion organisms.
- 5.68** These organisms coexist with pathogens. There are good bacteria and bad bacteria. It's actually not that simple, but a lot of people talk about it in that way. The gut microbiota helps regulate the immune and endocrine system. Stress and sleep deprivation are known to increase cortisol, and cortisol is known to change your gut microbiome. So one of the biggest challenges when we talk about the gut microbiome is that everybody assumes probiotics are the answer. If there's a problem with my gut microbiome, then I just need a pill. We're always looking for pills, right? I just need a pill to take so that—put the good bacteria in there.
- 5.69** Well, I just told you, there are 100 trillion organisms in our digestive tract. You taking one pill with one type of bacteria is a drop in the bucket, so to speak. You're probably not going to make a huge difference. Most of the studies that have been done so far on probiotics for mental health, metabolic health, have confirmed that. Single bacterial probiotics, for the most part, don't work. Instead, you need to look at living organisms, which means refrigerated probiotics and multiple different species. Some of those early studies are showing some promising results, so those might be helpful.
- 5.70** As a reminder, 95% of the serotonin is produced in our gut. But one of the reasons that I wanted to make the point that stress and sleep deprivation affect the gut microbiome is because taking a probiotic isn't going to be the answer if you're not sleeping and you're still stressed. Some of the basic commonsense lifestyle strategies that we're talking about may, in fact, be the primary way to help regulate your gut microbiome. You may not need to do anything with diet or with probiotics. Instead, you may just need to get better sleep or exercise more. That might be the way to better regulate your gut microbiome.
- 5.71** One of the problems with the gut-brain connection is that sometimes people can get what's called intestinal permeability or leaky gut. The gut, actually, normally—gut cells have these junctions in between them that are tight junctions. They prevent things from getting in or out. But sometimes these junctions can open up and allow for bacteria and other things to get into the nervous system. This is not always pathological. This may actually be one of the ways that we develop antibodies to pathogens in our gut and help fight them off. It's not always pathological, but clearly, if it's overdone or if it's done in an abnormal way, it can be pathological.
- 5.72** There is a hormone called zonulin that is released and that actually helps highly regulate whether these junctions are tight or not tight, meaning leaky. Zonulin release is stimulated by gluten and pathogenic bacteria. This helps us understand why maybe

some people are sensitive to gluten, even if they officially don't have celiac disease. The gluten may, in fact, be making their gut leaky, which is allowing potential pathogens into their bloodstream, which could be problematic.

- 5.73** What does this have to do with, say, something like bulimia nervosa? Like, bulimia nervosa, that's just like women wanting control, right? These are, like, women who—the paradigm right now is that bulimia nervosa is a psychological disorder. It's women who have either been traumatized or they've grown up in families where their parents were hypercritical. That's the stereotype, hypercritical parents. These poor women are just trying to be good enough. They're restricting and then bingeing and restricting and bingeing. What's the treatment for that? Talk therapy. Talk therapy will fix everything, right?
- 5.74** I know that what I just said sounds a little facetious, but actually that is the primary thinking of bulimia nervosa and anorexia nervosa. These are psychological issues. This research suggests otherwise. This research was initially done in animals, and what they found is that alterations in intestinal microbiota are responsible for binge-eating of palatable foods, and that the combination of stress and dieting history can alter your intestinal microbiota. They found one specific molecule, kynurenic acid—I'm probably not saying that right—and that that acid is metabolized by your gut microbes, or are produced by your gut microbes. And then that acid actually helps regulate binge-eating.
- 5.75** What they found was that when they supplemented the mice with either a specific probiotic or kynurenic acid, the mice stopped overeating. They then did this same research in humans with bulimia nervosa and found similar findings, that they appear to have lower levels of kynurenic acid. We don't yet have clinical trials of this or probiotics in people with bulimia nervosa. But wouldn't it be shocking if we could change the gut in some way to get people with bulimia nervosa to stop binge-eating? More importantly, wouldn't that be tragic and heartbreaking? That we subjected those women to years or decades of talking, when in fact talk therapy was not the answer that they needed?
- 5.76** What do mitochondria have to do with any of this? A lot more than you probably realize. One of the important things is that there are mitochondria in the intestinal cells. If those mitochondria are not being fed properly by things like short-chain fatty acids or butyric acid, if they are not being fed the proper fuel, they will not function properly. If they do not function properly, those gut cells will not function properly. If those gut cells aren't functioning properly, you may get dysregulation of the communication between the gut and the brain. That can then impact brain function.
- 5.77** I'm going to highlight another study in the next lecture that is going to explicitly address this when we talk about the ketogenic diet. But this research actually links mitochondrial function in intestinal cells with major depression. The mitochondria in your intestinal cells might be playing a critical role in whether you think, "I'm worthless. My family doesn't love me anymore. I think maybe I'd rather be dead. I can't get out of

bed in the morning. I can't think. I don't want to eat. Why do I even exist?" That state may be due to what's happening in your gut, and more specifically, what's happening in the mitochondria in your gut cells.

- 5.78** Although that sounds wild and although that sounds really complicated, all of the strategies that we're talking about may be commonsense lifestyle strategies that can address the disruption or dysregulation in that process. Can diet actually be a treatment? Some of the evidence that I've shared with you so far is just correlation. It just says, people who happen to eat a lot of ultra-processed foods happen to have higher rates of mental illness.
- 5.79** Yet again, we're faced with the question, well, is it just that mentally ill people eat junk food? Junk food is comfort food after all. So, mentally ill people are probably more likely to eat a lot of junk food. So that's it. Mental illness forces people or causes people or gets people to eat more junk food. There's nothing there. The junk food can't possibly be treated. We're still faced with these correlational dilemmas of cause-and-effect. The more important question is, can diet, if diet is driving brain function, if diet is playing a role in brain function, can changes in diet be a treatment? And most importantly, can they be a treatment for severe chronic mental illnesses?
- 5.80** Can they be a treatment for mild mental illnesses? I think most people agree. Yes, of course they can. If somebody's got mild anxiety and they clean up their diet and they start exercising, most people can understand and accept. "Well, yeah, sure, sure." Because they say that person didn't really, they didn't really have anxiety. They didn't have a real anxiety disorder. They were just a little off. They were just a little off and they feel better now. That's great. But again, the psychiatrists will say, "But we treat real mental illness, real ones. We treat severe chronic, unrelenting mental illnesses, and those mental illnesses need medications. There is no way around it."
- 5.81** Can diet do something for those people? First and foremost, I want to just reiterate, there is not a one-size-fits-all dietary approach. There are several different goals that dietary interventions can serve, and sometimes they are the exact opposite. One is that some people need to lose weight. Some people are obese, and the obesity itself may be playing a role in their brain function, and helping them lose weight might improve their metabolic and mental health.
- 5.82** But I've already shared with you the other end of the spectrum. There are people who are emaciated. People with anorexia nervosa, they are grossly underweight, they are malnourished, they are starving to death. There are people in third-world countries starving to death not by choice. There are people with, say, a man with chronic, unrelenting depression who also has cancer. He's got cancer and severe depression, and he has lost a tremendous amount of weight. All of those people would benefit from gaining weight. The metabolic intervention for them is, we need to get calories in you and we need to get nutrients in you. We need to restore your health.

- 5.83** Those are polar opposite interventions, right? We might want to address nutritional deficiencies. We might want to remove allergens, toxins, chemicals, if you are sensitive to these things, or if we have evidence, like with aspartame, that maybe it's not so good for us after all. People talk about generic healthy diets. I've shared with you some of the research on the gut microbiome, and I've shared with you this concept of changing metabolism through dietary interventions.
- 5.84** Let's talk about the Mediterranean diet. The Mediterranean diet is mostly fruits and vegetables, whole grains, legumes, nuts, fish, white meats, and olive oil. It includes moderate consumption of fermented dairy products and red or white wine during the main course. There's a large PREDIMED trial done where they did a randomized controlled trial, 7,400 people assigned to a Mediterranean diet. Some of them got free olive oil, some of them got free nuts, and the others were asked to do a low-fat diet. The ones who were assigned to the Mediterranean diet group had a 30% reduction in cardiovascular events.
- 5.85** The Mediterranean diet, in particular, the consumption of fats. This is less about fruits and vegetables and it is more about healthy fats. The consumption of healthy fats reduced actual cardiovascular events. They saw improvement in obesity, metabolic syndrome, diabetes complications, and even breast cancer outcomes. So the Mediterranean diet, in particular consumption of healthy fats, can improve metabolic health.
- 5.86** Can it do anything for mental health? Unfortunately, all we've got are two relatively small trials, but they both suggest yes. The HELFIMED trial looked at 152 people who had self-reported depressive symptoms, and the ones who got the Mediterranean diet intervention group had improvement in depression symptoms compared to the control group. Then the more rigorous trial was called the SMILES Trial. 67 participants with diagnosed major depressive disorder, half of them got a 12-week Mediterranean diet education group, and half of them got a control social support group.
- 5.87** The patients who were assigned to the Mediterranean diet group, 32% achieved remission, compared to only 8% who got the social support group. That is not the end of the diet story. As I said, there are so many other facets to think about. So we're going to be talking a lot more about other dietary interventions. But in the next lecture, I'm going to be focused specifically on one intervention called the ketogenic diet. Thanks.

Lecture 6: The Ketogenic Approach

At a glance — The ketogenic approach and its striking clinical effects.

- 6.1** I shared with you the story of Mike and his amazing response to a ketogenic diet. He had a dramatic reduction in hallucinations and delusions, and it really changed his life in profound ways. That one experience with him has now led to hundreds, if not thousands, of people sharing their stories with me about how the ketogenic diet has saved their life or dramatically improved their lives. The way that I think about the ketogenic diet is not a healthy diet that everybody in the world should do, but instead, I think about it as a metabolic and mitochondrial intervention.
- 6.2** For those of you who don't know, the ketogenic diet is high in fat, low in carbohydrate, and moderate in protein. It was developed—unbeknownst to most people, most people have heard of it; they think it's a fad diet; they may I think it's a dangerous diet — it was actually developed in the 1920s by a physician for one and only one purpose, and that is to treat epilepsy. It results in the body using fat instead of carbohydrates as a primary fuel source, and this includes the brain.
- 6.3** Ketogenic means the production of ketone bodies from fat. That is what ketogenic means. There are three types of ketone bodies in the human body—beta-hydroxybutyrate, acetoacetate, and acetone. These are measurable, respectively, in blood, breath, and urine. One of the great things about doing clinical and research work with this diet is that we can objectively test people for compliance with the diet. So unlike the Mediterranean diet or any other diet out there, I can tell within one minute whether a patient is compliant with the diet or not.
- 6.4** It mimics the fasting state, and most adults lose weight with this diet, but people can maintain weight, and they can even gain weight on a ketogenic diet, and that is particularly important when this diet is being used in children because they need to continue to gain weight, grow, and thrive. What is the ketogenic diet? It is not one thing. There are numerous variations of ketogenic diets. The classic one developed in the 1920s is usually referred to as a 4:1 or 3:1 ratio diet, which means 4 grams of fat for every 1 gram of everything else, protein, carbs, and anything else like alcohol.
- 6.5** That means a large percentage of your calories are coming from fat. But there is fasting and intermittent fasting; the absence of food can produce a state of ketosis. There's the MCT ketogenic diet. MCT stands for medium chain triglycerides, which are usually coconut oil. Those are more easily converted into ketones. So if people add MCT oil to a ketogenic diet, they can usually get higher levels of ketones and/or do a lower ratio diet.
- 6.6** There's the Atkins diet, sometimes abbreviated LCHF, low carb, high fat. There's the modified Atkins diet, which is actually an epilepsy treatment developed by Johns Hopkins, in which you do the Atkins diet, but essentially add more fat. Do the regular

Atkins diet, but add even more fat than you would normally think to add. There are calorie-restricted versions of the ketogenic diet used in cancer treatment. There are different types of foods.

- 6.7** I want to be clear, the ketogenic diet can be done as a vegan diet. The ketogenic diet can be done as a vegetarian diet. It can be an omnivore diet, which is what most people do. So they're eating both animal-sourced and plant-sourced foods. But there is the carnivore diet, where people are only eating animal-sourced foods. And Mikhaila Peterson and Jordan Peterson have popularized what they call the lion diet, where you are primarily eating beef, salt, and water.
- 6.8** There are ketone supplements that can increase levels of ketosis. This graphic shows you what exactly a 4:1 ratio diet looks like. This comes from the Charlie Foundation. What we see are four different meals. They all contain the same ingredients, and they all have the same number of calories, but they look very different from each other. The meal in the green plate—chicken breast, broccoli, mayonnaise, and a splash of olive oil somewhere on that. But what you see in the green plate is a 1:1 ratio diet, and that is what most people think of as a low-carb diet.
- 6.9** The 4:1 ratio diet you see in the red plate has a paltry serving of chicken, a paltry serving of broccoli. You still have that ramekin of mayonnaise that you have to consume, and you have a syringe of olive oil that you also have to consume. When I first started doing this work, I saw that syringe and I was so turned off. I was like, why would they do that? That's so unappetizing. The reason they put the syringe on the plate is to make the point. The 4:1 ratio diet is not a weight-loss diet. The 4:1 ratio diet is a medical prescription, usually prescribed for serious medical conditions. It has to be done religiously, but you should think of it as a medical prescription.
- 6.10** But instead of syringes of olive oil, what else can you eat on a ketogenic diet? Well, here are some staples—salmon, steak, butter, coconut. Coconut comes in coconut cream and all sorts of stuff. You can have bulletproof coffee, which you have coffee with added butter and MCT oil. You can have eggs and bacon, steak, low-carb vegetables, spinach, high-fat fruits like avocados. But you can get creative and make pizza and ice cream that are ketogenic.
- 6.11** I want to now tell the story of the ketogenic diet for epilepsy. How did this even come to be? Well, it turns out that fasting can stop seizures. Going without food can stop seizures. Hippocrates knew about this. Jesus used fasting to help stop seizures in someone in one of the New Testament versions. It's not in every version, but in one of the versions, Jesus used fasting to stop seizures. This has been known for millennia, and yet it was largely thought to be religious folklore, because more often than not, it was attributed to religion. That the reason people were fasting was not a medical reason. It wasn't a science-based reason. It was a religious reason.

- 6.12** Fasting pleases God. If somebody is struggling and suffering, we should fast them because they should make a sacrifice to God. If God is merciful, God will heal. And shockingly, a lot of people got healed. Now, the religious people thought that this was proof that God was healing. The conventional medicine people thought this was religious quackery and folklore. Science has now caught up to say, "Oh, my gosh, there's something to it." Now, if you want to believe that fasting pleases God, that's still acceptable. And maybe that's the mechanism through which God works. But fasting, in fact, can stop seizures.
- 6.13** In 1921, Dr. Geyelin reported the use of intermittent fasting to treat a child with epilepsy. The problem with fasting is that, as soon as people start eating again, the seizures usually come right back. It was the ingenious Dr. Russell Wilder from the Mayo Clinic who first developed the ketogenic diet. He did this to try to see, can we trick the body into thinking that it's fasting? Can we get the physiological benefits of fasting without starving people to death? Because if you fast too long, obviously, you die, and that's not a really good treatment either.
- 6.14** So, shockingly, it worked. Eureka. He figured it out. Early results were impressive. 50% were seizure-free. Another 35% were markedly improved. But by the 1950s, the ketogenic diet fell out of favor because we had a whole armamentarium of anti-seizure medications coming on the market. And make no mistake, these do work for a lot of people, and they are so much easier than doing a diet. If you can take a medication, have few side effects, and stop your seizures, I am still going to recommend, go with that, because that is the easiest way to stop seizures.
- 6.15** However, if you want long-term healing, if you want a different strategy, maybe I'll take back what I just said, because you might actually be able to shockingly heal your epilepsy. It was rediscovered in the 1970s at Johns Hopkins because, sadly, 30% of patients with epilepsy don't respond to any treatments that we have to offer. Medications, vagal nerve stimulators, even surgery. It was really Jim Abrahams, a famous Hollywood producer, who is responsible for the resurgence of the ketogenic diet. He had a son named Charlie who had treatment-resistant epilepsy.
- 6.16** They were basically told that "there's not much more we can do for you or your son. You're just going to have to accept that your son is having numerous seizures every day. He's often unconscious. He is clearly cognitively impaired and will probably live a shortened life. That's all we can do. We're really sorry." Jim Abrahams was unwilling to accept that. He did his own research, read about this crazy thing called the ketogenic diet that used to be used in the old days to treat epilepsy. He found the one and only one center in the world that was using it, Johns Hopkins. They flew Charlie there. Charlie started the ketogenic diet.
- 6.17** Within four days, his seizures stopped. Within four years, within about four or five months, he was off all of his anti-seizure medicines and remained seizure-free. Within four to five years, he was able to stop the ketogenic diet and remained seizure-free.

Charlie has never had a seizure since that fourth day in Johns Hopkins. Jim Abrahams was actually furious, furious with the medical field. "Why didn't somebody tell us about this? Why were they just watching my son seize 20 times a day, and they didn't even have the heart to mention this to us?" Well, most of them probably didn't even know about it.

- 6.18 So Jim Abrahams set up what's called the Charlie Foundation, which is focused on the medical scientific use of ketogenic therapies for neuropsychiatric and other conditions. There are now over 250 ketogenic diet centers around the world, and there's a Cochrane review. This is evidence-based treatment, if you believe in medical evidence. If you believe in research, if you believe in publications, Cochrane reviews are the gold standard meta-analysis in the medical field.
- 6.19 There was a Cochrane review done in 2020. They looked at 13 randomized controlled trials and 913 participants. At three months, seizure-freedom rates in children were as high as 55%. This is in children who had treatment-resistant epilepsy. They've already failed to respond to medications, surgeries, and other things, and this diet can stop seizures in up to 55% of those people.
- 6.20 Now, according to the research, no adults achieved seizure freedom, but that is because we don't have much research, even to this day, of the ketogenic diet for adult epilepsy. Nobody wants to do this research. Nobody wants to fund this research. People are willing to fund studies for children. Adults with epilepsy, well, that's just on them, or they should be able to figure it out on their own. The reality is that it is easier for children to do this diet when their parents are forcing it, and their parents are forcing it because they have seen them seize, unrelentingly, 20 times a day, every day for the last five years. Parents are like, "We are doing this diet one way or another."
- 6.21 An adult individual figuring out how to do it on their own and having the willpower, discipline, tenacity to do it, is a different thing. It can be hard. People do need support. They need help. They need education. Seizure freedom rates as high as 55%, but another 30% had a 50% or greater reduction in seizures, so that up to 85% are having a clinically significant response. The modified Atkin's diet in the research study so far was not found to be as effective as the 4:1 ratio diet. In one study, after one year, about 55% of the patients remained on the diet, and 45% were no longer on the diet, either because the diet didn't work for them or they couldn't do it.
- 6.22 Is this diet safe? That is probably one of the most frequent questions I get. There are some rare contraindications. There are rare fat metabolism disorders, porphyria, in which patients should not do the diet. If a woman is pregnant or breastfeeding, that is a relative contraindication. I know some in the keto community contest that, but I'm going to stay with standard traditional medical practice.
- 6.23 The ketogenic diet is mimicking fasting. Pregnant women should not be fasting. Pregnant women don't want the hormonal state of fasting because they are trying to

grow another human being, and you need calories and nutrients in order to do that. That's the commonsense quick version of why I do not recommend the ketogenic diet for pregnant and breastfeeding women. There are vitamin and supplement considerations. People doing a ketogenic diet may want to consider supplementation with a multivitamin, with sodium, potassium, and magnesium early on. You can get electrolyte disturbances. Some recommend calcium with vitamin D, although I think calcium is even a little more controversial.

- 6.24** The first month of the ketogenic diet can be very uncomfortable, and for people with severe mental illness, can be dangerous. I'm going to say that out loud and warn people, that people can get, if you've got diabetes and you're on medication, you can get hypoglycemic in a severe way because your medications may need to be altered. It's not because of the ketogenic diet. The ketogenic diet is rapidly going to lower your blood glucose, but your medications are going to lower it even further in a dangerous way. You need to work with a healthcare professional to adjust your medications.
- 6.25** Same deal for those on blood pressure medicines and same deal for those who have epilepsy or serious mental illnesses or cancer. But what about all that fat? That fat has to be bad for us. The American Heart Association says, yes, fat is bad for you. They still say that. The American Heart Association dietary guidelines are, everybody should eat a variety of fruits and vegetables, whole grains, low-fat dairy products, skinless poultry and fish. Notice how we're removing all fat. Nuts and legumes. OK, nuts maybe have a little fat. That's OK. Non-tropical vegetable oils. Those are their healthy fats. But we should limit saturated fat, trans fats, sodium, red meat, sweets, sugar-sweetened beverages.
- 6.26** What I want to say in response to this is, the American Heart Association is primarily concerned with LDL cholesterol. And LDL cholesterol does, in fact, increase in some people on ketogenic diets, and we need to at least have that conversation or people need to have those conversations with their health care providers. But it doesn't increase in all people on ketogenic diets. I've had many patients do ketogenic diets who had very high LDL cholesterol levels, and they came down after doing a ketogenic diet.
- 6.27** The second point is that not all LDL cholesterol is the same. There are large puffy LDL particles that are largely considered to be benign. Small, dense LDL particles are the ones thought to increase risk for cardiovascular disease. If you simply get an LDL test, that doesn't tell you what types of particles you have. There are tests where you can look at different types of LDL particles or apoB particles and others and get more refined information. But LDL is not the primary, and certainly not the only story, with heart disease.
- 6.28** The primary story with heart disease is insulin resistance and type 2 diabetes. That is unequivocally the biggest risk factor for heart attacks and strokes. It's not LDL cholesterol, but HDL cholesterol, triglycerides, inflammatory biomarkers all play a role. Increasingly, we are getting more and more evidence that saturated fat itself may not

cause heart disease. The Women's Health Initiative randomized controlled dietary modification trial, was the largest ever randomized controlled dietary trial. Over 40,000 women were assigned to either a low-fat diet or diet as usual.

6.29 At the end of that study, the women who lowered the fat intake, they increased their fruit and vegetable intake, they increased their grain intake, there was zero difference in heart attack or stroke rates. Now, some just say, well, that was a negative study. Another interpretation is that the low-fat diet is no better than the standard American diet. That is what that evidence tells us. Low-fat diet, no better than the standard American diet. And as a result of that study, that is why we all hear about the Mediterranean diet now, because there is wide recognition that there are healthy fats. Fat is essential, and some healthy fats are essential.

6.30 There was a pure study, which was the largest epidemiological study, 135,000 people from 18 countries followed for seven years. Here's the shocker from the largest epidemiological study ever done. High carbohydrate intake was associated with higher risk of total mortality, whereas total fat and individual types of fat were related to lower total mortality. Now, the experts in the United States are bashing and trashing this study because they don't like that conclusion. That conclusion just flies in the face of their recommendations.

6.31 So I'm not at all trying to say, "Problem solved, no more controversy. We should now have unanimous agreement." But what I am saying is that we have extremely large, well-done studies that are challenging this dogma that fat is bad for you and carbs are good for you. There's at least one study of the ketogenic diet on weight loss and diabetes in 262 patients that found improvement in overall cardiac risk compared to the control group.

6.32 But what about cancer? Doesn't eating fat cause cancer? I get all these kind of questions. This actually comes directly from the Harvard School of Public Health website. "Researchers once suspected an association between dietary fat in common cancers. However, in adults, the percentage of calories consumed from total fat appears to have no significant association with cancer risk. There is currently no clear evidence linking any specific type of fat, including saturated fat, with cancer risk." The confusion, however, is that obesity or being fat does significantly increase your risk for cancer.

6.33 How does this diet work? How on earth does a diet stop seizures? So shockingly, we know more about the effects of the ketogenic diet on the brain than we do any other dietary intervention, including omega-3 fatty acid supplements or vitamin E supplement or Mediterranean diet or vegan diet. We know more about the effects of the ketogenic diet on the brain than any other dietary intervention. Why is that? Because the ketogenic diet is an evidence-based treatment for epilepsy. It's been used for 100 years.

- 6.34** Researchers, biotech companies, neuroscientists, others have been studying this diet relentlessly, trying to figure out, what is this diet doing? How is it stopping seizures? We want to develop new pills. If we can figure out what this diet is doing, maybe we can develop new pills. That's their motivation. We actually know a tremendous amount. I'm not going to go through this complex graph; instead, I'm going to break it into simple-to-understand bullets. Understand, although I'm not giving you citations in these bullets, understand every bullet here is backed by literally at least 10, if not hundreds, of scientific studies published in the literature. All you have to do is Google it and you'll find it.
- 6.35** The ketogenic diet lowers blood sugar and insulin levels. That improves insulin signaling. It produces ketone bodies, which are an alternate source of energy. Ketone bodies themselves have anti-seizure effects. This diet changes neurotransmitter systems, including increasing GABA, decreasing glutamate, increasing adenosine. It also helps better regulate ion channels. It increases mitochondrial function and production, and that is central to my thesis. I actually believe, that is the primary mechanism of action of all of these even effects, is that the ketogenic diet is improving mitochondrial health.
- 6.36** It increases circulating polyunsaturated fatty acids, which are neuroprotective. It decreases leptin, which helps with appetite satiety. It decreases inflammation. So this is an anti-inflammatory diet. It changes the gut microbiome, and I'll share a little bit of evidence on that with you in a bit. Changes DNA methylation and gene expression. It increases autophagy, increases NAD, and activates Sirtuin genes, which are all thought to be involved in an anti-aging process.
- 6.37** I want to just share this study that was just published last year, that looked at—so some researchers are actually focused on the effect of the ketogenic diet on the gut microbiome. They think that is the primary mechanism of action. The ketogenic diet is changing the gut microbiome. Somehow that's doing something. Well, so how does that connect with what I'm asserting or with the brain energy theory?
- 6.38** This is a great study. They took human children who had epilepsy and were about to go on a ketogenic diet for their epilepsy. They took stool samples from these children before they started the treatment and transferred those stool samples into mice, and the mice were more likely to have seizures. If you get feces from children who are seizing a lot and transfer that to mice, there's something about the microbiome in those children that is increasing seizure risk in the mice.
- 6.39** Then they put the kids on the ketogenic diet because they were going to do it anyway. These kids were already signed up for ketogenic diet therapy. So these researchers figured, we'll just add these things on, take some stool from them. Now the kids are on the ketogenic diet, and they take stool samples again and transfer them to mice, and the mice become more seizure resistant from the poop, from the stool samples. But how does that work? What was happening?

- 6.40** The researchers looked at the brains of the mice, and what they found was that the mice colonized with these ketogenic diet microbes, exhibited altered hippocampal transcriptomes, meaning the hippocampus is changing gene expression, and what were the changes? Pathways related to ATP synthesis, that's mitochondria. Glutathione metabolism, that's mitochondria. Oxidative phosphorylation, that's mitochondria. The changes in the gut microbes were conferring a clinical benefit, reduced seizure threshold. The primary mechanism that at least these researchers could measure was that the hippocampus is changing gene pathways related to mitochondria and energy production.
- 6.41** It's hard to talk about the ketogenic diet without talking about weight loss because that's how most people know it. So low-carb and ketogenic diets are not fad diets for weight loss. Let me say that again. Low-carb and ketogenic diets are not fad diets for weight loss. Why do I say that? Because they've been around for over 200 years. We have a tremendous amount of evidence published in leading journals, leading medical journals, documenting that these are highly effective and safe weight-loss strategies.
- 6.42** In 1825, people noticed this connection between high-carb intake and obesity. William Banting, in 1863, wrote this letter on corpulence in which he really described what many say is the first low-carb diet. Now, he was a really wealthy man. He had been to so many physicians trying to figure out how to lose weight. Nobody could help him. He finally came upon this low-carb diet that worked for him. It really worked. He wrote this letter on corpulence, which was really just a pamphlet. It is largely considered the first ever weight-loss book or pamphlet.
- 6.43** He gave it away for free. He was rich. He wasn't trying to get money from it. He didn't need money. He actually just felt like, "I'm so frustrated that nobody could ever help me figure out how to lose weight. I just want to share this secret with the world. Going on a low-carb diet just helped me lose so much weight. This is great." Dr. Pennington wrote about carbs and obesity in the New England Journal of Medicine way back in 1953. Dr. Irwin Stillman wrote The Doctor's Quick Weight Loss Diet, published in 1967. 1972, the first Atkins Diet Revolution book. Dr. Agatston developed the South Beach Diet, a low-carb, low-saturated fat diet in 2003. Now we've got lots of people talking about the ketogenic diet.
- 6.44** Anybody nowadays who says, "I figured this out, I developed the ketogenic diet," no, you didn't. No, you're standing on lots of other people's shoulders. Anybody who says this is just the latest fad diet, they actually don't understand medical history or history. But doesn't fat make you fat? Doesn't eating lots of fat make you fat? That's what we all think. Time publishes, "Eat butter. Scientists labeled fat the enemy. Why they were wrong." What evidence do we have? We actually have hundreds of randomized controlled trials of low-carbon ketogenic diets for type 2 diabetes and weight loss. I'm going to share with you just a couple.

- 6.45 This one compared the Atkins, Zone, Ornish, and LEARN diets for change in weight and related risk factors among overweight premenopausal women. It was a randomized controlled trial of 311 women. They were all randomized to eat any of these diets. The Atkins diet group had significantly more weight loss than any of the other diet versions and more favorable metabolic effects than the other groups. It wasn't a dangerous diet. They were improving their metabolic health. This study was a two-year, randomized, controlled trial of 322 obese participants, randomly assigned to either a low-fat diet, a Mediterranean diet, or a low-carbohydrate diet.
- 6.46 Some health gurus who are proponents of plant-based diets have said this, single-handedly, is the best ever conducted weight-loss scientific trial ever. So what did they find? The people assigned to the low-carb diet lost the most weight. Now, you'll notice in this chart, they rapidly lose a lot of weight, and then something changes at about four months, and their weight starts to go back up. What changed? In this trial, the people on the low-fat diet were actually told, "Restrict calories. We want you to do low fat and calorie restriction." The people on the Mediterranean diet were told, "Restrict calories." Mediterranean and restrict calories.
- 6.47 The people on the low-carb diet, calories weren't mentioned. They just said, "Restrict carbs to 20 grams a day until four months. Then at four months, go up to 120 grams of carbs if you want." As soon as they went up to 120 grams of carbs, the weight started coming back on. Yet at the two-year time mark, the clear winner was the low-carb diet. In fact, other researchers have noticed this. A meta-analysis of very low-carb ketogenic diets versus low-fat diets, low carb wins for weight loss, for improvement in triglycerides, improvement in diastolic blood pressure, improvement in HDL cholesterol, and yes, slightly higher LDL cholesterol.
- 6.48 A meta-analysis of low-fat diets versus other diets for long-term weight loss, 53 studies, 68,000 subjects. Low-carb diets lead to more weight loss than low-fat diets. Higher-fat diets, get this. Higher-fat diets lead to more weight loss than low-fat diets. Fat does not cause people to be fat. Fat actually results in weight loss, even though fat has more calories. How can that be? Many lines of research. I share just a few of them on this slide. I won't go into the details, but it suggests that ketogenic and low-carb diets may actually change metabolism.
- 6.49 And I believe that is through mitochondrial changes, that ketogenic and low-carb diets are changing mitochondria. That changes metabolic health, which includes how many calories you're burning. So that even if you're doing nothing, even if you're not exercising, even if you're not doing anything, you're just sitting there, you may burn more calories by being on a low-carb or ketogenic diet compared to other types of diets. Can this be used for type 2 diabetes? I've already shared with you the sad and tragic news that 50% of Americans, adult Americans, have prediabetes or diabetes. Low-carb diets are known to lower glucose, and restricting carbohydrates and eating only moderate amounts of protein can be more effective for diabetics.

- 6.50** Do they really work? We've got this meta-analysis. Ten randomized trials, 1,300 participants. Low-carb diets reduce hemoglobin A1C more than high-carb diets. The greater the carbohydrate restriction, the greater the glucose lowering effect. At one year, however, hemoglobin A1C was similar in the two groups that they looked at. That's because most of these studies are not one-year studies. Most of these studies are 12-week or 20-week studies. Then they stop talking about the intervention. People aren't getting support. People are told, "Go do whatever you want now. You're done." People aren't sticking with the intervention, but they're also not necessarily being encouraged to stick with the intervention, so it's not surprising.
- 6.51** What if people stick with the intervention? Well, there's this service called Virta, out of California, where they follow people longitudinally, primarily people with type 2 diabetes and obesity, and they support them virtually with healthcare clinicians. There was an open-label, non-randomized controlled trial. 262 were assigned to the intervention. 87 people were a control group. They were just getting usual care. The people who got the ketogenic diet had significant reductions in hemoglobin A1C. Many of them had such a significant reduction that they could no longer be classified as type 2 diabetic.
- 6.52** On average, they lost 12% of their body weight or about 30 pounds. They reduced their diabetes medicine use. 94% of insulin users either reduced or eliminated their insulin use. Sulfonylureas, which are a class of diabetes medicine, were eliminated in all of the patients. 25% of the patients, one in four, were no longer on any diabetes medicines. And another quarter or another 35% were taking only Metformin. But is it safe? This study suggests absolutely, yes, almost all of the biomarkers that are associated with safety and heart disease risk improved, except for LDL cholesterol. LDL cholesterol was the one thing that got a little bit worse in the ketogenic diet group.
- 6.53** But again, the overall, when you plug all of the biomarkers into the American College of Cardiology 10-year risk assessment profile— how likely is this human being to have a heart attack?—it was dramatically better. It was significantly better, because their glucose got better, their blood pressure got better, their weight went down, all of their other biomarkers. So, please don't throw the baby out with the bathwater because of a little increase in LDL cholesterol. Please treat the whole human being.
- 6.54** In fact, the American Diabetes Association now acknowledges low-carb and very low-carb diets are actually among the most studied eating patterns for type 2 diabetes. "Oh, what do you know? Now we're going to admit it." These have been around 200 years. "Actually, yeah, we do know more about these. In fact, they are good for just about all of the metabolic biomarkers." What does any of this have to do with mental illness? I've told you about neurological uses, weight loss, diabetes. Do we have any evidence that the ketogenic diet can be helpful for mental illness?
- 6.55** For depression, right now, all we've—the strongest signal we have for depression is through studies in which the ketogenic diet was used for another condition. The

ketogenic diet was used for weight loss, but the researchers also measured depression symptoms as a secondary outcome. Or the ketogenic diet was used for epilepsy, and the researchers also measured mental health or depression. We've got some lines of evidence. We've got a couple of animal models showing an antidepressant effect in animals put on ketogenic diet. There are clinical trials now underway for major depression.

- 6.56** What about Alzheimer's disease? I've told you that Alzheimer's disease appears to be a metabolic disorder. Some are calling it type 3 diabetes. Can the ketogenic diet play a role? We actually have several animal models now. Some of them I share with you on this slide. We have trials of ketogenic agents versus placebo in older adults, without even having to change their diet. We have trials of 23 people with mild cognitive impairment randomly assigned to a low-carb or a high-carb diet, and the low-carb diet group had memory improvement. The memory improvement correlated directly with levels of ketosis.
- 6.57** Weight loss, calories consumed, even insulin levels did not seem to matter. It was the presence of ketones that appeared to be correlated with memory improvement. Others have tried to do this diet. We have trials of the ketogenic diet in humans showing improvement in the biomarkers of Alzheimer's disease. This is probably the best study that we have. It's only 26 patients, but they were randomized to 12 weeks each of the ketogenic diet and the low-fat diet, separated by a 10-week washout. That means some patients got the ketogenic diet first, and we measured things, and then 10 weeks off, and then 12 weeks of a low-fat diet; and others got low-fat diet, and then the ketogenic diet.
- 6.58** What the researchers found is that when these Alzheimer's patients were put on a ketogenic diet, they had improvement in activities of daily living, improvement in quality of life, and even improvement in cognition. The cognition score was not statistically significant, but it was there. And in a larger sample, maybe we would see a difference, but the other two were statistically significant. There are larger controlled trials underway now of the ketogenic diet for Alzheimer's disease, to see if we can slow the progression or possibly even prevent or reverse some of the symptoms.
- 6.59** What about a disorder like alcoholism? Well, that has nothing to do with—alcoholism, that's just people drinking too much, right? That has nothing to do with metabolism. Certainly has nothing to do with diet. Certainly a dietary intervention doesn't even make sense for alcoholism. Well, shockingly, that's not true. Alcohol use disorder has been associated with brain glucose hypometabolism. Alcoholics are not getting enough energy in their brain. That may actually be one of the reasons they are now alcoholic.
- 6.60** Animal models have suggested reduced alcohol intake when, say, rats are placed on a ketogenic diet. Under the leadership of Dr. Nora Volkow, who is the director of the National Institute of Drug Abuse in the United States— she is a world-renowned neuroscientist— under her direction, she recognized alcoholism is a metabolic brain

disorder. There's this thing called the ketogenic diet that might actually restore metabolic function in the brain. We've got all this neurology evidence and other stuff.

- 6.61 So she and her colleagues did this study, and they took alcoholics, and they put them in a detox unit, a government-funded detox unit, and they all got a detox. But when they were admitted, half of them were started on a ketogenic diet, and half of them were just given the standard American diet, which is what they would get in any other detox. Then they measured a lot of things.
- 6.62 What they found is that the alcoholics who got the ketogenic diet ended up getting about 50% fewer benzodiazepines for their detox. And nonetheless, they had fewer withdrawal symptoms from the alcohol, even though they were getting less medicine. They reported fewer cravings for alcohol. When the researchers did brain scans, they noticed improvement in brain metabolism and reduction in brain inflammation in the people on the ketogenic diet. The ketogenic diet can provide an alternate source of fuel that can restore brain function and help people potentially beat alcoholism.
- 6.63 What about the more severe, serious mental disorders, like Mike's? We now have so much more evidence than Mike's story and the others that I published early on. Schizophrenia and bipolar disorder, we have numerous animal models published in the literature, in which they have animal models of psychosis in particular, in which they look at behaviors. This is how they actually test new antipsychotic medications. When new antipsychotics are being developed, they have to start in these animal models to show that the antipsychotic reduces psychotic symptoms or at least behaviors that are associated with psychosis. They have to demonstrate it.
- 6.64 We've got several studies now, showing that the ketogenic diet appears to be similar to the antipsychotic medication trials. The human trials date all the way back to 1965, in which 10 patients, 10 women with schizophrenia, who were hospitalized, they were getting medications. Many of them were getting electroconvulsive therapy. They added on a ketogenic diet. The researchers noticed that, within two weeks, they noticed significant improvement. When they stopped the diet, most of the symptoms returned.
- 6.65 But that was 1965, and then there was this long lapse. In 2017, I published the first two case reports of the ketogenic diet working for schizoaffective disorder. In 2019, I published two more case reports. An older woman and a woman who's now in her 30s who had schizophrenia or schizoaffective disorder for, one of them 20 years and one of them for 53 years, both of them achieved long-lasting remission of illness, off of antipsychotic medications. They were able to stop antipsychotics and remain free of psychosis.
- 6.66 The woman, unfortunately, the older woman, she was 70 years old when she started the diet. She was treated by Dr. Eric Westman at Duke University, initially just for weight loss. She was just trying to lose a few pounds. She ended up losing 150 pounds.

Got rid of her schizophrenia as a side effect. She, unfortunately, passed away at the age of 85, of COVID pneumonia, a couple of years ago.

- 6.67** The other woman is alive and well. She had suffered from schizophrenia for 20 years, had also suffered from anorexia nervosa and other psychiatric disorders. And she achieved full, lasting remission from her schizophrenia. She's off antipsychotic medicines. She got her master's degree in social work. She's now married. She works as a social worker in a mental health clinic, and nobody knows her story there.
- 6.68** She is not interested in publicly sharing her story because she's got a life. She's got a family, she's got a job. It's not easy to come out and say, "I used to have schizophrenia for 20 years." But she has a new life, and very willingly and openly allowed me to publish her case report and tell her story in an anonymous way. So if you know any social workers who work in mental health clinics, you may now have to have a level of suspicion. "Is it you?" I'm joking.
- 6.69** Do we have other evidence, though? I think those stories, in my mind, are the most hope-filling stories because they say it is possible. Even if it's not happening for you or someone you love, those two stories say it is possible. It is possible to suffer from unrelenting schizophrenia for 53 years and get a recovery, get a remission. If it's possible for her, then maybe it's possible for others. We, as a field, need to rapidly figure this out because there are tens of millions of people desperate for better solutions. This opens up an entirely new way to think about schizophrenia and to treat it.
- 6.70** But we have more evidence. This study, the bottom one, 31 inpatients admitted to a French psychiatric hospital. They had treatment-resistant bipolar disorder, major depression, or schizophrenia. They were all placed on a ketogenic diet. Only 28 patients were able to do the diet. So I just have to put that out there up front. Three of 31 patients could not do the diet, didn't want to do it, didn't want to comply. Let's remove those 10%. Of the 28 patients who were able to do the diet, 100% had at least some symptom improvement. 43% achieved clinical remission. That doesn't happen with standard treatments. It does not happen with standard treatments. 43% clinical remission.
- 6.71** 64% were discharged on less medicine than they went in on. It wasn't that the hospital was just loading them up with more meds. They were actually removing meds, taking them away, and patients were getting better. And oh, what a nice side effect. 96% of them lost some weight, too. This study out of the UK was a trial of either 6 to 8 weeks, somewhere in that range, depending on the patient, of the modified ketogenic diet for patients with bipolar disorder. 20 patients ended up completing this study.
- 6.72** Now, one of the common complaints is that nobody can do a ketogenic diet. It's too hard and too restrictive. People with bipolar disorder, there's no way they can do a ketogenic diet. Well, actually, this study says otherwise. 91% of their daily ketone

readings were positive, demonstrating a very, very high level of compliance. On this front, I just want to say, for the skeptics or for the people who are like, "I know lots of people who tried keto and they couldn't stick with it," well, you know what? If you're just doing it to lose a few pounds, you're not really probably all that highly motivated.

6.73 When you're fighting for your life, when you're fighting for your sanity, when you're fighting for your independence, guess what? You are highly motivated. You will do anything. You will do anything to get better, including a ketogenic diet. What happened to these 20 people? Well, they lost some weight on average, little more than eight pounds each, just in 6 to 8 weeks. Systolic blood pressure went down. Brain scans showed improvement in, essentially, brain metabolism. And the symptoms of bipolar disorder that they measured, mood, impulsivity, energy levels, anxiety levels, all improved. Patients got more improvement the higher the level of ketones. There appeared to be a correlation between the level of ketosis and the improvement in symptoms.

6.74 Then this study was just published out of Stanford. It actually got international headlines. Ketogenic diet in 21 participants who had treatment-resistant bipolar disorder and schizophrenia, all placed on a ketogenic diet, 79% of the participants had clinically meaningful psychiatric improvement. Of the ones who had metabolic syndrome, the ones who came in and had metabolic syndrome, all of them, 100%, reversed their metabolic syndrome. On average, they lost 10% of their body weight. What they found was that the greater the adherence to the diet, the greater the symptom improvement.

6.75 Again, I'm not saying the keto diet is easy. I'm not saying, "Just prescribe the keto diet and leave people to their own devices to do it." People need education and support and guidance, especially people with chronic severe mental illnesses. They are already down and out. They are already vulnerable. But we can do that. We can help them. We can support them. The one thing I do need to say, as much as I wish I could say, "Everybody, go out, try this on your own, wing it, see how it goes," if you are a person with a severe mental illness—what do I mean by severe? I mean hallucinations, delusions, suicidality. If you've ever been hospitalized, that counts as pretty severe. You don't get hospitalized unless it's pretty severe.

6.76 For the people with severe mental illness, I am asking you, please don't do this on your own. Please get professional help. To all the professionals out there, I'm begging you, please step up. Start helping. Learn about this treatment and start offering it to all the people who want it. But why am I saying that? I'm saying that because this keto adaptation period in the first month can, in fact, be dangerous. Some people can develop insomnia. If they go without sleep, the ketogenic diet can activate people. It can just rev up your metabolism, including your brain metabolism. Transiently, people can have way too much energy, so that they're not sleeping.

- 6.77** That can rapidly turn into mania and psychosis in people who are already vulnerable. They need help and they need a safety system in place to get them through that. Those symptoms pass. It's not that I'm saying we shouldn't do the ketogenic diet. Almost every other treatment in psychiatry, most classes of medications, electroconvulsive therapy, transcranial magnetic stimulation, they can all induce hypomania and mania and psychosis. So this is no different. This is like them. But you shouldn't go do ECT on your own. You shouldn't be taking psychiatric prescription medicines on your own. And I would say, if you're trying to treat a severe mental illness, you shouldn't be using the ketogenic diet on your own. You should be getting medical help and supervision.
- 6.78** Additionally, some of the medications that we prescribe can interfere with the diet's effects, and somebody needs to help you manage those medications. Treatment includes more than just the diet. I am not here to say, "Everybody go on the keto diet, and it will be the magic cure-all, fix-all." Patients might need adjustments in their medications. They still need to sleep. They need a daily routine, maybe adding exercise, therapy, stress management, other tools. If they're using drugs and alcohol, they might need help getting off those. All of those are going to play a role in a comprehensive treatment plan.
- 6.79** Should everybody be on a ketogenic diet? My answer is no. I know there are some gurus and some of even my good friends in the keto community will say, "Yes, everybody should be on it." I am going to respectfully disagree with them and say, no, I don't think everybody needs to be in ketosis. I don't think everybody needs a ketogenic diet. I think a ketogenic diet can play a very powerful role in improving people's health and helping people maintain health, maybe even in helping people prevent severe illnesses.
- 6.80** So if you're in the middle age and you've got some risk factors for Alzheimer's disease, maybe doing ketogenic diet intermittently or for a period of time or intermittent fasting or other things could play a role in you preventing Alzheimer's disease. But should every single human being be on a ketogenic diet? Absolutely not.
- 6.81** So what do I recommend for diet? I've talked a lot about different dietary patterns, Mediterranean. A lot of people ask, "So, which one is it, Chris? Should people do a Mediterranean or a keto diet?" There is not a one-size-fits-all diet for all human beings. I want people to assess their current mental and metabolic health and even decide if a change in diet is warranted or might play a role in their treatment.
- 6.82** What are the universal principles I am willing to recommend? Eat real, whole foods. Try to get away from ultra-processed foods. Try to get away from artificial ingredients. Avoid added sugars and artificial sweeteners. I'm sorry to say, but I'm saying, avoid sweet things. If you avoid added sugars and artificial sweeteners, at some point fruit is going to start tasting sweet to you. You can have sweet things, but it's going to be what you consider boring now.

- 6.83** For the people—I remember, I was like a sugar addict when I was growing up. I drank Kool-Aid every day. Every day I drank Kool-Aid. I hated the taste of plain water. I couldn't drink water. It was disgusting. It was so bland. Why would anybody drink water? Everything I drank was sweet, and a lot of the foods I ate were sweet. I can tell you, I remember thinking fruit, that's like a bland health food. It doesn't have much taste, and it's certainly not sweet.
- 6.84** Now, I don't eat a lot of sweet stuff. I really don't have things too often with added sugars or artificial sweeteners. I'm not militant about it, but for the most part, that's not what I eat. I can tell you, eating fruit actually is sweet. It's shocking to me. It's shocking because I'm not one of those people that I thought I would ever be one of these people. I'm just like everyone else. What I'm here to say is, if it can work for me, if I can do it, you can do it, too.
- 6.85** You might want to consider elimination diets, intermittent fasting, or ketogenic diets if you have a condition that might respond. That is the end of our diet discussion. Next, we're going to talk about substance use and how that can affect your metabolism, mitochondria, mental health, and metabolic health. Thanks.
- 6.86** Throughout the years, I have seen many of my patients do very well on different type of diets. So I do agree that there's no one-size-fits-all. And I'm wondering, because there's so much information, and I love it because my patients come and say, "Oh, doctor, what about this? What about that? What about this?" It helps me move me into learning a little bit more about all these things that are coming up. But besides biomarkers in bloodwork, what other tools could a provider use—for example, the microbiome test, the micronutrients, genetic testing—to help tailor the nutrition plans for each person?
- 6.87** Those are really important questions. There's a lot of controversy in the answers to those questions. Depending on who you talk to, you'll get wildly different answers. If you talk to people who are solidly 100% in the functional medicine community, that is what they do. They're going to do genetic testing, microbiome analysis, all sorts of biomarkers, hundreds and hundreds of blood tests. Then based on that, they're going to prescribe something for you.
- 6.88** Now, there are a lot of people in allopathic medicine who say that's all quackery. There's no scientific basis for any of that. I will be honest with you, I am somewhere solidly in between them. I do not do hundreds and hundreds of tests. I actually don't think microbiome tests right now are clinically useful. That doesn't mean that the microbiome isn't playing a dramatic, powerful role in these disorders. I mean, I shared with you the study about the microbiome from epileptic children appears to increase or decrease seizures in mice, but we don't know what it is in those microbiota. We don't know.

- 6.89** Maybe next year we'll have some breakthrough study that gives us the magic answer; maybe it'll take 10 years. Hopefully at some point we'll have more answers. Hopefully at some point we will be analyzing microbiota to be able to say, "Oh, this is why you've got binge-eating disorder, or this is why you've got bulimia. Clearly, there's a problem with your microbiome. Here's the plan. We don't need to talk to you about whatever happened to you as a kid, whatever. It doesn't matter. Let's fix your microbiome and we'll fix your bulimia." I'm really hopeful that day is going to come, but we're not there yet.
- 6.90** Instead, I am a really big fan of some basic commonsense biomarkers and assessments, the ones that I'm sharing with you through this course. Why am I so committed to those? Because those we have tremendous amount of evidence for. We have hundreds, if not thousands, of studies backing those biomarkers being highly associated with all bad things, with higher rates of cancer and premature mortality and diabetes and cardiovascular disease and everything else. I think those are unequivocal.
- 6.91** Are there other biomarkers that should be included? Absolutely. I have included some of them. A lot of primary care docs are going to hate me recommending fasting insulin. I think the time has come. Fasting insulin needs to be added. Some people will disagree with me on homocysteine. Why are you measuring homocysteine? That doesn't matter. I think the time has come. I think we have more than enough evidence. Homocysteine matters, and there are things we can do about it. But I'm hopeful. I'm hopeful that we'll evolve. The basics that I believe in, I'm presenting in this course.
- 6.92** Does that mean those are the only ones I would ever consider? Absolutely not. Depending on the person in front of me and the symptoms they're having and the severity of those symptoms, I might want a lot more. A lot of people might need an autoimmune workup, if we have reason to suspect an autoimmune disorder. But I'm hopeful. I'm hopeful that enough of the science is coming together, connecting metabolism, mitochondrial function, with brain function.
- 6.93** There are researchers who are doing biomarker studies already for patients with severe depression, autism, and others. Some of those early studies are proving really valuable. High levels of sensitivity and specificity, but we need more research before everybody's going to start doing it in clinical practice. More importantly, we need wide availability and accessibility, like affordability of those tests, because not everybody can afford 400 blood tests out of pocket.
- 6.94** That's one of the biggest barriers that patients have when they're trying to assess. "Oh, I don't get this test because it's too expensive. Therefore, I'm not doing anything, or I'm afraid to go and reach out because it's too expensive to get the testing done."
- 6.95** But that's where I want to say, all of the strategies that I'm talking about, many of which relate to lifestyle medicine, which are free or low-cost interventions, I don't know that we need the tests. I'm convinced that when I put a patient on the ketogenic

diet and their schizophrenia gets better, I have no doubt that their microbiome is changing. I actually don't care how. There's a part of me that just says, but I don't need a test to figure that out. I don't need a test to measure that or document that. It doesn't really matter clinically. All I need to know is, put them on the diet, see their symptom improvement, and, hey, they're on their way to healing. That's great.

- 6.96** Did their microbiome change in the process? Yeah, probably. Can I document it? No. Did I need to document it? No, not really. I hope researchers will, and I hope, so that we can, again, better understand the mechanisms and maybe we'll come up with a pill instead of a ketogenic diet. I'm doubtful that will happen, but anytime soon. But yeah.
- 6.97** Can you explain why certain people, like your patient Mike, need deeper levels of ketosis, like the 4:1 ratio, and why, as soon as they go off, even for one meal, their symptoms return?
- 6.98** The issue of going off and having symptoms return is much more pronounced in the first six months to a year of the diet. The very first case reports that I published were Mike's and then another female patient's with schizoaffective disorder. Part of both of those publications were that their symptoms of schizoaffective disorder got dramatically better when they were in ketosis, and both of them broke the diet multiple times, and their symptoms came back with a vengeance each time they broke the diet. Then we reinitiated ketosis and symptoms got better.
- 6.99** But that, so first of all, we see that in epilepsy. If somebody starts a ketogenic diet for treatment-resistant epilepsy and their seizures stop, if they break the diet within the first six months, they will often have a seizure within 24 to 48 hours. It's not uncommon at all for them to have a seizure. However, if they do the diet for, say, five years and then stop the diet, oftentimes they seem to be healed. They can stop the diet and their seizures never, ever come back. And Charlie, of the Charlie Foundation, is a perfect example of that. He had unrelenting seizures, like sometimes 20 seizures a day, every day. Ketogenic diet stops them within four days. Four or five years later, he stops the diet, and he's never had a seizure since.
- 6.100** I can tell you that, in a lot of the patients that I've treated and many of the patients that I've heard from, it seems to follow a similar course. That early on, they're still very fragile, and there's something about their metabolism. I would argue the cells don't have enough healthy mitochondria yet to function normally, somewhere in the brain or body; and that the system needs more time to reset, repair, heal, however you want to think of it.
- 6.101** But I've had many patients, one patient who has actually spoken publicly with me about her experience with ketogenic diet, treating her very severe schizoaffective disorder. She also had opioid use disorder. She had been diagnosed with borderline personality disorder, all sorts of things. She did the ketogenic diet for three years, has gotten essentially a full recovery. Full remission, recovery, getting straight As in grad

school, is no longer on a ketogenic diet and still recovered. So I think it's going to be highly variable.

- 6.102** The unfortunate news is, we don't have a lot of long-term data in people with severe mental illness. We don't know how many people will be able to stop the diet and transition to a different diet and remain well. Will it be the exact same as what we see in epilepsy? Maybe, but maybe not. They are different manifestations of illness or different manifestations of a mitochondrial process or a metabolic process. So time will tell. But again, even if people have to stay on the ketogenic diet for life, with no breaking it, for a lot of them, that is a much better life. It is a much better quality of life than living with chronic, unrelenting severe mental illness.
- 6.103** The question I have relates to nutritional psychiatry. When you spoke about that earlier, you mentioned things like niacin and vitamin C, and how early on there seemed to be some promise about those benefits. And I wanted to see if you could elaborate a little more about why it didn't live up to the promise of people seeing remission early on?
- 6.104** There's no doubt that those two examples, niacin and vitamin C, fit into the mitochondrial theory of mental illness. Niacin is essential for mitochondrial function. It plays a key role in the transfer of electrons and energy production within mitochondria. Niacin very nicely fits in to the mitochondrial theory of mental illness. If somebody has niacin deficiency, that could very well explain why they would develop a neuropsychiatric disorder and why restoring niacin will completely cure them.
- 6.105** Vitamin C is an antioxidant, and when people have high levels of oxidative stress, it can further impair mitochondrial function, and relieving that oxidative stress with high doses of antioxidants, in theory, should work. But I shared with you all the evidence. We've really spent millions and millions and millions of dollars pursuing antioxidants for a wide range of neuropsychiatric disorders and obesity, diabetes, cardiovascular disease, Alzheimer's disease, and they just don't seem to work all that well for all that many people. It doesn't mean that there can't be rare people for whom it is like a miracle cure, but it doesn't seem to work a lot for a lot of people.
- 6.106** Niacin in particular is interesting because niacin derivatives like NMN, NAD supplements, others, they're all the rage nowadays. Everybody's taken them because it's supposed to be good for you. It's interesting how some things just never seem to go away. At the end of the day, there is no doubt that there are people who have vitamin deficiencies and develop severe chronic neuropsychiatric disorders, and replacing those vitamin deficiencies can be life-saving. Vitamin B12 is a very clear unequivocal example. Vitamin B12 deficiency is common. If people are following a vegan or vegetarian diet, they are at high risk of vitamin B12 deficiency.
- 6.107** If people are taking Metformin, that puts them at risk. There are other medications, other conditions. But by one estimate, 50% of people in India, which is a highly vegan,

vegetarian country, 50% of people in India are vitamin B12 deficient. If you diagnose vitamin B12 deficiency, replacing vitamin B12 can change everything. Unfortunately, if people go too long with severe vitamin B12 deficiency, they can suffer from permanent neurological damage. So you may not be able to reverse some of the damage that gets done. So recognizing and diagnosing it early on is important.

- 6.108** I think that's part of the confusion. I think the reason the niacin story didn't play out well, I have no doubt that the researchers who first promoted it probably did see some patients that got dramatically better with niacin supplementation. Maybe those patients were deficient somehow or something, or the niacin was just serving an antioxidant or other function, restoring metabolic mitochondrial health somehow or another. But the clinicians became too exuberant and over-promised. And that's the challenge.
- 6.109** Schizophrenia, just like all of the other neuropsychiatric disorders, are heterogeneous disorders, which means you can have many causes of it. And I'm talking about a lot of those causes in this course. So one person could have genetic or epigenetic differences. Another person could have trauma contributing. Another person might use substances. I'll talk about in the next hour. Another person might have dietary factors. Some might have nutrient deficiencies, but it could be any of the nutrient deficiencies. Some might be deficient in magnesium; another might be deficient in B12; another might be deficient in folate. They all converge at mitochondria. They can all produce mitochondrial dysfunction, but the treatments are very different.
- 6.110** Giving somebody niacin isn't going to help magnesium deficiency at all. That person needs magnesium. I think that's one of the reasons I end being so cautious in what I say. It's one of the reasons I actually said not everybody should be on a ketogenic diet, because I don't want to be like the other people and over-promise. I want to be realistic, that there are lots of things that can play a role. Because if we can stay credible, if we can stay open-minded to some of the complexity, but also some of the simplicity—on one hand, I'm saying, yeah, it's more complex.
- 6.111** If people are like, well, what's the one solution you're recommending for all people with mental illness, to cure every mental illness on the planet? I'm like, I don't have that. I'm sorry. Then people say, well, then it's so confusing and overwhelmingly complicated. I'm like, no, it's not. This course walks you through the basics, and we'll probably get 70% of people with mental illness better. If people follow the recommendations that I'm making in this course, I really do believe easily 50% of people suffering from mental health conditions will get better.
- 6.112** Will they be cured? I'm not going to go that far. Will they get a complete unrelenting remission? I'm not going to go that far. But will they get significantly better? I will go that far. Yes, I think they will, because there are lots of factors at play. I'm literally hearing from thousands of people who are sharing their success stories. I know it's not just me. I'm not making this up. It's not in my head. Some people right here in studio

audience have shared their stories with me about how things are dramatically better.
So I know it's possible. Hope that helps.

Lecture 7: The Brain on Substances

At a glance — The brain on substances — substance use disorders in the metabolic frame.

- 7.1** Now we are going to dive into the topic of substance use. Substance use disorders represent a whole class of disorders unto themselves. But if you don't already know, people with mental health disorders also have very high rates of substance use disorders. In fact, this ties in beautifully with the metabolic and mitochondrial theory of mental illness, and, in fact, helps us answer so many questions that our field has struggled to answer over centuries, literally.
- 7.2** Substances that impact metabolism and mitochondria include a lot of really common household ones. Prescription medications, a whole class of them, and I've already talked about some of them earlier, but alcohol, marijuana, tobacco, smoking cigarettes, vaping, and the substances that people often vape. Nicotine and caffeine. All of these impact mitochondria for better or worse. I just want to say at the outset, on the surface, this actually seems impossible to believe, right? It's like, how can that really be?
- 7.3** Honestly, when I was first developing this model, this theory, I had the same thought. I was like, this isn't possible. How can everything connect so beautifully through mitochondria and metabolism? I've never been taught this. Every substance that I looked for, I remember one of the key moments as I was developing this theory is, I thought, wait, this is impossible. And I'm going to test something out. I know that marijuana can make some of my patients more psychotic. And marijuana has nothing to do with mitochondria, but I'm going to look into it and see.
- 7.4** I simply put in the search terms, marijuana, mitochondria, and hundreds of articles began coming up about the intricate connections between THC, marijuana, and mitochondrial function. I thought, wow, wow. So how is that possible? It's possible because metabolism and mitochondrial function are, in fact, the foundation of living organisms. They are the foundation of cellular function. When cells malfunction more often than not, it somehow impacts metabolism or mitochondria, because if metabolism and mitochondria are functioning optimally, they can deal with just about anything.
- 7.5** They can roll with the punches. They can adapt. Cells can adapt. Living organisms can adapt. We call that resilience. That is why these are so foundational. What does it have to do with mental health? If you don't know, substance use disorders are associated with higher rates of subsequent mental disorders, but people who initially develop a mental disorder have much higher rates of later substance use disorders on top of their mental illness. Again, they feed each other. It is a bi-directional relationship, or it is a vicious cycle, a feedback loop.

- 7.6 The mental disorders are pretty much all of them. All of them pretty much are associated with increased rates of substance use disorders. This includes depression, anxiety, PTSD, bipolar, psychotic disorders, personality disorders, eating disorders, essentially all of them. It is already widely known and understood in our field. What I'm saying, this is nothing new. I'm telling you one of the bases of the mountain. We already know this. This is established in the mental health field.
- 7.7 The current dogma is that several mechanisms may explain the common co-occurrence of substance use disorders and psychiatric disorders. One is pre-existing psychiatric disorders may increase the risk of developing substance use disorders, in part because substances are often used to cope with symptoms of the psychiatric disorders, even if they ultimately make those disorders worse. At the same time, substance use, especially adolescent substance use and long-term exposure, may predispose individuals to developing psychiatric disorders.
- 7.8 How exactly that works? Our field is at a loss. Our field doesn't offer any concrete answers, and more importantly, it doesn't offer answers or solutions that are actionable. What the brain energy theory does is, it offers concrete answers and, more importantly, concrete solutions. Then others in our field will say, substance use disorders and other psychiatric disorders share minimal genetic risks and environmental vulnerabilities, such as trauma and adverse childhood experiences. We've already talked about that. Trauma and adverse childhood experience, increased risk for all of the mental disorders and substance use disorders.
- 7.9 But again, exactly how does that work? Our field is like, well, yeah, no one knows. No one knows. We offer symptomatic treatments that work for lots of people, but as we've talked about, fail to work for far too many. Is there any proof, do we have any proof, that substance use is directly impacting metabolism and, more importantly, impacting mitochondria? This review article, published again in the Proceedings of the National Academy of Sciences, pretty prestigious journal, "Impaired energy metabolism in the brain is a characteristic feature of exposure to drugs of abuse."
- 7.10 Mitochondria themselves actually have receptors for marijuana, nicotine, alcohol, benzodiazepines, right on their membranes, and they are directly impacted by these substances. Now, these substances also affect receptors on the cell membrane, which is what our field is largely focused on, but we have largely ignored, well, what happens after that substance impacts a nicotine receptor on the surface of the cell? All I'm saying is, let's go deeper. Yes, nicotine affects nicotinic receptors. What happens next?
- 7.11 And what happens next, invariably, are changes in metabolism and mitochondrial function. Why? Because you're either stimulating cells or inhibiting cells. Those are the two principles. What does that mean? At the end of the day, stimulating cells means increased mitochondrial function, increasing the engine of the cell. Inhibiting those cells means inhibiting mitochondrial function, slowing the engine down. Just like when a car goes faster or a car goes slower, you can talk about all the different things that

play a role in whether cars go fast or slow, but you always have to implicate the engine in the process.

- 7.12** The engine is almost always—even if you're putting the car in neutral or idling, you're disengaging the engine. The engine is always implicated, just about always implicated. It's not 100% always implicated. And so I would say the same thing with mitochondria. It is just about always. Can I think of some exceptions to the rule? I probably can, but that is the broad message. Recent studies, again, I quote, "Recent studies have broadly implicated mitochondria as targets through which drugs of abuse influence neuronal function and behavior."
- 7.13** I am not going to have the time, and you probably don't have the interest, for me to go through the hundreds, if not thousands, of research articles that support this. But for the skeptics, for the scientists who are listening to this, all you got to do is Google it or do a PubMed search, and you will find everything. But I'm going to give you a tiny snapshot. We're going to go through different substances of abuse, and I am going to hopefully get you to think about those substances of abuse in a different way. We're going to start with alcohol.
- 7.14** Tragically, for any of you who like a good drink every now and then, like to relax and socialize, I have a message for you from the World Health Organization. "No level of alcohol consumption is safe for our health. Alcohol is a toxic, psychoactive, and dependence-producing substance and has been classified as a Group 1 carcinogen by the International Agency for Research on Cancer, decades ago. This is the highest risk group, which also includes things like asbestos, radiation, and tobacco." That's where we're at with some large international organizations.
- 7.15** What does alcohol have to do with metabolism and mitochondria? Quite a bit. Mitochondria are involved in the conversion and detoxification of alcohol to acetate. So alcohol goes through this conversion process. It gets converted into acetaldehyde and then converted ultimately into acetate. And acetate, it turns out, is a fuel for mitochondria, especially in brain cells. So alcohol gets processed in the liver, converted into acetate, and that acetate travels up to fuel mitochondria in brain cells.
- 7.16** This was work done by none other than Dr. Nora Volkow. This is what led her to further conclude that alcoholism is a disorder of brain metabolism, because the reward for alcoholics is that, when they drink alcohol, they get a boost of acetate and that fuels their brain and they feel normal. It's not that they feel great. It's not that they're looking for a high. They are just looking to feel normal. They are looking to feel not anxious. They are looking to feel not brain fogged. They are looking to feel not unmotivated. They want to be motivated. They want to feel alive.
- 7.17** When they drink, they say, if you talk to an alcoholic, they'll say, "That's when I feel good." This research helps us understand that they are not lying. They aren't making it up. They're not rationalizing their use; they are, in fact, identifying that they have a

brain energy problem when they are sober. The only way that they have figured out to fix it is to drink alcohol. So that is what led Dr. Volkow to do the study of the ketogenic diet with alcoholics, thinking maybe we can fix their brain metabolism in another way that will help them not need alcohol to survive and do OK.

- 7.18** Obviously, if we employ brain energy treatments or metabolic treatments over time, over several years, we might actually be able to restore normal brain metabolism so that alcoholics can more easily give up alcohol and function optimally in their lives. One of the problems with alcohol, in addition to it being a fuel source and a complicated fuel source, is that excess alcohol consumption leads to rising levels of acetaldehyde. If you binge-drink, your liver can't process the alcohol fast enough, and you end up getting high levels of acetaldehyde, which is toxic to cells, and you'll never guess where that toxicity seems to be most pronounced. It's at the level of mitochondria.
- 7.19** Acetaldehyde is a mitochondrial poison. It changes the morphology of mitochondria, it reduces ATP production, and it increases oxidative stress. We have large epidemiological studies showing that, if you look at people who drink alcohol, do they have increased risk for health conditions? The clear unequivocal example are people who are heavy drinkers. People who are heavy drinkers, which is defined, at least some definitions, for women, four or more drinks on any day or eight or more drinks per week. For men, five or more drinks on any day or 15 or more drinks per week.
- 7.20** Heavy drinking is associated with increased risk for mental health conditions, essentially all of them, and obesity, diabetes, cardiovascular disease, and premature mortality. So heavy drinking, I would argue, through the mechanism of mitochondrial toxicity, is resulting in all of the neuropsychiatric disorders and all of the metabolic health disorders. Here's the conundrum. When we look at these large epidemiological studies, light to moderate drinking is often associated with improved health. That's where we get these recommendations. One or two glasses of wine a day is good for you. Drink up, but don't overdo it. But drink up in moderation, and that's good for you.
- 7.21** How can we understand that? I want to let you know, there is passionate debate in the field right now about this. Some people are clinging to that research and that evidence base, saying, "Yes, it's true. We are not giving up alcohol. We're going to keep drinking. We want to drink low levels of alcohol, and it's good for us. You can't tell us it's bad. It's good for us." Others are saying, "No, no level of alcohol is safe." Again, I'm not making stuff up. This is the World Health Organization that says that. No level of alcohol consumption is safe.
- 7.22** But how can we understand these epidemiological studies then that say, if you drink a little bit, maybe you do better than people who don't drink at all? Well, there are some possible benefits. The possible benefits include, alcohol can reduce blood pressure and reduce stress and anxiety in some people, and lowering blood pressure even for a short period of time and lowering stress and anxiety might be beneficial because we've talked about the toxicity of those on their own.

- 7.23** There's this thing called mitohormesis, in which, if you stress mitochondria, you might actually get more mitochondrial biogenesis or the production of more mitochondria. So, tiny doses of toxins can actually be beneficial to human health. It's like when you work out. If you don't push yourself at all, you're not really going to get many benefits. If you push yourself to failure, what you're doing is, you're overstressing the mitochondria. Actually, whether it's lifting weights or running or whatever you're doing, if you push yourself to failure, the mitochondria at some point are overstretched. They are the cause of your failure. They are the reason you can't do another rep or you can't run more.
- 7.24** They are the reason. They can't produce enough ATP to keep up with you. What happens is that they develop high levels of reactive oxygen species, and that signals the cell, "Make more mitochondria. She needed to run farther. Like, next time, we need to make sure she can run farther, so make more mitochondria so that the next time she's in that situation, she can run farther. He needed to lift that really heavy boulder or that really heavy weight in the gym. He couldn't do it. Make more mitochondria, so that he can lift that heavy thing the next time."
- 7.25** These are all adaptive, and they're really based on survival. They are based on, do you need to run fast or not, in order to survive in the environment you happen to be living in, or the tribe that you happen to be living in? If you need to be a fast runner, then your body is going to expend energy and resources on making you a fast runner. If you don't need to be a fast runner, your body is going to say, "Hey, you don't need to be a fast runner. You can just sit around in comfort and safety and peace, and you don't need to run that fast." It's not going to waste resources on giving you more mitochondria to be a fast runner because you don't need that skill or survival trait.
- 7.26** So mitohormesis is one possibility, that maybe stressing mitochondria with low levels of a mitochondrial toxin, alcohol, could in fact be adaptive. That is a possibility. There are others who argue, no, it might actually be a healthy user bias. What does that mean? That means that people who become ill stop drinking. They stop drinking because they can't handle it anymore. It makes them sick. It makes them weak. When you look at these large epidemiological studies and you see people who don't drink at all, you have to question, well, why did they stop drinking? Did they stop drinking because they were already becoming ill and vulnerable? And is that why they have worse health outcomes than people who drink one drink a day?
- 7.27** We can also add to that bias, adverse childhood experiences. People who don't drink at all and who never started drinking, there's a reason. There's a lot of pressure in our society to drink. There has to be a reason why somebody is not a drinker. One of the more common reasons that people are not drinkers is because they had an alcoholic parent. If you had an alcoholic parent, you already get one adverse childhood experience just from that alone. There's a good chance you have more than one

adverse childhood experience because that alcoholic parent was probably a jerk, at least on some occasions, and did things that they should not have done.

7.28 Now we're looking at the group of zero drinkers and we're saying, "Oh, wow, they're a traumatized group, or they've had a lot of adverse childhood experiences, and no wonder they have worse health outcomes than people who drink very low levels of alcohol." So that's another possibility. Social drinking. Some people only drink in groups with friends. They only drink when they go out. They would never consider drinking at home alone. We have to take that into account because that's a bias.

7.29 We've already discussed, if you have friends, if you have a social community, that is protective. You're part of a tribe. You feel valued and respected and wanted and loved. If drinking wine over dinner around a fire is part of that process, then those wine drinkers are going to have benefits, maybe having nothing to do with the alcohol. Maybe the alcohol is still a poison for them, but it's the process of, "I have friends, I'm connected, I'm talking with them." That is actually the health advantage.

7.30 So, alcohol is complicated. And yet, if we look at the toxicity of alcohol, there are two clear unequivocal circumstances in which alcohol is toxic to the liver and to the brain. This is well known. Liver pathology from alcoholism leads to cirrhosis, alcoholic liver disease. We've got this other term, non-alcoholic fatty liver disease. Well, the primary one is the alcoholic fatty liver disease. Alcohol poisons the liver, and it can result in cirrhosis and death for a lot of people. Alcoholism is the primary cause of cirrhosis. How does that work? How exactly? What does the science tell us? This science dates back to the 1960s.

7.31 It's mitochondria. Alcohol is poisoning the mitochondria and liver cells, and that is how it is poisoning the liver. This has been known for decades. Alcohol is a mitochondrial liver poison. What about the brain? The severe consequence of alcoholism in the brain is something called Wernicke-Korsakoff syndrome. Wernicke's disease is dementia, and Korsakoff's syndrome is actually psychosis. These are people who have dementia and psychosis as a result of their alcoholism.

7.32 How does that work? It appears to be through two mechanisms. One is the brain damage itself from the alcohol, and I would argue that that is mitochondrial toxicity. I think some would argue that has not been definitively proven, but I think we have more than enough evidence. The liver is not—liver and the brain, human cells are human cells. But there's one camp. It's brain damage from the alcohol itself. But the thing that throws people over the cliff is thiamine deficiency on top of it. And alcoholics are more vulnerable to developing thiamine deficiency.

7.33 And so the treatment is give people thiamine. We might ask the logical question, well, what does thiamine do? Thiamine is an essential nutrient that serves as a co-factor for a number of enzymes, mostly within mitochondria. Thiamine deficiency induces mitochondrial dysfunction because thiamine is an essential vitamin primarily for

mitochondria. That is what makes it essential. What is the treatment for Wernicke-Korsakoff syndrome? Give people intravenous thiamine and glucose. That is a metabolic treatment.

- 7.34** Let's go to tobacco. Tobacco and mental illness. Mental illness is associated with higher rates of smoking and higher levels of smoking among smokers. People who have any mental disorder are much more likely to smoke. Then when you look at the amount of cigarettes they consume, it's higher than non-mentally ill people. The CDC estimates that nearly one-third of all cigarettes consumed in the United States are smoked by people with mental health conditions.
- 7.35** A significant portion of smokers have mental illness. 23% of US adults with any mental illness reported smoking cigarettes during the past month in 2020, compared to only 14.5% of adults with no mental illness. So almost double the rate of smoking in people with mental health conditions versus not. Lifetime smoking rates are even higher in people with mental health conditions. 60% of people diagnosed with major depression have been a smoker at some point or another. 83% of people with bipolar disorder have been a smoker at some point or another. 90% of people with schizophrenia have been a smoker at some point or another, compared to only 32% among adults who don't have a mental illness.
- 7.36** Well, that's tobacco and mental health. What about tobacco and metabolic health? Smoking is the leading cause of premature preventable death in the United States. Smoking can disrupt metabolism. It raises triglycerides, lowers HDL cholesterol, makes blood more likely to clot. It damages cells that line the blood vessels, increases the buildup of plaque in blood vessels, and causes thickening and narrowing of blood vessels. Smokers are 2-4 times more likely to develop heart disease than nonsmokers. Not shocking. But here is something that is shocking to most people.
- 7.37** Smokers, on average, are more likely to be thinner than nonsmokers. In the 1950s and '60s, tobacco was marketed specifically to women for one and only one reason. Have a slim figure. If you want a slim figure, smoke. Slim Jims, brand name. Smoking slims you down. In fact, the studies show this. Smoking results in weight loss, weight suppression. Yet people who are smokers are twice as likely to develop insulin resistance and type 2 diabetes. How does that work?
- 7.38** Most of us think insulin resistance and type 2 diabetes are disorders of overeating or disorders of obesity. This, along with lots of other examples, says that is not the right way to think about them. Insulin resistance, type 2 diabetes, are not disorders only of overeating. They can be a disorder of overeating, but they are a disorder of mitochondrial toxicity. Because we have evidence that tobacco is directly poisoning mitochondria. We've got tobacco, smoke, has over 7,000 chemicals. At least 250 are known to be harmful, including hydrogen cyanide, carbon monoxide, and ammonia.

- 7.39 Everybody knows cyanide, that's a poison. Guess what it poisons? It doesn't poison the whole cell. It poisons mitochondria. Carbon monoxide, what does that do? That prevents oxygen from being carried. What is oxygen for? Mitochondria. Cigarette smoke is toxic to the human body in other ways, but I would argue mitochondrial toxicity is its primary mechanism of action. And we've got lots of evidence that it increases reactive oxygen species, increases oxidative stress. Actually, cigarette smoke results in dose-dependent inhibition of mitochondrial complex I and II. So we actually see reduction in ATP production as people are smoking.
- 7.40 That leads us to vaping. Everybody should vape, right? Vaping is fine. As long as we're not smoking, vaping is good. Let me be clear. If somebody is a cigarette smoker and is trying to stop and cannot stop, and vaping helps them stop smoking cigarettes, I am all for it. Vaping is so much safer than inhaling cigarette smoke. That is fairly well established now. Vaping is a harm reduction practice. Vaping is safer than smoking. However, vaping is not safe. Safer than smoking does not equal safe.
- 7.41 So vaping, you're getting the nicotine or the THC or whatever it is you're vaping, and I'll talk more about those molecules and how those molecules can adversely impact human health. But vaping also includes other ingredients, chemicals. There are rare lung problems, massive lung inflammation. There are all sorts of problems with vaping. So, again, if somebody is a cigarette smoker and they choose to go to vaping to try to reduce exposure to cigarettes, I'm all for it. But I would still say, as a general health and wellness, if you want to optimize your health, I'm going to say, try to get off the vaping. Try to reduce it or eliminate it.
- 7.42 Now we turn to cannabis. Cannabis is everybody's friend. It is the substance that will cure what ails you. It's good for seizures, it's good for PTSD, it's being legalized across the country. Why would they legalize it if it wasn't good for you? It must be good for you. Doctors are prescribing it. It's great. Unfortunately, THC is very active at mitochondria, in particular. Cannabinoids work at CB1 and CB2 receptors, and these are located throughout the brain and body. CB1 receptors are also located on cell membrane. So CB2 receptors are mostly on cell membrane. It's influencing cell biology across your body.
- 7.43 CB1 receptors are also located on cell membranes, but they are specifically also located on mitochondria themselves. THC, when it binds to CB1 receptors on mitochondria, increases oxidative stress and induces cerebral mitochondrial dysfunction. This has been well established. Cannabinoids and mitochondria and glucose metabolism. So glucose metabolism, the bottom line is, we've got a complex slide, and I'm not going to go through it because I just want you to understand. THC, cannabinoids, are directly influencing mitochondrial function. That directly influences brain metabolism, and that can influence the effects that we see.
- 7.44 What are the effects of cannabinoids? The desirable effects of cannabinoids are to chillax. I want to chillax. To treat anxiety or PTSD. Some people say, "It really helps with

my insomnia." In extreme cases, people will say, "It stops my seizures." CBD or THC stop seizures. There have been national news stories on that, how some children with treatment-resistant epilepsy, nothing was helping them get better, and they started using a THC product or a CBD product, and it stopped their seizures. That was, in large part, one of the big stories that helped get it legalized.

- 7.45** How does that fit into the brain energy theory? It does. Impairing mitochondrial function can reduce hyperexcitability. All of those things that I just told you about— anxiety, PTSD, seizures—they all represent hyperexcitable brain circuits. If you slow mitochondrial function, if you slow metabolism in those cells, you can reduce symptoms. The challenge is, what are you doing over the long run if you use this every day, even multiple times a day? What I would argue is, you are actually further impairing the function of those cells, further impairing mitochondrial function in those cells, and making that human being more likely to develop a chronic disorder and more likely to develop new mental health and metabolic disorders over time.
- 7.46** I want to at least briefly mention cocaine and opioids. Is there any evidence that cocaine or opioids—they work through different receptor mechanisms. Yes, cocaine, dopamine, reward pathways. Opioids, opioid receptors, the mu-opioid receptors. This isn't replacing that. This is simply asking the question, what happens after those molecules attach to the opioid receptor? What does that do to the cell? How do those effects—like, something attaching to that one receptor, that doesn't explain much of anything. That's one part of the pathway.
- 7.47** How does that result in addictive behavior? How does that result in all of the other consequences, in all of the other neuropsychiatric and physical health conditions that we see develop in people who become addicted to cocaine? Well, we've got good evidence that cocaine directly, in both humans and nonhuman primate studies, that intravenous cocaine administration significantly reduces glucose metabolism in key regions of the reward system of the brain. It immediately disrupts brain metabolism. Withdrawal from cocaine persistently impairs metabolic activity. So these reward pathways become metabolically injured.
- 7.48** That's the way I want you to think about it. These drugs of abuse are injuring brain cells. The injury then makes them more vulnerable to malfunction, and it drives people to keep using. Cocaine can actually lead to mitochondria-mediated cell death. So if you're really a heavy cocaine addict, you can actually develop permanent brain damage from it.
- 7.49** What about heroin? Is there any evidence? Yes. This was a study in 36 individuals with heroin dependence compared to 30 healthy controls. Abnormal cerebral metabolism, abnormal brain metabolism, and metabolic connectivity in individuals with heroin dependence, set them apart from healthy controls. Heroin is doing the same thing. It is damaging brain energy, metabolic circuits, pathways, mitochondrial biology. It is

ultimately leading to impaired brain energy metabolism, which leaves people vulnerable and addictive.

- 7.50** This study in animals actually looked specifically at what drives the addictive behavior. What they found is that it appears to potentially be due to mitochondria themselves; that mitochondria themselves are playing a role in energy sources going from astrocytes to neurons, and that when you impair mitochondrial trafficking, that you can actually—you get these addictive behaviors, and that when researchers blocked the effects, they could actually restore, with a specific molecule, the researchers gave these heroin, morphine-addicted mice a specific molecule that would block the effects on mitochondria.
- 7.51** And not only were the mitochondria improved, but the animals no longer exhibited addictive behavior to the morphine. They didn't appear to be addicted to the morphine anymore. Preventing the mitochondrial dysfunction can prevent the addictive behaviors. Conversely, what I would argue and what Nora Volkow pursued in the ketogenic diet trial is, if we can improve mitochondrial function through a completely different mechanism, other than just abstinence, if we can improve mitochondrial function through a dietary intervention, then maybe we can help people recover.
- 7.52** This all raises the question, can drugs cause mental illness? We know that drug use is associated with mental illness, but is that just because people who are already vulnerable, maybe genetically, they were going to develop both, or they had childhood trauma, and that was just putting them at risk for both? Does drug use itself cause mental illness? This is a highly contentious question still. I would argue, yes, based on the mitochondrial science that I've shared with you. I would argue, yes, it's easy to understand and it's obvious.
- 7.53** And all of the animal models that have been done support this. The problem is, animal models are models. We can't ask animals, "How are you feeling right now?" We can't ask mice, "Are you having hallucinations or delusions?" We have to guess, based on behaviors, what those animals might be experiencing. We would never be allowed to ethically do any of these studies in humans. I could never take a group of 100 children and give 50 of them high doses of cocaine and placebo to the other 50, and see which ones develop mental illness. That would never be allowed ethically, but that would be what would be required to prove it beyond a shadow of a doubt.
- 7.54** Do we have any research to support this? Well, this was a study of almost 10 million people. It's a cohort study. What they found is that, if a person went to an emergency room reporting psychotic symptoms that were supposedly due to substance use—they used marijuana, they used some cocaine, they took psilocybin, they developed psychotic symptoms, and it became a crisis, and they went to the emergency room. The researchers looked at that group of people, and those people within three years had a 163-fold increased risk of new diagnosis of schizophrenia. If you take drugs and they

make you psychotic, you have a 163-fold increased risk of being diagnosed with schizophrenia over the next three years.

- 7.55** Well, but wait, Chris Palmer, you're saying that the drugs are actually impairing mitochondria and that there are more symptoms other than just psychosis. So what about everyone else? If you went to the emergency room for any substance use associated issue— a DUI. "Oh, I got a little drunk and I fell and broke my ankle. I'm here for a broken ankle, but the doctor reported, yeah, they were drunk." Do you have an increased risk for developing schizophrenia over the next three years? Yes, you do. Tenfold increased risk. If you are in an emergency room for anything related to a substance, tenfold increased risk of schizophrenia.
- 7.56** Now, I'm using schizophrenia because that's the disorder people are most terrified of. That's the disorder that everyone unequivocally agrees, that's a real disorder. That's a real thing. Are these people at increased risk for anxiety? Oh, of course. Are they at increased risk for depression? Oh, yeah, no brainer. Are they at increased risk for bipolar disorder? Oh, sure. But the drugs are probably making them bipolar, or the bipolar is making them use drugs. People rationalize all of the other symptoms. But schizophrenia is a hard, serious, life-changing diagnosis. People who use drugs are much more likely to develop it.
- 7.57** I want to talk about drug use in this context of this brain energy theory. The two principal consequences of mitochondrial dysfunction are overactive brain regions and underactive brain regions. Both are happening. If you get impaired energy production in your cells, impaired mitochondrial function in your cells, the paradox is that that cell can be underactive 95% of the time, and it can become hyperexcitable 5% of the time. That is the important paradox of this theory, because it helps us understand, then why would people use drugs? If it's hurting them in the long run, why would they use them?
- 7.58** It gets into actually this really complex situation. When you talk to somebody who's using, they are going to resist. If you go home and you got a son who's using a lot of marijuana and you listen to this lecture, you're going to be like, "Hey, son, stop using. That's really bad. Dr. Palmer says that you might develop schizophrenia. Stop doing it." Your son is going to say, "No, Mom. I'm fine. I don't know who that stupid Dr. Palmer is, but I'm fine. And marijuana helps me chillax. It makes me feel good. Do you really want me to feel bad?"
- 7.59** We need to talk about the different phases of substance use and how they make sense. So people initiate substance use, and when they very first use, when they are naive to the substance, they've never used it, the first dose can be powerful. For some people, it can be powerfully rewarding. How does that work? We're really talking about two broad categories. There are others, like, opioids don't fall neatly into stimulants or sedatives, but most of the other substances of abuse do fall neatly into, this is either stimulating or inhibiting.

- 7.60** How does that make sense? When people first use a substance, if they have underactive brain regions, they have brain fog, they just don't have much motivation, they don't feel very good, stimulants are going to make them feel alive. Stimulants are going to make them feel great and they're going to think, "Wow, where has this been all my life? I didn't know I could feel this good. I didn't know my brain could work this way. This is amazing. I am going to be capable of such great things if I just use this substance more and more."
- 7.61** That may sound facetious but think about people who get addicted to cocaine. The whole socioeconomic range is influenced by it, affected by it. You have, in the 1980s and '90s, in particular, successful, wealthy business people were all snorting cocaine. It was the greatest thing in the world. If you have any impairment of brain function, initially, using a stimulant is going to increase brain metabolism and it's going to make you feel great.
- 7.62** What about sedatives? How can they be helpful? Well, if you have any hyperexcitable brain regions, if you have anxiety, if you have ruminations, if you cannot relax, if you have psychosis, anything that stops brain metabolism may, in fact, bring you relief. You may think, "Oh my gosh, thank god. Thank god that marijuana just—finally, after weeks of suffering, I have relief. Why didn't my doctor tell me about this? I'm going to treat myself. I know what to do." And you're going to start using.
- 7.63** Once you start regularly using, your brain is going to adapt. You're going to start to develop tolerance. You may need higher doses to get a similar effect. At some point, the system will level out, and your brain cells will actually upregulate or downregulate receptors to try to counteract the effects of the substance. And that sets you up for being trapped. Now you're trapped. Good luck getting out of that trap. Because when you try to go without it, you have the exact opposite reaction.
- 7.64** So if you have gotten hooked on stimulants, your body is trying, your brain is trying to counteract those effects. And as soon as you stop the stimulant, you're going to crash and burn, and you're going to feel horribly depressed. Your brain is not going to be functioning properly. You're going to say, "I have brain fog. I can't think. I'm not motivated. I'm depressed. This is a living hell." And sure enough, the only thing that's going to get you out of that is a stimulant in the short run.
- 7.65** If, on the other hand, you're hooked on sedatives, alcohol or other sedatives, they are suppressing brain function. Your brain is making adaptations to counteract that. If you try to go cold turkey without them, you're going to have symptoms of hyperexcitability. That means, when an alcoholic stops drinking alcohol, he is more likely to have anxiety, ruminations, tremors, seizures, and he may even die. Those are all symptoms of hyperexcitability. So, that is the trap of substance use.
- 7.66** Now I want to do a quick self-assessment. You all probably don't want to do this self-assessment now. Here's the deal. Alcohol. One drink is defined as 12 ounces of beer, so

a can of beer, 5 ounces of wine. We're not talking the extra-large pour that you might ask for from the waiter or waitress, 5 ounces of wine, 1.5 ounces of distilled spirits, so a tiny shot of distilled spirits. That is one drink. How often do you have two or fewer drinks in one day? So you're drinking something, but it's one or two or even a half. And how often do you have more than three drinks in one day? So again, if you make really heavy good martinis that include 4.5 ounces of distilled spirits, that one martini is three drinks.

- 7.67** Do you smoke cigarettes? Do you vape nicotine? Do you use nicotine in other forms, pouches, gum, patches? Do you use THC or marijuana in any form? Smoking, vaping, edibles, anything? If so, how often? Do you use CBD products? Do you use any other recreational or illicit drugs, opioids? I mean, there are so many recreational drugs nowadays. Lots of new ones, so I'm not going to list them all. Just, do you use something? Because I probably haven't even heard of them.
- 7.68** Do you drink coffee, more than two cups per day? Or do you drink coffee after noon, 12 noon? If you drink two cups of coffee in the morning, don't even put it down as a potential red flag, but if you drink more than two cups per day or you're drinking coffee throughout the day, put it down as a potential red flag. And do you use caffeine in other forms, pills, energy drinks, or others? What if you have a substance use disorder? About 10% of American men have a substance use disorder. Most of them don't know it. A lot of them don't think about it. These are really, really common, and most people don't know it.
- 7.69** Most of the time, when we think about substance use disorders, these are the traditional recommendations. Alcoholics Anonymous, other self-help groups, outpatient treatment, go to a mental health professional. Maybe consider checking yourself in for a detox recovery program. There's medication-assisted treatment. You can take buprenorphine, Suboxone, for your opioid dependence. There are medication-assisted treatments for alcoholism, and there are halfway houses. If things are really bad, check yourself in to a halfway house. Live in a sober community.
- 7.70** These are traditional treatments. I am not at all opposed to any of these traditional treatments. These can work for a lot of people, but they also fail to work for far too many people. Can we think about a metabolic approach to substance use treatment? On this front, I just want to mention a nonprofit organization. I have no relationship with them whatsoever. I don't even sit on an advisory board. They don't give me any money, nothing. But they are a nonprofit organization founded in 2006, focused on helping people overcome substance use disorders through a model of metabolic health strategies. That's not the way they talk about it, but that's the way I'm going to talk about it.
- 7.71** They have over 560 classes per week, including yoga, art, CrossFit groups, bicycling clubs. They do running groups. They have music events. They have in-person local programming. So some cities have actual gyms that you can go to for free. They have

on-demand programming. They've got music, community. They are located throughout the United States. 83% of their members report remaining sober after three months.

7.72 The only reason I plug the Phoenix is because it is free, it is available. If you are poor, if you're in prison, you can access this. I just want to give people at least one concrete resource. There are numerous others. That is it on substance use. Next, we're going to talk about, how do you put this all together and create a treatment plan for yourself or a loved one? Thanks.

Lecture 8: Recovery Begins with Hope

At a glance — Recovery begins with hope — implementing strategies, tying it together.

- 8.1** All right. So now we are going to try to tie everything together and talk about, how do you actually implement some strategies in your own life? Or if you know somebody, have a loved one who's struggling with mental illness, how can you help them? What can you actually do? First and foremost, I am going to advocate, we need integrated human health care, whole person health care. That means the three pillars, so to speak, are, we need to help people improve metabolic health, we need to help people improve mental health, and we need to focus on longevity and healthy aging. That those are three interrelated pillars.
- 8.2** And although this sounds so obvious and simple, to many people, they're going to say, "Come on, Chris Palmer, this is nothing new. Everybody already knows this. We already do this. This is standard practice." I want to point out, we in the mental health field routinely prescribe pills that cause obesity, that cause type 2 diabetes, that increase risk for cardiovascular disease, and that ultimately increase risk for premature mortality. That is standard routine mental health treatment. We sometimes impair metabolic health and threaten longevity and healthy aging with mental health treatments. What I am saying is that we need a paradigm shift. That is no longer acceptable, and we, in fact, can do better.
- 8.3** The first step is assessment. You need to understand, where are you at with your mental and metabolic health? I have already asked you to do some fairly straightforward assessments on your metabolic health, your psychosocial health, your sleep health, circadian rhythms, your exercise, habits and abilities, physical abilities, and we did a substance use self-assessment. You're going to recall all of that, or refresh your memory on all of those assessments, but we need to do more. We're going to wrap up the basic assessment.
- 8.4** The next thing I'm going to ask is, do you have any of the following mental health symptoms? Now, is this an exhaustive list? It is not, but it's going to get at a lot of the common neuropsychiatric disorders. Do you have depression, anxiety, obsessions or ruminations, compulsions, fatigue for no clear reason? Poor concentration, poor memory, unexplained mood swings? Do you often get irritable or agitated? Do you have hallucinations or delusions? Granted, people who have them often don't see it themselves, so maybe a loved one or a professional is going to have to help with those.
- 8.5** Any suicidal thoughts or acts? Have you ever harmed yourself on purpose, even if the intention was to not kill yourself? Do you have anger, aggression, or any violence, even if you think there's a good reason for it? Are you unable to get things done? Have you withdrawn from friends or activities, even if you think there's a good reason why you

should? Any significant changes in eating patterns or weight recently? Do you have flashbacks or nightmares? Do you have low self-esteem?

- 8.6** Again, one of the challenges with these symptoms is that we can talk about normal, extreme, and disorder. If you tell me that "my spouse just filed for divorce, and my daughter is really struggling with it, so I have some depression," that's not a disorder of your metabolism. That is you being a human being. And yet it's important to know that you're going through this stressful period that is resulting in stress and depression, because that is taking a metabolic toll on you. Even though it is normal, we know that you are at risk. You are being taxed. You are being threatened.
- 8.7** Your spouse leaving you threatens your status in society. It threatens your finances. It threaten lots of things. Threatens maybe even your own identity, and those threats are causing a stress response. That stress response is taking a metabolic toll. So even if you have some of these symptoms and you think, "Well, I know why I have these symptoms," it's important to note them because they are taking a metabolic toll on you. If you have any of these symptoms and you think, "I have no idea why I have those. I just can't snap out of it. I'm chronically depressed and I don't know what's wrong with me," then that suggests that you potentially have metabolic dysfunction in your brain causing those symptoms.
- 8.8** Next, we're going to talk about stress. How frequently do you feel stressed or burned out? Everybody feels stressed a little bit, so I don't want you to put down every micromoment of stress that we all experience, because we all experience stress. If you don't have any stress at all, that's probably a problem. You're probably in denial about your life and about the challenges that you're facing. We should all experience at least a little bit of stress. But how do you manage your stress? This is probably the more important question. Here are some examples of things people can do to manage their stress.
- 8.9** Do you exercise? Do you meditate? Do you talk or vent with others? If you're really stressed, do you get on the phone with a family member or friend and start ranting, and that's really helpful, catharsis? Do you overeat? Do you binge-watch Netflix? Or do you start bingeing on social media, eating some cookies and chips, maybe smoking some marijuana? Do you retreat and hide in bed? Do you just hide from the world when you're stressed? Do you use other substances, alcohol, prescription medications, other things?
- 8.10** Are there any things that you might want to change? If you've written down some things that you're like, "Yeah, that's probably not the best strategy," maybe circle those as, "Yeah, maybe I could work on one or two of these things." Now, I'm going to ask you some generic health questions that didn't fit neatly into some of the other categories. Are you underweight or overweight? By definition, that means a BMI greater than 25 or less than 18. But in reality, I'm really just going to ask more generically, do you think you are underweight or overweight? Most people, when they

are, they know it. People have told them. You can have a BMI greater than 25 and be lean and ripped. Arnold Schwarzenegger at his peak had a BMI much greater than 25, and yet he had very low body fat.

- 8.11** Do you get headaches, seizures, chronic pain, or other neurological symptoms? Do you often feel hot or cold when others don't? Do you have any gastrointestinal problems, upset stomach, nausea, vomiting, diarrhea, constipation, inflammatory bowel disease, irritable bowel syndrome? Do you have any autoimmune disorders? Are you aware of any hormone imbalances, like thyroid hormone? Or if you're a woman, do you have reason to believe that hormones might be changing for you? Do you have excess signs of inflammation, such as bloating, swelling, joint pains, anything that makes you think you might have higher levels of inflammation?
- 8.12** Do you have allergies that are frequent or for which you take allergy medicine regularly? Even seasonal allergies are associated with increased risk of depression and other neuropsychiatric disorders. So any allergy, if you know that you're allergic to mold and dust, that puts you at higher risk right away. Do you have cavities, oral lesions, or do your gums bleed when you floss, if you floss? Have you ever had an infection that required hospitalization? Even though this is beyond your control, if you had an infection that required hospitalization, that probably took a devastating toll on your mitochondrial health.
- 8.13** It is a double whammy. The double whammy is that the infection itself, the inflammation itself, was harming mitochondrial function and may have left cells more vulnerable. Why do I say that? We have an abundance of data. People who get hospitalized for infections have much higher rates of neuropsychiatric metabolic disorders after that hospitalization, all the way from children to adults. In adults, the rates of delirium are through the roof. A lot of adults who get hospitalized for an infection, a lot of them are delirious, meaning they are having wide-ranging neuropsychiatric symptoms. They have much higher rates of progressing immediately to dementia, and they have much higher rates of mortality.
- 8.14** The second whammy in getting being hospitalized for an infection are all of the antibiotics and everything else that they are pumping you with. Trust me, let them pump you with antibiotics if you are that sick and on death's door. Please don't deny the effectiveness of antibiotics. Please don't refuse antibiotics. They will save your life. I am all for antibiotics, but those antibiotics are massively disrupting your gut microbiome. They are making lots of other changes, and that is a toll that needs to be reconciled once you recover. Getting hospitalized for an infection is a serious risk for metabolic and mental health or neuropsychiatric disorders.
- 8.15** For women of reproductive age, do you get significant pain around the time of menstruation? Do you notice changes in mental health symptoms around the time of menstruation? This would indicate that you are particularly sensitive and/or vulnerable to the changes in estrogen, progesterone, and other modulators of brain metabolism. It

suggests that you are definitely in a more vulnerable category than women who don't get those symptoms. And that just puts it on our radar. There's no blame, shame. There's no judgment. You're not doing anything wrong. You can't help it. But it's a warning sign. It's just part of the assessment that you're in a more vulnerable state than other people.

- 8.16** Women, again, have you ever been pregnant? If so, how many times? Even if you miscarried or had an abortion, any pregnancy, every pregnancy takes a metabolic toll on a woman's body. Why? Because you are growing a human being, and that requires a lot of your nutrients. That new human being is taking whatever it needs to grow and thrive, and that can leave women's bodies depleted of essential nutrients. Do we have evidence for this? Yes, we do. The more pregnancies women have, the more likely they are to suffer from metabolic disorders.
- 8.17** And most people just brush it off like, "Well, she's had 10 kids, or she's had four kids. Of course, she's overweight. How could she not be overweight? She's had four kids. Of course, she's developing gestational diabetes." They're not small things, and there are things we can do to help women get through that process. I'm not at all advocating for nobody should be pregnant. I'm not at all advocating for, let's just let the species die off. Not saying that. What I'm saying is that pregnancy takes a metabolic toll on a woman's body. We just need to face that, understand that, accept that. Then we need to develop strategies to help women get through it, to help women be more resilient or recover if they have challenges.
- 8.18** We see high rates, as all of you know, high rates of neuropsychiatric disorders during pregnancy and in particular after pregnancy. The neuropsychiatric disorders are across the board. Across the board. It's not just postpartum depression, it's also postpartum OCD, postpartum psychosis, postpartum mania. So women need help. Women need at least monitoring or women need to advocate for themselves and maybe do a little extra self-care when they're pregnant.
- 8.19** Do you take any prescription medications? If the answer is yes, obviously, there are thousands of prescription medicines. There's no way I could go through every class of them or every one of them and tell you, "Well, this one's OK, that one's not OK." I just want to put on your radar, in the same way that I put on your radar with substance use, sometimes it can be beneficial and sometimes it can be harmful. It's important for you to consider that. Most importantly, I want you to just understand, what impact are these medications having on my metabolic health or, more specifically, on my mitochondrial health?
- 8.20** Sometimes you can Google the medication and mitochondria, and you get an immediate answer. Sometimes that answer is fine and great, and other times that answer is not so great. But knowing is better than not knowing. I am not advocating for everybody stopping their medications. Even if you find bad news or concerning news,

please do not stop your medications on your own, but work with your health care provider and let them know.

- 8.21** "I'm a little concerned about some of these prescription meds that I'm on, and I want to see, can I do something different? Could I use lifestyle strategies, for example, and try to get off of some of these medicines? If I can reduce my blood sugars by going on a specific diet and incorporating more exercise and prioritizing sleep, will you, my health care provider, work with me and get me off this medicine? Let's see if I can get off of it."
- 8.22** That is a very reasonable, solid approach. Your healthcare provider may pooh-pooh you initially and say, "Well, this is going to be work, and I don't have time for that." Push back. Stand up for yourself. Let them know, "No, I want to do this, and I would like you to work with me. If you're unwilling to work with me, then, OK, thanks for letting me know." You don't have to say anything right there because maybe it'll take you a little bit of time to find a new doctor. Keep them on for emergencies until you locate another doctor. But you can say, "OK, thanks for letting me know," and go try to find a health care provider who will work with you.
- 8.23** So, some diet, nutrition questions. Again, there is no one-size-fits-all dietary pattern for all humans, but I want you to think about your typical eating patterns, the foods, the type of meals that you eat now. What time do you eat meals and snacks? Are you pretty regular? Is it erratic? Is it all over the place? Do you eat close to bedtime or in the middle of the night? Those are universally not good. So universally, no matter what diet you're following, no matter what dietary pattern you're following, if you are eating at night, that is not good for metabolic health, and there is emerging evidence that that is also not good for mental health.
- 8.24** Why? Because you're disrupting your circadian rhythm. Circadian rhythms are based on two things—light and dark, and feeding and fasting. When it is dark out, humans are supposed to be sleeping and fasting. When it is light out, we are supposed to be awake, alert, out exploring and eating. Not all the time, but at least some of the time. Do you consume processed foods, ultra-processed foods, excess sugars, artificial sweeteners, beverages that contain many calories? That would be another rule of thumb.
- 8.25** In general, it's better to not drink calories. In general, it is better to drink—Harvard medical school now is recommending coffee, tea, and water. Coffee, tea, water, unsweetened, nothing in the coffee or tea. Coffee, tea, water seem to be the safest beverages. Do you eat at restaurants or order takeout food often? This is a big one, and it's a challenge because some people have very active lives. They travel all the time. It is not going to be easy for them to fix. But I can tell you, most restaurants, 95% of restaurants, are not looking out for your health. They are simply trying to sell people highly palatable, rewarding meals because that is a good business model. If people don't leave full and happy, they're not going to come back.

- 8.26** Most people don't go out to eat looking for a health experience. They're going out as a treat. They're going out for fun. That means if you're eating out a lot, it's probably not going to be ideal for health. There are some strategies that you can employ to change that. Certainly, there are options on the menu, and you can look at that, but that's a risk factor. Are you restricting calories or macronutrients too much? So crash dieting, trying to go long periods of time with fasting, even though you're hungry, irritable, feeling weak or dizzy, that is not good. That is not good. You're not doing your metabolism any favor. Your metabolism is actually plummeting, trying desperately to save your life.
- 8.27** If you have symptoms of serious metabolic or mental health disorders, I would say only if you have symptoms, I don't think these need to be routine labs, but if you do have symptoms, you should probably get levels of folate, vitamin B12, thiamine, vitamin D, and magnesium checked. Low levels of these vitamins, nutrients, can contribute to neuropsychiatric and metabolic disorders. If you have moderate or severe symptoms of mental or metabolic disorders, you might want to consider interventions like intermittent fasting or a ketogenic diet.
- 8.28** I want to come back to the first two questions I asked you all, which is, how's your mental health on a scale of 1 to 10? I want you to look back at the answer if you don't remember it. What did you say? How's your mental health on a scale of 1 to 10? And what did you say it would take for you to rate yourself one point higher? I simply want you to reflect on, would you change anything now? After all of the information we've just talked about, how would you rate your mental health now?
- 8.29** Same with metabolic health. How did you rate your metabolic health when we first started this course? And how would you rate it now, knowing everything you know? And has anything changed? And now I want to talk about treatment. There are many treatment options. If you have a serious mental health condition, I am not throwing out all of the traditional treatments. They are still on the list. That means psychiatric medications can play a role and can be life-saving for some people. Psychotherapy can, too. If things are dire and severe, electroconvulsive therapy, TMS, VNS, ketamine, psychedelics can play a role.
- 8.30** But let's also start thinking about more comprehensive—some people would call them holistic, but I'm going to call them effective— metabolic mental health treatments, like changes in diet or nutrition, social connection, stress reduction, managing substance use, managing sleep, considering vitamins and supplements, exercise, structure and routine, hormones, purpose in life. We're going to summarize them in the nutshell of lifestyle treatments—nutrition, exercise, sleep, substance use, stress reduction, and relationships. Those are the pillars of improving mental and metabolic health, I would argue.
- 8.31** But again, I deviate from the American College of Lifestyle Medicine on the one-size-fits-all nutrition recommendation and advocate for personalized nutrition

interventions, which can be very different than "just eat more fruits and veggies." Fruits and veggies aren't going to help schizophrenia. Next, I want you to create your plan. So all of these assessments, I want you to look at everything that you've written down, all of the assessments.

- 8.32** I want you to first understand that, even if you say, "Well, I don't really care about this. My sleep is, yeah, according to Dr. Palmer, my sleep is not great, but I don't care about my sleep. I'm good with sleeping five hours a night, that's perfectly fine. But what I really care about are my panic attacks. Somebody had better do something about my panic attacks." What I need to let you know is that your poor sleep may be contributing to or causing your panic attacks. And even though you say, "I don't care about sleep, or I'm even going to resist making changes in sleep," I want you to understand that all of the things that you have just identified in these assessments are all interrelated.
- 8.33** There's no way around it. Even if somebody does a microbiome analysis and says, "Your microbiome is off. Changing your sleep and stress levels might get your microbiome set," you may not need to do anything with diet or probiotics or anything else. These things are all interconnected and interrelated. But now what I want you to do is, I want you to identify two things that you care about. Two things on all of the assessments that you've done, two things that you think, "My life might be better if I didn't have so much depression, or I'd really like to be able to do 40 pushups." Or whatever, whatever those are.
- 8.34** I want you to just pick two things that you think, "I care about this, and I'd like these to be better." These two things are going to be the start of your journey. They may be the end of your journey if you can manage them and get them better. You may be able to work on these two things, get them better, and you may actually say, "I'm really happy now. I'm pretty good. I think I'm pretty good now." And that may be the end. Over time, things may drift, and then you want to reassess again and think, "OK, what do I need to work on now?"
- 8.35** But even if these two things aren't the end of your journey, these two things can be the start of your journey. And that's why I want you to start with two things that you care out. Then next, we're going to make them SMART goals. For each one, you're going to make them a SMART goal. What is a SMART goal? A SMART goal means specific, measurable, achievable, relevant, and time-bound. What do I mean by that? I mean that if you say, "I'm drinking way too much, and I, yeah, I think he's right. I probably need to do something about it. I'm going to stop drinking, and I'm never going to drink again," that is a goal. That is not a SMART goal.
- 8.36** It is a goal. If you can accomplish that goal, great. But what I'm telling you now is that, if you don't make it a SMART goal, you are much less likely to succeed. So even if you are an out-of-control alcoholic, simply this all-or-nothing thinking, "I need to stop drinking, I'm never going to drink again," almost no one can do that. Instead, what they need to do is create a specific, measurable, achievable, relevant, time-bound goal.

- 8.37** Now, that goal for that person could be, in the next week, I'm going to look up that Phoenix program that he talked about because I'm on a budget and I don't have any money for anything. I can't find a mental health treater to treat me. I'm going to look up that Phoenix program and I'm going to try to attend one of their online groups. That is specific, measurable; it is realistic and achievable; it is highly relevant, and it is time-bound. In the next week, I'm going to do this thing. That is going to start you on a journey to achieving this goal. Even if the goal is, "I am never going to drink again," you've got to get on a journey to that goal.
- 8.38** Again, if you want to try just complete abstinence, cold turkey, go for it. I'm not going to stand in your way, but it's unlikely to work. If your goal is, "Oh, my gosh, I'm obese. I need to lose probably 250 pounds. I'm going to lose 250 pounds," That's a goal. That is not a SMART goal. That is a goal, but it is not easily achievable. Losing 250 pounds is extraordinarily difficult, and it's going to take a lot of time, a lot of time, a frustrating amount of time. If you want to embark on a journey to lose 250 pounds, I still want you to turn it into a SMART goal.
- 8.39** And the SMART goal is not even going to be, "I'm going to lose weight," because that is the desirable outcome that you can measure. So that is the measurement that you are going to use. But you have to be more specific and make it achievable. If you're going to go on a low-carb diet, it could be, in the next week, I am going to read a book or read some information about low-carb diets and learn more in the next week. So I'm going to prepare. I'm going to learn. The next week, I am going to try to reduce my carbohydrates by this much, for a week, and that's it. That's what I'm going to do. That is my specific goal.
- 8.40** I'm going to measure the effect on my weight, in the hopes that it will make a way. But what I'm really measuring is my carbohydrate intake. I'm measuring my carbohydrate intake. It's very specific. It is achievable. It is highly relevant because you have reason to think this might help me lose some weight, and it is time-bound. I'm going to do it for a week, and then I'm going to reassess. That can be the start of a journey toward 200 pounds of weight loss. But saying, "I need to lose 200 pounds," and then beating yourself up when you go to your friend's wedding next weekend and binge-eat on all the wedding cake and drink extra, that's a setup for failure.
- 8.41** It's a setup for failure because it's not specific, measurable, achievable. It's not time-bound. And you are setting yourself up to feel like you're a loser, to feel like, "I can't do anything. Other people can lose 200 pounds. Why can't I?" Well, it's not an easy thing. Each of your goals, I want you to just, over time, that even could be your SMART goal. Over the next week, I am going to list out two SMART goals based on the two things that I've identified that I want to change.
- 8.42** Next, I want you to think about an accountability partner. Is there someone that you can share this with to be your accountability partner? What could you ask of them that would help you follow through with the steps that you need to take? This could be a

romantic partner, a spouse, girlfriend, boyfriend, family member, brother or sister, just a regular friend. If you really don't have anyone, you could consider hiring professional help. Therapists can help with this, health and wellness coaches. This is what they do. There are lots of people available for hire that will be an accountability partner.

8.43 But I really want you to think critically about, how can they help and how exactly will it be helpful? My preference is that the accountability partner is not a punisher. That you're not going to ask your accountability partner to yell at you if you don't do it because that's not going to be very helpful. And then you're going to be really hesitant to work with this accountability partner going forward because it just becomes a really negative, toxic situation that just, again, allows you to beat yourself up, allows you to feel like a loser. And now what? Now you're going to invite somebody else to attack you, too? That's not a very good setup. That's not a setup for success.

8.44 What could an accountability partner do? They can do lots of things. If nothing else, they could maybe send you texts or call you or check in with you and just say, "How's it going?" That can be highly motivating for people, because now they've put themselves out on the line. "I'm going to do this thing over the next week. Gosh, now I ask this person that I really like to check in with me. I haven't really done anything. I'm getting kind of procrastinating or I'm having second thoughts, or I just got really busy. Life got busy. Life gets busy." Well, if this person is checking in with you every now and then, like, "Hey, how's that thing going?", it's going to motivate you, hopefully, to do it.

8.45 You're going to talk with them ahead of time. "If I'm not doing it, please don't yell at me. Maybe, be helpful. Help me solve the problem of what's getting in the way. How can I do it? How can I find the 10 minutes that I need to do this? What needs to change?" They could be a coach or a mentor. Again, there are professionals you can hire, but I'm a huge fan of using your own community. Use your community. These people are going to do this for you, and guess what? You're going to give back. When they need help, you're going to help them.

8.46 Maybe they need help moving to a new apartment. You're going to be the friend that actually steps up and helps them move to that new apartment because that's part of connection and community and purpose. And that's good for your health, too. This process is going to include other people. You're not on this journey in isolation. You're not doing this yourself. You're going to include other people. You're going to ask for help, but you're also going to be prepared to offer your own help.

8.47 If they see you improving your health, they're probably going to get curious. Like, "What's going on? Hey, what about me? I could stand to lose a few pounds." That's a great accountability partner. "You want to do it with me? Hey, we'll both do it." That's amazing. And that's when things really take off, if you can find other people who are pursuing the same goal. So accountability partners, it could be somebody in an AA group, somebody in some other self-help group. You can go online and find all sorts of community in social media groups.

- 8.48** I want you to prepare for plateaus and setbacks. They're going to happen. It's going to happen. When you implement your plan and experience setbacks, or you don't achieve the results that you anticipate, what are you going to do? Let's say your goal is, "I'm going to go on a low-carb diet and I'm going to try to lose weight," and the first three months go phenomenally well. In three months, you lose, let's say, 30 pounds. Then you keep doing the exact same thing for the next month, and the scale doesn't budge. That's called a plateau.
- 8.49** And I'm telling you now, that is normal. That is your body's adaptation to try to save your life. Your body is sensing food scarcity. Body weight is changing rapidly. Lower metabolism, lower metabolic rate. Save this person. Save this person's life. This person is starving to death. Rapid weight loss, starving. How are you, psychologically and pragmatically, going to manage that situation? So, many people will give up in frustration, and that's it. That's the end of it. "A month, I haven't lost any weight. I'm doing everything right. This doesn't work for me," and then they're done. Game over. They go off the diet, they regain all the weight, sometimes even more, and then they feel like even a bigger loser.
- 8.50** What can you do to ensure this doesn't happen to you? This is one of the really important ways that your accountability partner can help. Because, again, I want you to tell them, "I'm going to experience plateaus and setbacks, and you're going to be my trusted friend in that situation. I'm going to be really frustrated when that happens, and I'm going to want to give up, and you're going to be the friend that doesn't let me. You're going to be the friend that listens to me whine and complain and talk about how unfair it is, and you're going to stay with me."
- 8.51** The same thing goes for somebody who's trying to give up alcohol. "I'm an alcoholic. I need to stop drinking." They give up drinking for three months. Their girlfriend breaks up with them. What do they do? They go on a bender. "Well, I was giving up alcohol because I thought she was really mad at me, and that was, she was the whole reason I was giving up alcohol, and now she dumped me, and she's moving on, and I'm such a loser, and I have no love anymore. She was the only reason I was really doing it. So, screw it all. I'm going to go on a bender." That's called a setback or a relapse. It's expected.
- 8.52** Your accountability partner is going to help you pick up the pieces when you come out of your bender. Your accountability partner is going to say, "Yeah, that happens. And yet, you got three months under your belt. How did you do it? What did you do? Let's implement those strategies again. Let's keep going. You're unlikely to get another girlfriend if you keep drinking like this. You're probably going to be dead if you keep drinking like this. You need to keep going forward."
- 8.53** Are there resources that might be helpful? I really want you to think through these plateaus and setbacks, because these are probably one of the primary reasons people don't stick with it. They reach a plateau or a setback, they get frustrated, they feel like

it's not working, they feel like a loser, and then they go back to their old habits and say, "This isn't going to work for me." Are there other resources that can help, other than just this accountability partner? Do you have a healthcare provider that could be helpful? Are there experts that you know, even if they're just online experts?

- 8.54** Maybe there's a Peterson Academy course that you want to, like, "I'm going to go watch that course again when that happens." Maybe there are family or friends, there are books, there are websites. Think about resources that might be helpful in those times when you're going to struggle. Potentially, most importantly, do you know anybody who has done what you're trying to do? Do you know somebody who's actually lost 200 pounds? Do you know somebody who's given up alcohol, who was an alcoholic? Those are going to be invaluable resources, even if they're not available to you as the accountability partner.
- 8.55** Maybe you don't know them that well, but you might just want to let them know, "Hey, could I check in with you every now and then?" Or maybe you'd call them up when you're in the setback or in the plateau, and you then reach out for help. And I just want to say, every recovery journey begins here, and here is hope. Every recovery journey begins with hope. Nobody will start any of this if they can't see a path to a better life. If nothing else, if you yourself, if you're fine, if you think, "I'm good. I'm in a good place. I've got good metabolic mental health," but you know someone else who isn't, I want you to remember, if you want to try to help that person, their recovery journey begins with hope.
- 8.56** It does not begin with a lecture from you about how they need to go on a diet. It does not begin with a lecture from you about how they're drinking too much and they're an alcoholic out of control and they need to get their stuff together. That induces shame. Recovery journeys don't begin with shame. Recovery journeys begin with hope. You have to try to somehow let them know, "There is a better life possible for you. I actually see that you could potentially achieve it."
- 8.57** With a lot of people, it's really important, if you're trying to help them and you care about them, to let them know that you think they deserve it. They are worth it. They are worthwhile enough to fight for, even if they are the ones who have to do the fight. With that, I really want to just put in one final plug. I hope people might consider joining our grassroots movement. We want to change the mental health field. We want integrated mental and metabolic health. So please consider joining our group at brainenergy.com. And with that, I thank you.
- 8.58** If someone is on a ketogenic diet and they eat a large serving of protein, their glucose goes up, their ketones go down. Is that related to gluconeogenesis? Can you explain what's happening there?
- 8.59** More than likely, it is related to gluconeogenesis. Protein can be very easily converted into glucose by the liver. And so, if you are primarily eating protein, like a carnivore

diet, especially if it is low-fat, so if you're eating very lean protein, lots of turkey and chicken and lean fish, and that's all you're eating, you're unlikely to be in a state of ketosis. You're probably going to have fairly regular levels of glucose. That can be a really good diet. It's still a low-carbohydrate diet and can result in weight loss for a lot of people, but it is not a ketogenic diet.

- 8.60** The body's preferred fuel is, in fact, glucose. Glucose is the default fuel for most adult humans in most societies. If the body is given the option of glucose or ketones, it is almost always going to revert to glucose. Very low levels of ketones are always produced, but that's basically the gist of eating too much protein. For people who are doing low-carb or keto diets, it really does matter. I mean, if your primary goal is weight loss, you may, in fact, be able to eat as much protein as you want. And whether you get high levels of ketones or not may not be relevant for you. You might be able to lose a tremendous amount of weight, simply by reducing insulin signaling and reducing glucose levels and other pathways.
- 8.61** So you might be able to do an all-you-can-eat protein fast. And the original Atkins diet was that. It was just, eat as much protein as you want. It doesn't really matter. Just restrict your carbs. That was the rule. Restrict your carbs and do whatever you want with protein and fat. And literally for millions of people, that helped them lose weight. So we know that that's effective. And the weight-loss studies that I shared with you, some of them I shared with you, document in clinical trials, yeah, it actually works. Just telling people, reduce carbohydrates to 20. Don't give them any other instructions. They can do whatever they want. They'll lose weight.
- 8.62** That often will not, in and of itself, stop seizures. That's why the Johns Hopkins group developed the modified Atkins diet, which says, yeah, kinda sorta reduce carbs, eat as much meat as you want, but add fat. Add fat at every opportunity you can think of. That is no longer optional. We're really giving you two instructions. Reduce carbs and add fat. Make sure you get enough protein, but as long as you're getting enough protein, just keep adding fat every opportunity you can think of.
- 8.63** Pour some olive oil, put some butter on, have some heavy cream, have avocados, have nuts, whatever fat sources you like. For people using ketogenic diet for neuropsychiatric conditions, I'm not convinced that they can get away with the all-you-can-eat protein buffet and get the health benefits that they desire.
- 8.64** Throughout the lecture, you mention intermittent fasting and fasting, and wondering if you could expand upon that and maybe provide some insight for anyone who's interested in beginning that journey.
- 8.65** So, I guess I have to start that conversation with, there are some people that I think should not do intermittent fasting and certainly should not do prolonged fasting. The two groups are people who are already underweight or malnourished. They are having a metabolic problem because of deficiency of nutrients and calories. And so,

encouraging them to go without nutrients and calories doesn't make sense, and it can be really harmful. So telling somebody with anorexia nervosa, like, "Intermittent fasting could help you." They're already doing more than enough intermittent fasting, probably. And if anything, they need to stop doing it.

- 8.66** The other category of people that I would be reluctant to prescribe prolonged fasting for are young, growing, normal-weight children. If children are growing and they actually do need calories and they are trying to grow, and if you calorie-restrict them too much or nutrient-restrict them too much, you'll actually have an impact on their growth curves and other things. Children, underweight people, let's take it off the table as options for them. But as you probably know, there are millions and millions of people who could benefit from intermittent fasting and fasting.
- 8.67** That includes all of the people who have overweight or obesity, but even other people who are normal weight but may have metabolic dysfunction of one kind or another. You can be normal weight and have type 2 diabetes or have a neuropsychiatric disorder. I think that it really does need to be individualized for the person, so I don't have one-size-fits-all strategies. But the guidelines that I would give people to consider or think about, if they want to experiment with it, would be maybe start with maybe intermittent fasting, which for some people might be skip one of the meals that you normally eat. If you normally eat three meals a day, see if you can go without one of them.
- 8.68** Sometimes people will do time-restricted eating. If you normally eat beginning at 6 AM, as soon as you wake up, and you snack until 10 PM, right before you go to bed and you have a bedtime snack, maybe intermittent fasting, or it could include just try to get that eight-hour fast that you do from 10 PM to 6 AM a little longer. Get a little more time in there, whichever side of the equation you want. Stop eating at 9 PM, stop eating at 8 PM, stop eating at 7 PM. Slowly but surely, restrict your eating window, and that could be considered intermittent fasting.
- 8.69** Early on, people will be hungry, slightly irritable. They'll be put off. I would say that is normal and expected. Don't be deterred by that. Don't give up because of that. Don't think it's not working for you because of that. You're not going to feel amazing the very first day you go without food. You're actually going to feel quite not amazing. But don't do it for more than two weeks if you continue feeling really miserable. Most people, within about four, five days, will actually start to notice benefits when they do either intermittent fasting or fasting. Most people will start to notice benefits, and the benefits are, "I have a little more alertness. I have a little more energy."
- 8.70** Because you're trying to teach—so one of the benefits of that is that you're trying to teach your body to tap into other fuel sources. Instead of relying on carbohydrates, fats, and protein that you're eating, you're trying to teach your body, train your body, to tap into existing reserves. That actually is a process. Some people lose, not completely lose the ability, but the ability to tap into fat stores and other fuel sources is

diminished profoundly. Mitochondria play a role in all of that because mitochondria and fat cells are actually the things that are releasing the fat from the fat cells. You can't release fat from a fat cell without mitochondria. So certainly plays into it.

- 8.71** Do I ever use fasting? I don't use fasting a lot, but I do use it in some cases. In patients who are really overweight or obese, especially in patients with type 2 diabetes, I've used prolonged fasting, up to seven days, and it can sometimes be almost miraculous. I want to say, I am a medical professional supervising that, and I am carefully monitoring their blood glucose levels and their medications, and I am rapidly adjusting their diabetes meds when I'm doing that. So I'm not encouraging people to do that on their own and wing it because they'll probably get into trouble.
- 8.72** But I can tell you, it can feel like nothing short of a miracle. I mean, again, the first two to three days, people are irritable, hungry, weak, lethargic. If they can get through that, something shifts. They're really going into ketosis. And that's when everything starts to change. And at day four and five, a lot of people will start saying, "I'm not hungry anymore. It's shocking how I can get through the whole day without food, but I'm not hungry. Instead of thinking about food and preparing food and eating food, I'm doing things. I'm so productive now. This is amazing. I didn't know I could be this a person."
- 8.73** Much more importantly, people just notice tremendous improvement in mood, energy, motivation, other things. And so they just feel great. Again, in patients with severe neuropsychiatric disorders, I worry about triggering hypomania and mania in those cases. I would want those people to work with mental health professionals to make sure they're not becoming manic or hypomanic. But yeah, I think fasting and intermittent fasting can be quite powerful. We have people like Jason Fung in Canada, actually, but he's known in the US and Canada. He's like the fasting guru. And he's treating tens of thousands of patients for weight loss, diabetes control, and other conditions using fasting and intermittent fasting.
- 8.74** But in Germany, they actually have fasting clinics, people with all sorts of health conditions that aren't getting better with conventional allopathic treatment. They go to these fasting clinics. I've talked to many people who say it's nothing short of miraculous, that somehow doing like a—sometimes they'll fast people for up to a whole month. Sometimes doing these extended water fasts, they are medically supervised. So again, these are clinics. They're not, you're checking yourself in, you're getting medical supervision. But some people leave those with like, "My joint aches are going away, my GI symptoms are going away, my autoimmune symptoms are going away." It can really be powerful.
- 8.75** And although it sounds like quackery to most allopathic people, I just want to point out again, that treatment has been used in almost every culture on Earth for millennia. Let's just start with that. Were all of those humans really stupid or were they observing something that was actually real? Then secondly, the science has caught up. We have a tremendous amount of science published in the New England Journal, Nature, Cell, and

other prestigious journals documenting the physiological, biological effects of fasting and how it might help with a wide range of health conditions.

- 8.76** My question is about supplements, and you can find now online a lot of supplements that support cell health, mitochondria, help with this function. Is it safe and is it efficient to use and what can be harm? What's the danger?
- 8.77** It's a great question. It's a question I get asked a lot, and I don't have a really easy answer because there are literally hundreds, if not thousands, of supplements. And I will absolutely say up front, some of them may, in fact, or can be really beneficial to people who have metabolic or mitochondrial disorders. There's no doubt about it. We have published research studies documented. It seems that some supplements seem to be quite helpful to some people. And on the other end of the spectrum, with supplements, you never always know exactly what you're getting.
- 8.78** If you just go to a local drug store and buy a supplement off the shelf, you could be getting a supplement from China, and that could contain tainted molecules, sometimes poisons that get recalled. Because the FDA is not highly scrutinizing supplements. It's a different level of scrutiny. And you don't always know that you're getting what is advertised on the label even. And you certainly don't know that you're getting the dose that's advertised on the label. A lot of times you're not.
- 8.79** We've seen studies where the researchers go in, test the supplement. See, this is supposed to have 100 milligrams or 1 milligram, whatever, of melatonin, say, and they can be wildly all over the place. Some of them have 0.1 milligrams of melatonin, others have 8 milligrams of melatonin, all the same label package. On melatonin, there was recently a scandal, about some melatonin gummies that people were giving their kids had CBD products in them. And it didn't label it as CBD. They were just adding CBD. Why not? So, wide range.
- 8.80** I think that the thing that I would say is this. If people aren't doing the basics of health, supplements probably are not going to help at all. More often than not, people are not doing the basics of health. As a rule of thumb, I do not recommend supplements. As a rule of thumb, I recommend the basics of health that I've covered in this course. All of the lifestyle medicine things, but also, what medications are you taking, even if they're prescription medicines? That's where I diverge from allopathic medicine. I am open to the possibility that medications might be harming people in some circumstances and that sometimes we need to de-prescribe medicines. But that's, in my mind, part of the basics.
- 8.81** If all of the basics are in place, and a person is still struggling and not improving, then I am all for additional strategies to improve mitochondrial health, and that includes supplements. There are some supplements—urolithin A, for example, is available by a couple of companies. Urolithin A probably has reasonably good evidence that it improves mitochondrial function and that that actually can confer some health benefits

to older adults, for example. The health benefits are improved muscle capacity. Urolithin A can result in improved mitochondrial health in the muscles of older people, and that can result in higher exercise capacity as a result, just as a result of adding that supplement compared to a placebo.

- 8.82** Antioxidants can be helpful for some people, but by no means are they a panacea. There is a wide range of antioxidants. Melatonin can be great if you're suffering from jet lag and you need to reset. If you need something to reset, I'm not opposed to using melatonin. I'm a much bigger fan of taking a light box with you. If you're traveling around the world and you're going to be jet-lagged, I actually travel with a tiny little portable light box. Use the light at the time that you want to be waking up, and that will help you reset your circadian rhythm, in my mind, much more powerfully than any pill is going to do. And it's light. Very few side effects.
- 8.83** So I think that's the bottom line. I've covered some studies. There was one study done in patients with bipolar disorder where they use this, over 20 supplements that were all targeting mitochondrial health. That was the whole point of the study. The researchers know people with bipolar disorder have mitochondrial dysfunction. We think that that is driving their bipolar disorder. So we are going to flog them with all of these mitochondrial supplements. They didn't do anything. They didn't help.
- 8.84** And they probably didn't help because they were probably at the same time prescribing antipsychotics and mood stabilizers that were dramatically impairing mitochondrial function. Who knows what was going on with diet, sleep, substance use, and other things that were impacting mitochondrial function. So I would say that we do have pretty good evidence that supplements alone are never going to be the answer or are rarely going to be the answer. Hope that helps.